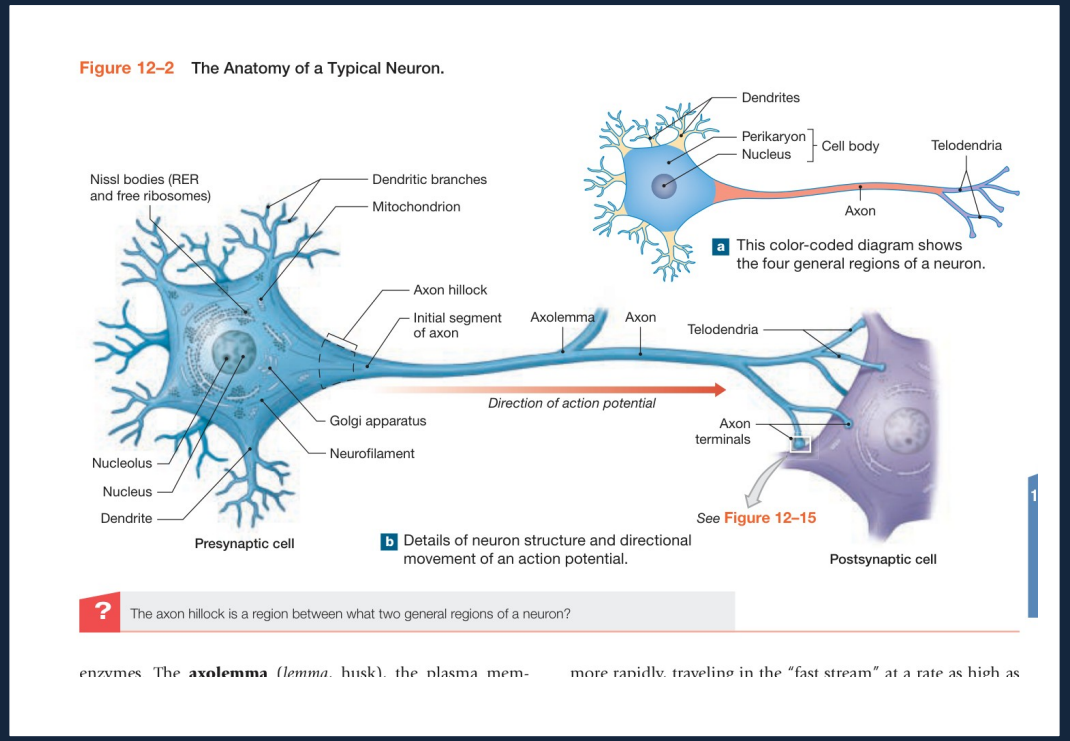
STUDY TOGETHER • TEACH IT BACK

BIO 233

Chapters 12 + 13

Nervous tissue. Spinal pathways.
Reflexes that protect and coordinate.


Figure 12–2 The Anatomy of a Typical Neuron.

Dendrites
Perikaryon
Nucleus
Cell body
Telodendria
Axon
a This color-coded diagram shows the four general regions of a neuron.
Nissl bodies (RER and free ribosomes)
Dendritic branches
Mitochondrion
Axon hillock
Initial segment of axon
Axolemma
Axon
Telodendria
Direction of action potential
Golgi apparatus
Neurofilament
Axon terminals
Nucleolus
Nucleus
Dendrite
See Figure 12–15
Presynaptic cell
b Details of neuron structure and directional movement of an action potential.
Postsynaptic cell
? The axon hillock is a region between what two general regions of a neuron?

Detailed visual review • Memory cues
Partner checks • Source cautions
Separate one-sided, text-only exam sheet

Prepared for Waleed and a classmate

117 slides

Use this deck as a conversation, not a script

EXPLAIN

One person teaches the slide in their own words; the other asks “why?” and “what happens if it fails?”

RECALL

Cover the labels. Name the structure, its location, its function, and the direction of information flow.

CHECK

Work the original practice questions before viewing their answers. Use the glossary and source notes for weak spots.

PRIORITIZE

LECTURE PRIORITY identifies explicit or repeated emphasis. BOOK DETAIL adds assigned reading; it is not a prediction of the exam.

REMEMBER Recognize it → name it → explain it → predict a change.

What this review can—and cannot—promise

COVERAGE

All 16 chapter learning-outcome sections, all six supplied handouts, and the four available transcript files are represented.

RECORDING LIMIT

The recordings are incomplete. The later Chapter 13 reflex material is included from the handout and textbook even where no recording is available.

SOURCE CAUTION

Speech recognition and some lecture/book statements disagree. Marked cautions preserve the distinction instead of silently treating everything as fact.

EXAM RULE

The recorded instruction allows one side of a standard sheet, typed or handwritten text. Any image must be drawn by hand—not printed or photocopied.

REMEMBER Comprehensive source coverage is not a guarantee of the exact questions your professor will choose.

C12/C13 learning outcomes; T-C exam instructions.

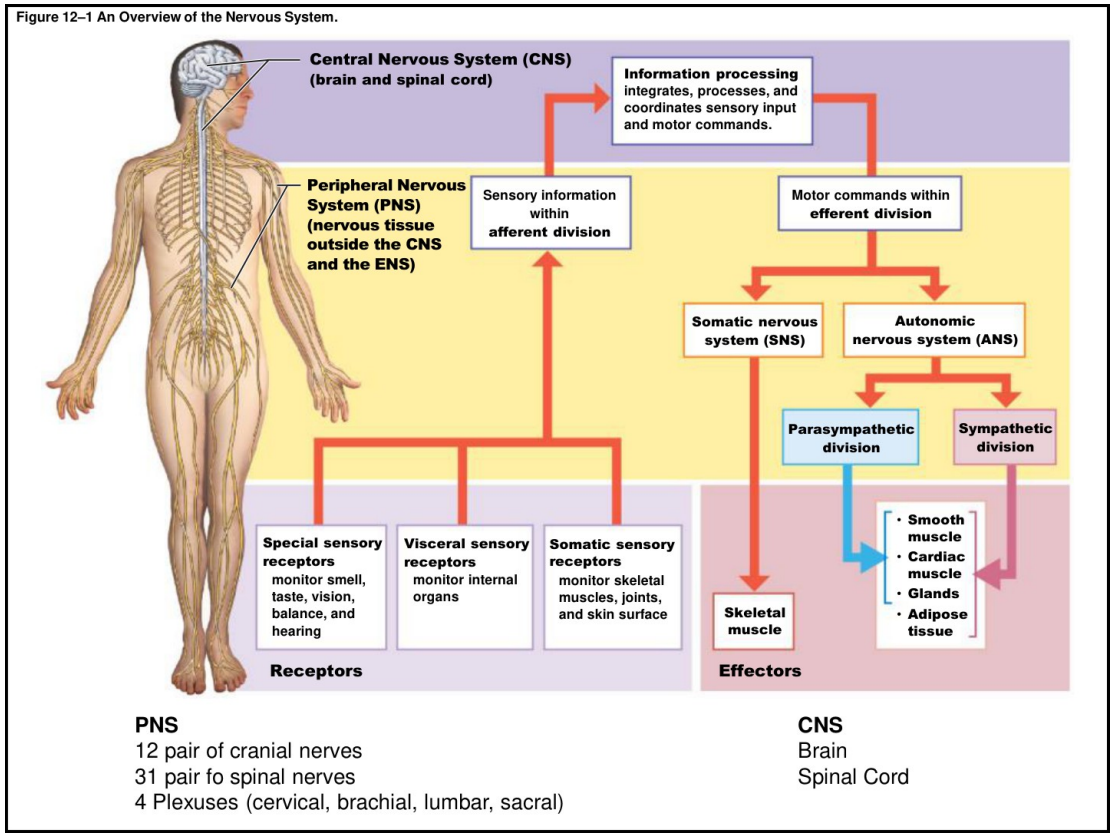
1

Organize the nervous system

Start with the map, then identify the cells and the structures that support them.

Use the memory cues, then explain the mechanism without reading.

The anatomical map: CNS, PNS, and ENS



CNS

Brain and spinal cord: integrate sensory input, process information, and coordinate motor commands.

PNS

Nervous tissue outside the CNS and ENS in this textbook's framing. Cranial and spinal nerves connect the CNS with peripheral tissues.

ENS

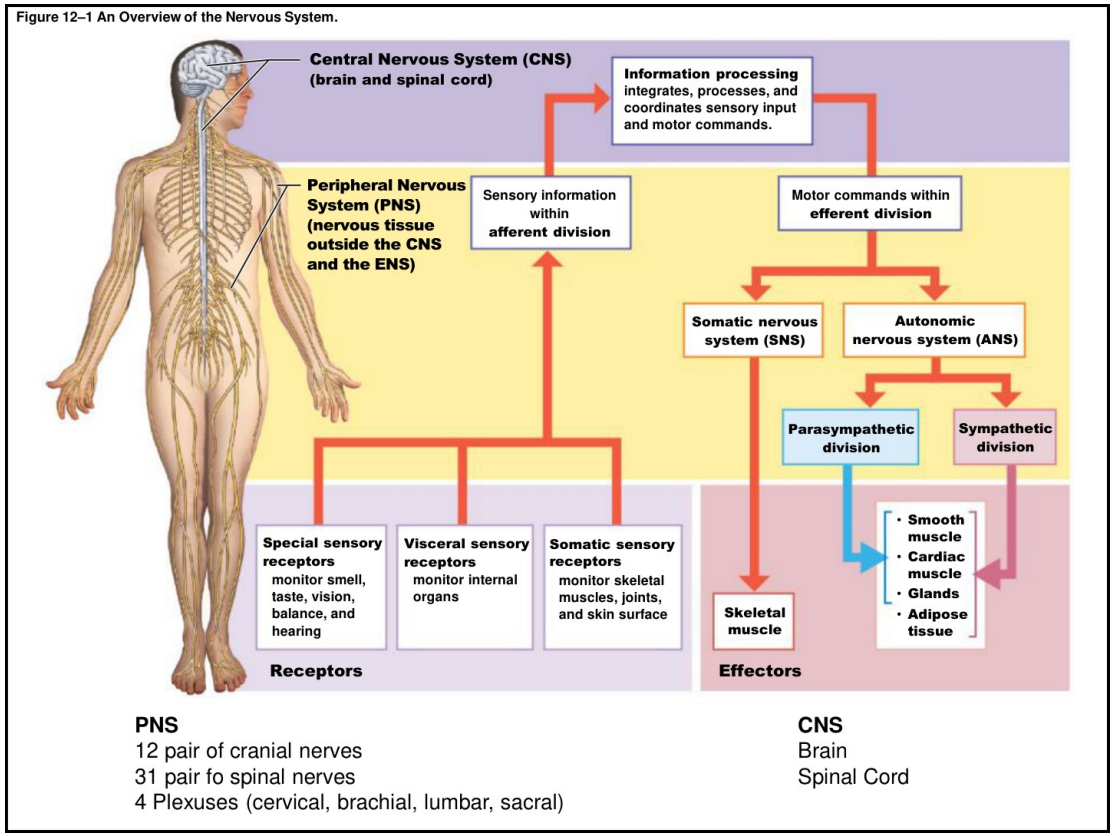
Neuronal networks in the digestive-tract walls. Can coordinate local visceral reflexes; sympathetic and parasympathetic activity influence them.

BIG DISTINCTION

Anatomical divisions describe structures. Functional divisions describe what information travels through those structures.

REMEMBER Central = control center. Peripheral = connecting paths. Enteric = intestine.

Afferent arrives; efferent exits



AFFERENT / SENSORY

Receptors send information toward the CNS: special senses, somatic sensation, and visceral sensation.

EFFERENT / MOTOR

Commands leave the CNS for effectors. “Motor” includes muscles and glands—not just visible movement.

SOMATIC MOTOR

Targets skeletal muscle. Includes voluntary movement and involuntary somatic reflexes.

AUTONOMIC MOTOR

Targets smooth muscle, cardiac muscle, glands, and adipose tissue. Sympathetic and parasympathetic are its major divisions.

REMEMBER SAME: Sensory–Afferent; Motor–Efferent. “A arrives, E exits.”

Three receptor groups: where is the information from?

Term	What it monitors	Think of…
Exteroceptor	External environment; touch, temperature, pressure and special senses	A hand touching a cold desk
Interoceptor	Internal organs; stretch, deep pressure, internal conditions	Distension of a visceral organ
Proprioceptor	Position and movement of skeletal muscles and joints	Touching your nose with your eyes closed

REMEMBER EXTERO = exterior; INTERO = interior; PROPRIO = your own position.

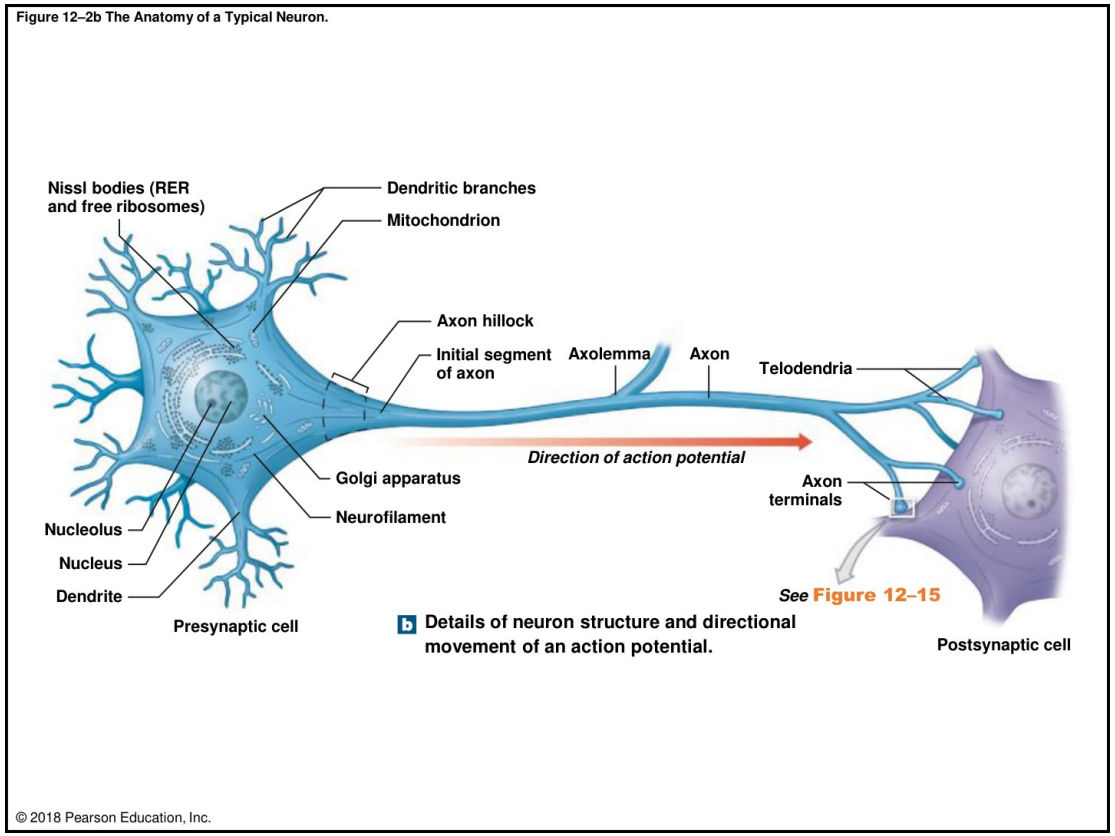
Do not confuse these building blocks

Term	Meaning	Location or scale
Neuron	A single signaling cell	One cell with processes
Nerve	Axon bundles with connective tissue and vessels	PNS
Tract	Axon bundle carrying related information	CNS
Ganglion / ganglia	Collection of neuron cell bodies	PNS
Nucleus / nuclei	Functionally organized collection of neuron cell bodies	CNS

REMEMBER A nerve is not a neuron. A ganglion is not a CNS nucleus.

Book pp. 390, 394–395, 440, 442; T-A/T-C.

Neuron anatomy: trace the signal



INPUT REGION

Dendrites receive information. The cell body is the soma; the perikaryon is its cytoplasm around the nucleus.

TRIGGER REGION

Axon hillock is the transition from soma to axon. The initial segment is the key action-potential initiation region in the chapter's model.

CONDUCTING REGION

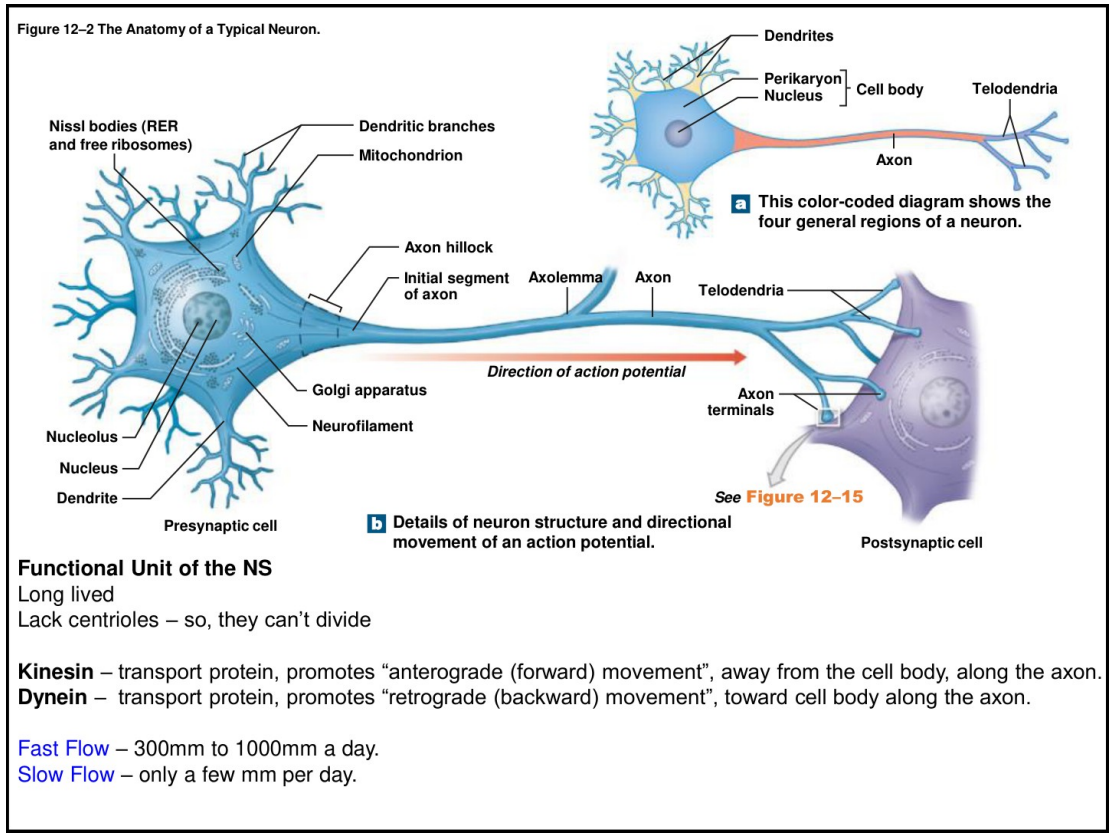
Axon carries the action potential. Axolemma = axon membrane; axoplasm = axon cytoplasm.

OUTPUT REGION

Collaterals branch along an axon. Telodendria are fine terminal branches ending in synaptic terminals.

REMEMBER Dendrites receive; axon carries; terminals communicate.

Axonal transport: supply line and return route



Functional Unit of the NS
Long lived
Lack centrioles – so, they can't divide

Kinesin – transport protein, promotes "anterograde (forward) movement", away from the cell body, along the axon.
Dynein – transport protein, promotes "retrograde (backward) movement", toward cell body along the axon.

Fast Flow – 300mm to 1000mm a day.
Slow Flow – only a few mm per day.

ANTEROGRADE

Kinesin carries material away from the cell body toward terminals along neurotubules.

RETROGRADE

Dynein carries material from terminals back toward the cell body.

ENERGY + SPEED

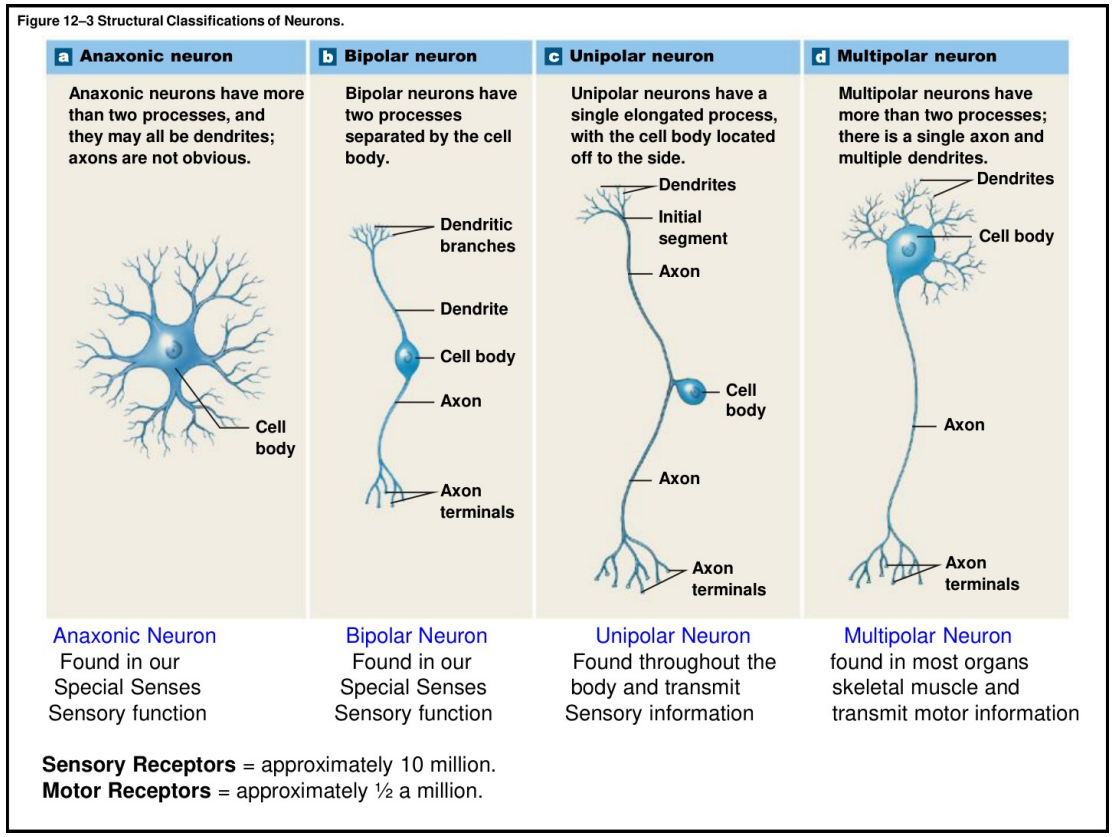
Transport uses ATP. The book gives fast transport up to 1,000 mm/day, averaging about 300; slow transport is a few mm/day.

WHY IT MATTERS

Terminals need proteins, enzymes and vesicle materials. Retrograde transport also returns debris and can carry harmful material toward the CNS.

REMEMBER Kinesin keeps going out; dynein delivers back.

Four neuron shapes



ANAXONIC

Many processes but no obvious axon; found in the brain and special sense organs in the textbook.

BIPOLAR

One dendrite and one axon, with the soma between them; associated with special senses.

UNIPOLAR

A single process with the soma off to the side; typical sensory neurons of the PNS.

MULTIPOLAR

Many dendrites and one axon. Includes skeletal motor neurons and many CNS neurons; shape alone does not prove a motor function.

REMEMBER BI = two processes. MULTI = many dendrites. UNI = one main process.

Functional classes answer a different question

Class	Direction / role	Key detail
Sensory neuron	Peripheral receptor toward CNS	Cell bodies commonly in sensory ganglia
Motor neuron	CNS toward a peripheral effector	Somatic targets skeletal muscle; visceral targets other effectors
Interneuron	Integrates, distributes and coordinates signals	The chapter emphasizes interconnected CNS neurons

REMEMBER Structure asks “what shape?” Function asks “what job?”

Book pp. 394–395, 452; C12 L1 slide 5; T-B/T-C.

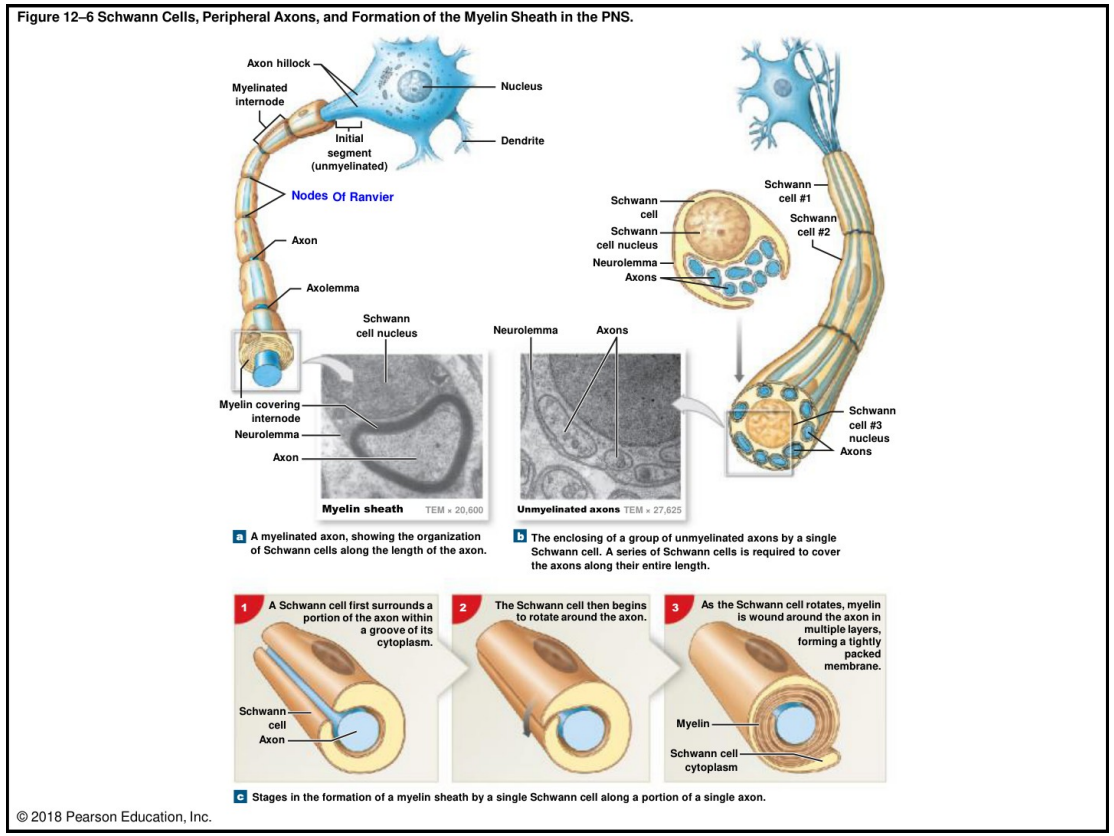
The four CNS glial cells

Cell	Main jobs	Memory cue
Astrocyte	Structural support; regulates local ions/nutrients; supports BBB; neurotransmitter recycling; repair/scar response	Astro = star-shaped support
Ependymal cell	Lines ventricles and central canal; helps produce, monitor and circulate CSF	Ependymal = lining
Oligodendrocyte	Myelinates CNS axon segments; supports organization	Oligo = CNS insulation
Microglial cell	Phagocytosis of debris, wastes and pathogens	Microglia = cleanup crew

REMEMBER A–E–O–M: Astrocytes, Ependymal cells, Oligodendrocytes, Microglia.

Book pp. 395–399; C12 L1 slides 6–8; T-A/T-B/T-C.

The two PNS glia—and the myelin comparison



SATELLITE CELLS

Surround neuron cell bodies in ganglia and regulate their local chemical environment.

SCHWANN CELLS

Enclose peripheral axons. A myelinating Schwann cell forms one internode on one axon; many cells cover the full length.

OLIGODENDROCYTES

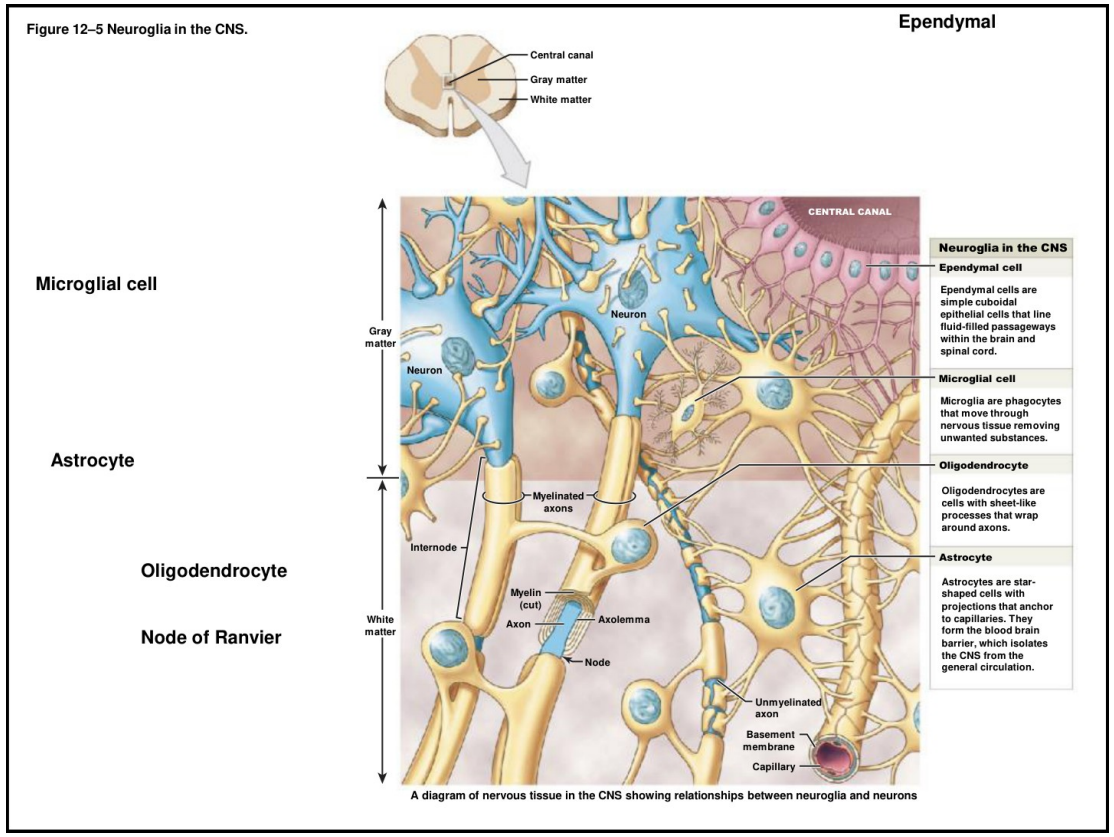
A CNS oligodendrocyte sends processes to myelinate segments of multiple axons.

UNMYELINATED ≠ UNPROTECTED

A Schwann cell can enclose several unmyelinated axons without winding a thick myelin sheath around them.

REMEMBER Schwann = peripheral sheath; oligodendrocyte = central, multiple segments.

Myelin, nodes, gray matter and white matter



MYELIN

Lipid-rich membrane wrapping that electrically insulates axon segments and increases propagation speed.

INTERNODE / NODE

Internode = myelinated segment. Node of Ranvier = gap between internodes where action potentials are regenerated.

WHITE MATTER

Rich in myelinated axons; organized mainly for moving information between locations.

GRAY MATTER

Rich in cell bodies, dendrites and unmyelinated processes; important for integration. In the cord it forms the central butterfly.

REMEMBER White = wires; gray = processing groups. These are helpful summaries, not exclusive contents.

Peripheral repair: degeneration comes before regrowth

Figure 12–7 Peripheral Nerve Regeneration after Injury.

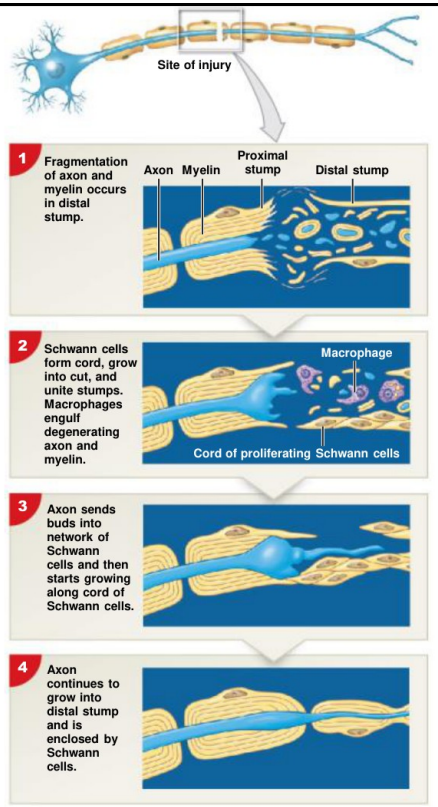
Wallerian Degeneration: the process of nerve regeneration following an injury

Limited Regeneration in the CNS

1. More axons likely to be involved.
2. Astrocytes produce scar tissue that can prevent growth across the damaged area.
3. Astrocytes release chemicals that block regrowth of axons.

Spinal Cord Injury

20,000,000,000 interneurons in the CNS, many are in the SC.



1 | DISTAL BREAKDOWN

After severe axon injury, the disconnected distal axon and its myelin degenerate: Wallerian degeneration.

2 | CLEANUP + GUIDE

Macrophages remove debris. Schwann cells proliferate and form a guiding cord.

3 | AXON REGROWTH

The surviving neuron can send sprouts into that pathway; Schwann cells enclose the regrowing axon.

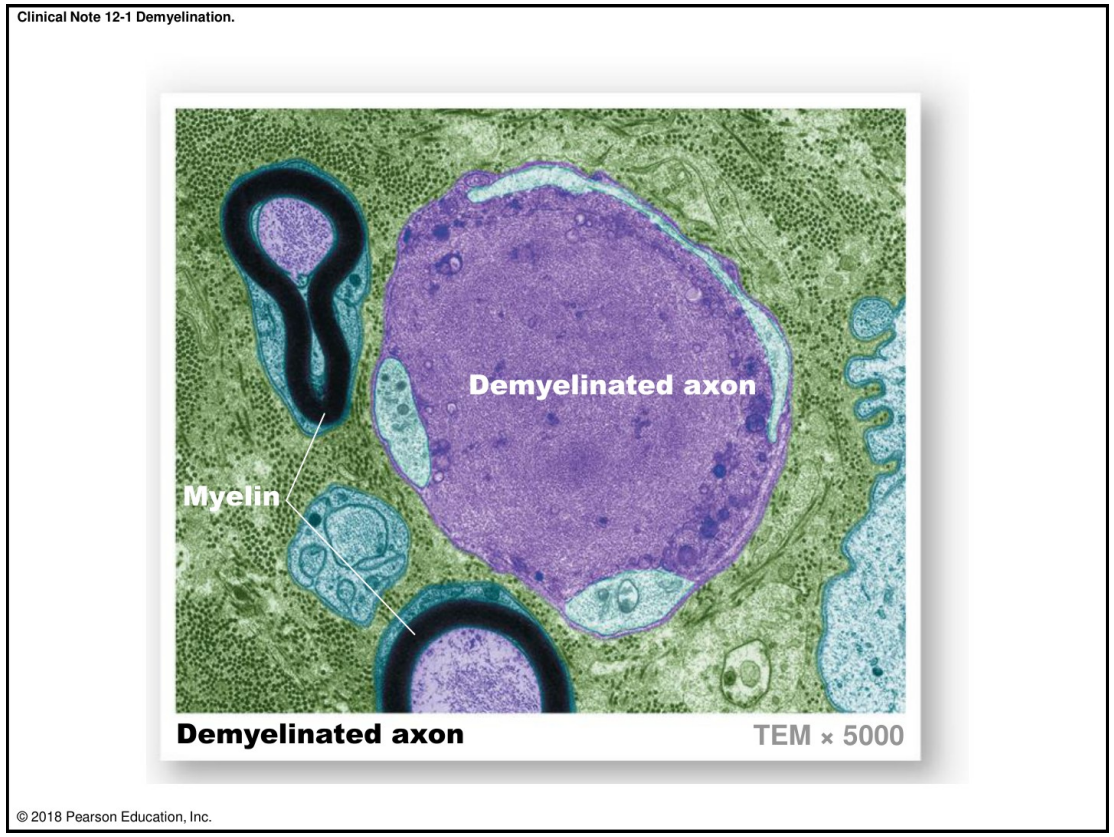
4 | FUNCTIONAL OUTCOME

Recovery depends on regrowth reaching the correct target. Misalignment, a large gap or failure to reconnect can prevent recovery.

REMEMBER Break down → clean up → guide → regrow → reconnect.

Book pp. 399–401, 442; C12 L1 slide 10; T-A.

Clinical connections: apply the cell biology



DEMYELINATION

Loss of myelin disrupts effective signaling. The chapter links this to sensory and motor deficits.

CNS VERSUS PNS

The book's main clinical note describes multiple sclerosis in the optic nerve, brain and spinal cord; it describes peripheral demyelination in Guillain–Barré syndrome.

CELL BODY SURVIVAL

A severed distal axon loses its normal connection to the soma's maintenance machinery.

SOURCE CHECK

One end-of-chapter question describes MS as peripheral. That conflicts with the chapter's own clinical note; use the note's CNS description.

REMEMBER Ask: which cell, which compartment, and which signal is disrupted?

Checkpoint 1 | foundations

1. Is a dorsal root ganglion a CNS nucleus? Explain both parts of your answer.

2. Which glia make myelin in the CNS and PNS? How does one cell's wrapping differ?

3. A neuron has one axon and many dendrites. Does that prove it is a motor neuron?

4. Why can a peripheral axon sometimes regain function after injury?

REMEMBER Take turns: answer without looking, explain why, then reveal the next slide.

Book pp. 394–401; C12 L1; T-A–T-C. Original practice questions.

Checkpoint 1 | foundations | answers

1. No. A ganglion is a collection of neuronal cell bodies in the PNS; a nucleus is a comparable CNS collection.

2. Oligodendrocytes in CNS: segments of several axons. Schwann cells in PNS: one myelinated internode on one axon.

3. It is multipolar structurally. The book and lecture also recognize multipolar interneurons.

4. A surviving soma can support axon regrowth along Schwann-cell guides, provided the correct target is reconnected.

REMEMBER A correct label is only half the job. Explain the structure, mechanism, or consequence.

2

Turn ion gradients into signals

Separate resting conditions, graded changes, action potentials, and propagation.

Use the memory cues, then explain the mechanism without reading.

Resting membrane potential: start at about −70 mV

Figure 12–8 Processes That Produce the Resting Membrane Potential.

RMP Resting Membrane Potential
The condition of the axon when the neuron is not conducting an impulse, but is prepared to do so.

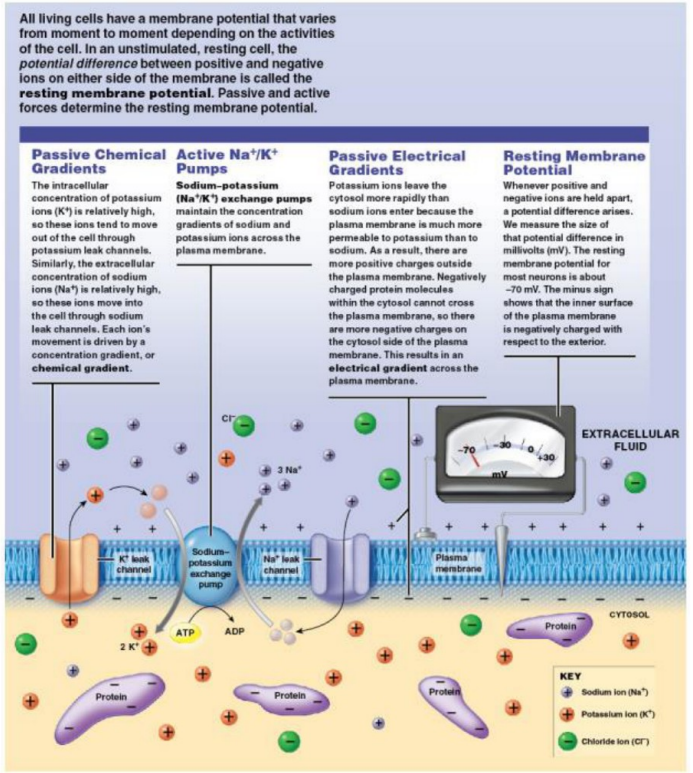
Measurement in the difference of positive and negative ions on either side of the membrane.

Extracellular Fluid: high [Na^+]
Intracellular (cytosol) **Fluid**: [K^+] and negatively charged proteins.

Leak Channels: ions move freely across the membrane.

Electrochemical Gradient
electrical gradient
chemical gradient

An axon in **Resting Membrane Potential** is said to be **Polarized.**
Polarized = (-70 mv)



WHAT IS MEASURED?

The voltage inside the cell relative to outside—not an absolute amount of electricity in the cell.

ION DISTRIBUTION

Na+ is high outside; K+ is high inside. Intracellular negatively charged proteins also matter.

SELECTIVE PERMEABILITY

At rest, the membrane is more permeable to K+ than Na+. Leak channels remain open.

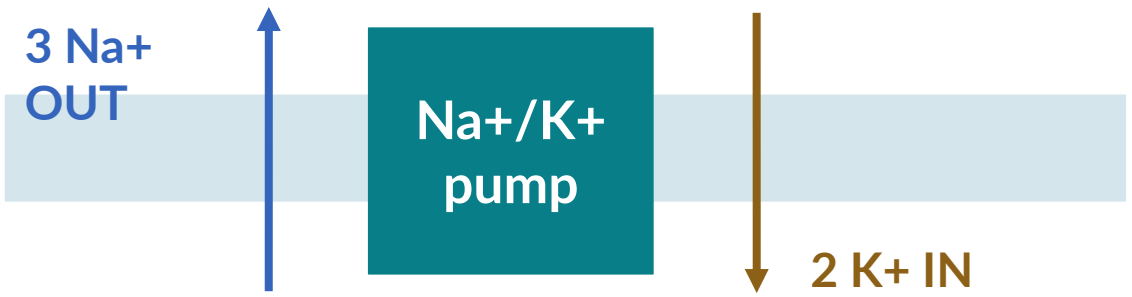
POLARIZED

A resting neuron has a charge separation and can respond to a stimulus; it is not currently firing an action potential.

REMEMBER Na+ mainly outside; K+ mainly inside; inside is negative at rest.

Book pp. 402–405; C12 L2 slides 1–5; T-A/T-B.

The sodium–potassium pump: 3 out, 2 in, 1 ATP



TRANSPORT RULE

For one ATP used, the pump moves 3 Na+ out and 2 K+ in, against their concentration gradients.

WHY KEEP PUMPING?

Na+ leaks inward and K+ tends to leak outward. Pumping maintains the gradients that signaling requires.

NOT THE AP SWITCH

Rapid action-potential depolarization and repolarization occur through ion channels, not by the pump instantly moving the voltage up and down.

OVER TIME

The pump restores/maintains ion concentrations. Only a small fraction of ions moves during one action potential.

REMEMBER “3 Na out, 2 K in, 1 ATP.” Say the direction every time.

Book pp. 403, 405, 412; C12 L2 slides 1–5; T-A–T-D.

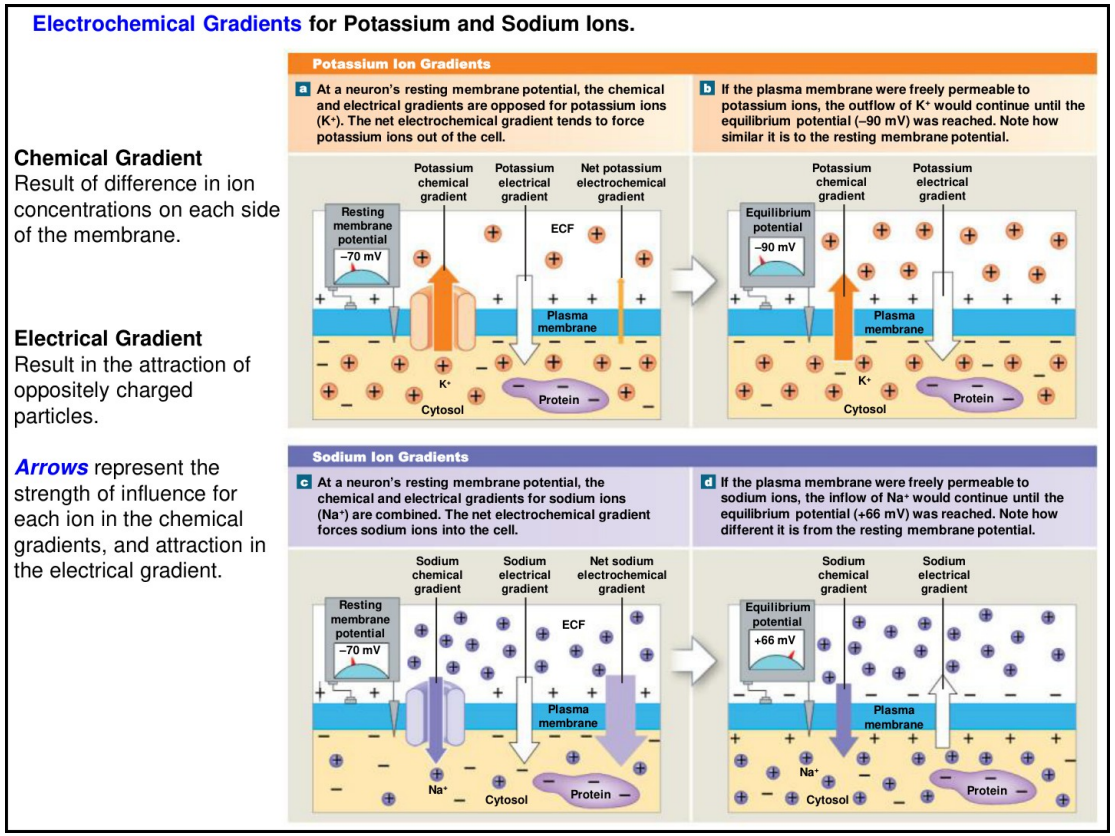
Electrochemical gradients: two forces, one net movement

Ion at rest	Chemical gradient	Electrical gradient	Net tendency
K+	Out: concentration is higher inside	In: negative interior attracts it	Outward tendency, weakened by electrical pull
Na+	In: concentration is higher outside	In: negative interior attracts it	Strong inward tendency

REMEMBER K+ has a tug-of-war. Na+ gets two pushes in the same direction.

Book pp. 402–405; C12 L2 slide 4; T-A/T-B/T-D.

Equilibrium potential is not resting potential



K+ EQUILIBRIUM

The book's K+-selective thought experiment reaches approximately −90 mV, where the inward electrical force balances outward diffusion.

Na+ EQUILIBRIUM

In the Na+-permeable thought experiment, equilibrium is approximately +66 mV.

REAL RESTING MEMBRANE

The resting value is about −70 mV because real membranes have more than one ion pathway plus active maintenance.

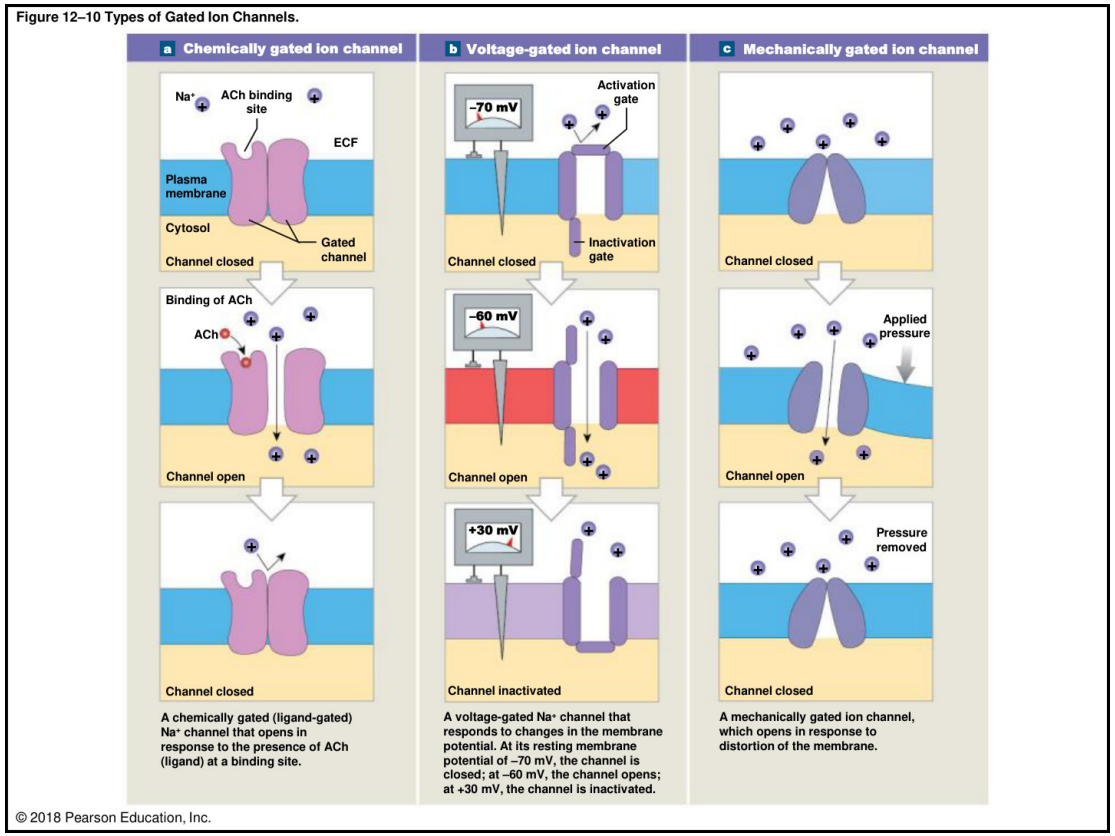
IMPORTANT CAUTION

Do not learn "turn the pump off and the cell must become −90 mV" as a general rule. The textbook's example assumes specific permeability conditions.

REMEMBER −90 = K+ equilibrium; −70 = typical rest; +66 = Na+ equilibrium.

Book pp. 402–405; C12 L2 slide 4; T-A/T-B.

Three ways to open an ion channel



CHEMICALLY GATED

A ligand binds a receptor/channel. Example: ACh at the neuromuscular junction or a cholinergic synapse.

VOLTAGE GATED

A change in membrane potential alters the channel. Na+ and K+ channels support axonal action potentials; Ca2+ channels matter at terminals.

MECHANICALLY GATED

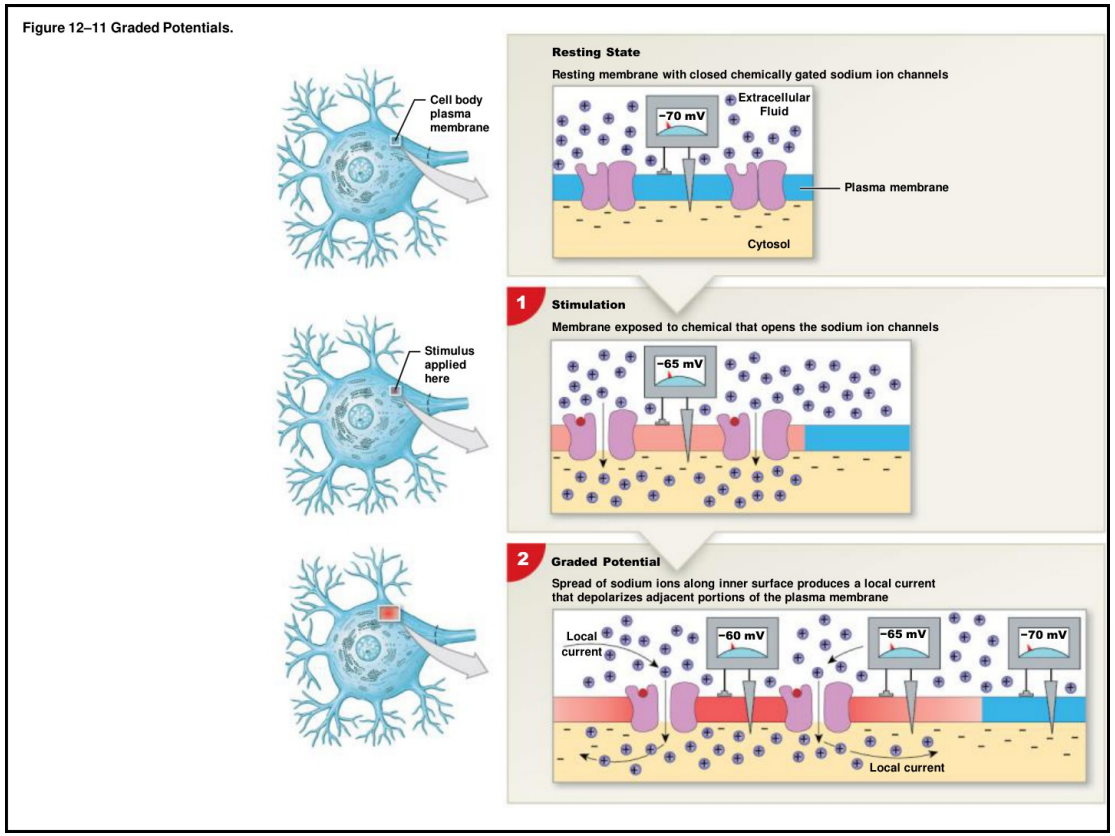
Physical deformation opens a channel, as in touch, pressure or vibration receptors.

LEAK CHANNEL

Different from the three gated categories: it is open at rest and contributes to resting conditions.

REMEMBER Chemical = key; voltage = charge; mechanical = stretch or push.

Graded potentials: local and variable



LOCATION

Common on dendrites and soma in the neuronal examples; the book also describes graded potentials in other cell types.

AMPLITUDE

A stronger stimulus produces a larger local change. There is no all-or-none threshold for the graded potential itself.

DISTANCE

The effect decreases as it spreads because of resistance and charge loss/spread; local current is passive.

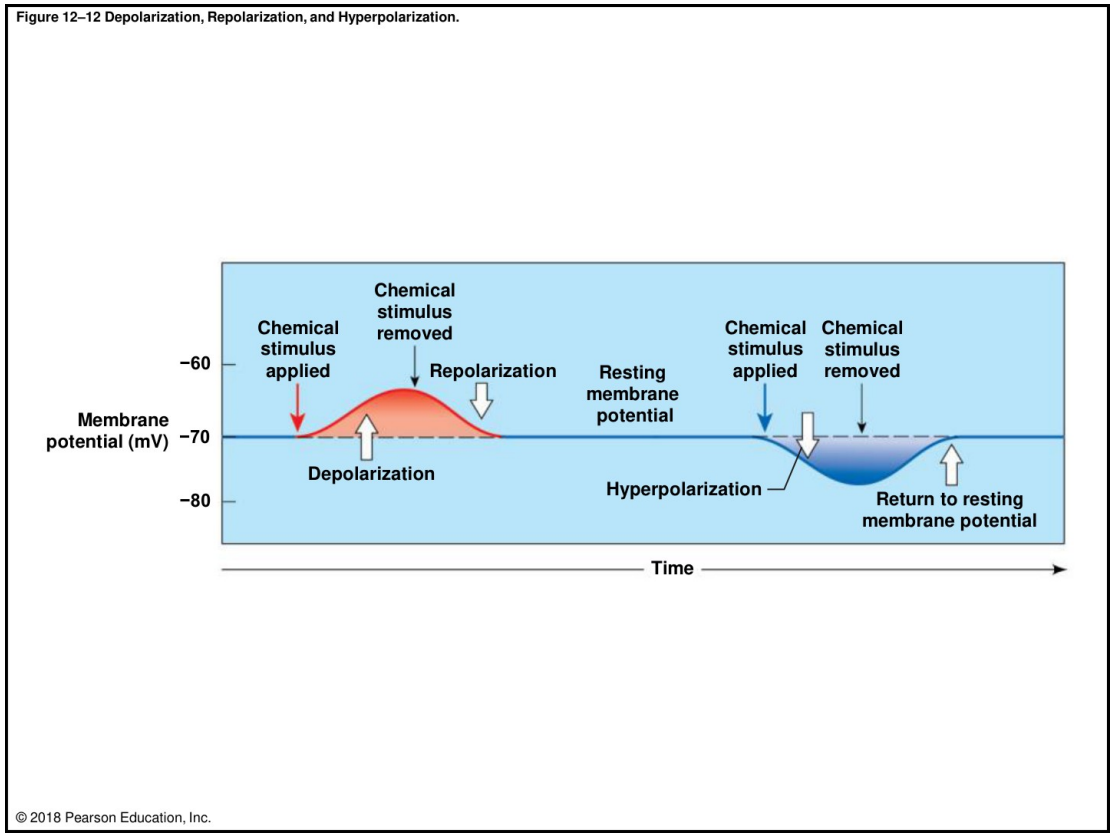
DIRECTION

A graded potential can depolarize or hyperpolarize. It may or may not bring the axon's initial segment to threshold.

REMEMBER GRADED = gets smaller with distance; size is graded to the stimulus.

Book pp. 407–409; C12 L2 slides 7–10; T-A/T-B.

Know the three voltage words



DEPOLARIZATION

The membrane becomes less negative / more positive. −70 to −65 mV is depolarization even without an action potential.

REPOLARIZATION

Movement back toward the resting membrane potential after depolarization.

HYPERPOLARIZATION

The membrane becomes more negative than its resting value, such as −70 to −85 mV.

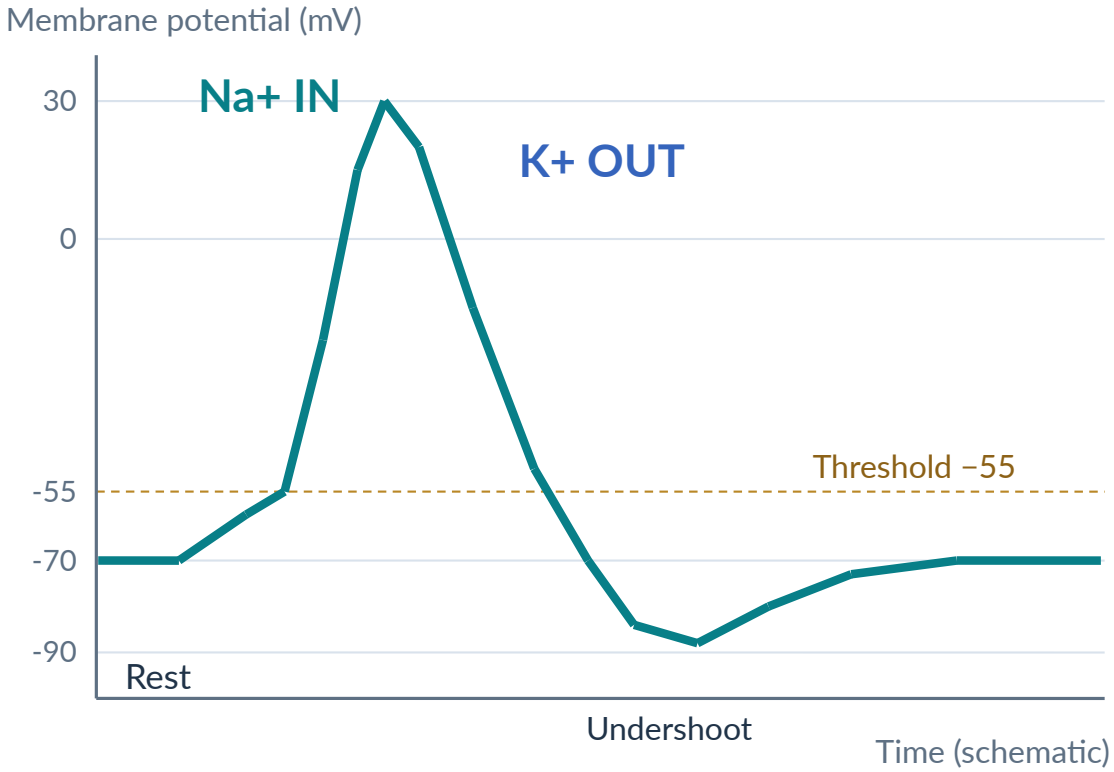
THRESHOLD

Your professor emphasizes approximately −55 mV at the initial segment. The book gives −60 to −55 mV and some figures use −60.

REMEMBER Depolarization is a direction of change—not a guarantee that a neuron fired.

Book pp. 407–409; C12 L2 slides 7, 10–14; T-A/T-B.

The action potential: one event, four phases



Course threshold shown; book figures may use −60 mV.

1 | REACH THRESHOLD

A graded depolarization brings the trigger region to about −55 mV in the lecture example.

2 | Na+ ENTRY

Voltage-gated Na+ channels activate; rapid Na+ influx produces the rising phase and a peak around +30 mV.

3 | K+ EXIT

Na+ channels inactivate; voltage-gated K+ channels allow outward K+ movement, repolarizing the membrane.

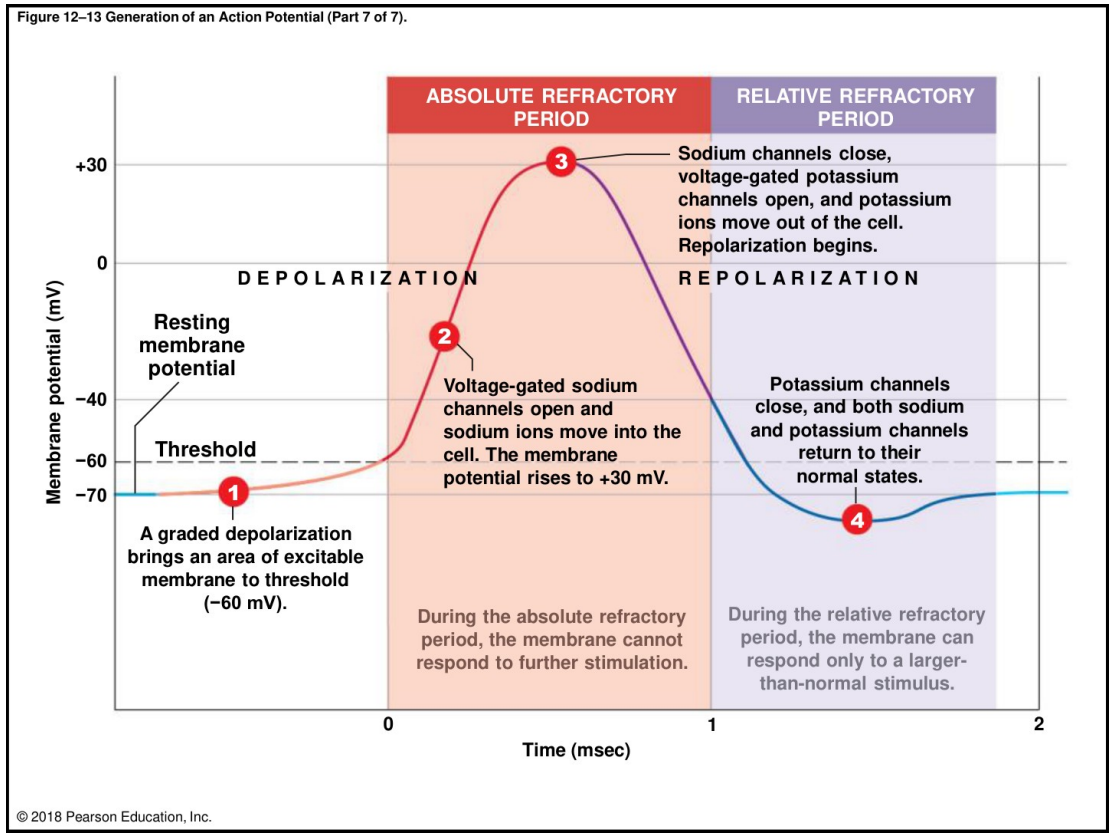
4 | AFTER-HYPERPOLARIZATION

Delayed closure of K+ channels allows a brief undershoot before return toward −70 mV.

REMEMBER Na+ IN brings voltage UP. K+ OUT brings voltage DOWN.

Book pp. 409–412; C12 L2 slides 11–14; T-A/T-B.

Refractory periods explain recovery and direction



ABSOLUTE

From Na+ channel activation until recovery from inactivation: another action potential cannot be triggered in that membrane region.

RELATIVE

Some Na+ channels have recovered, but K+ conductance/hyperpolarization makes another action potential require a stronger-than-normal stimulus.

DIRECTION

The recently active membrane is refractory, helping the normal impulse travel toward excitable membrane ahead.

MAXIMUM FREQUENCY

The absolute refractory period limits how rapidly a neuron can fire again.

REMEMBER Absolute = absolutely not. Relative = requires more.

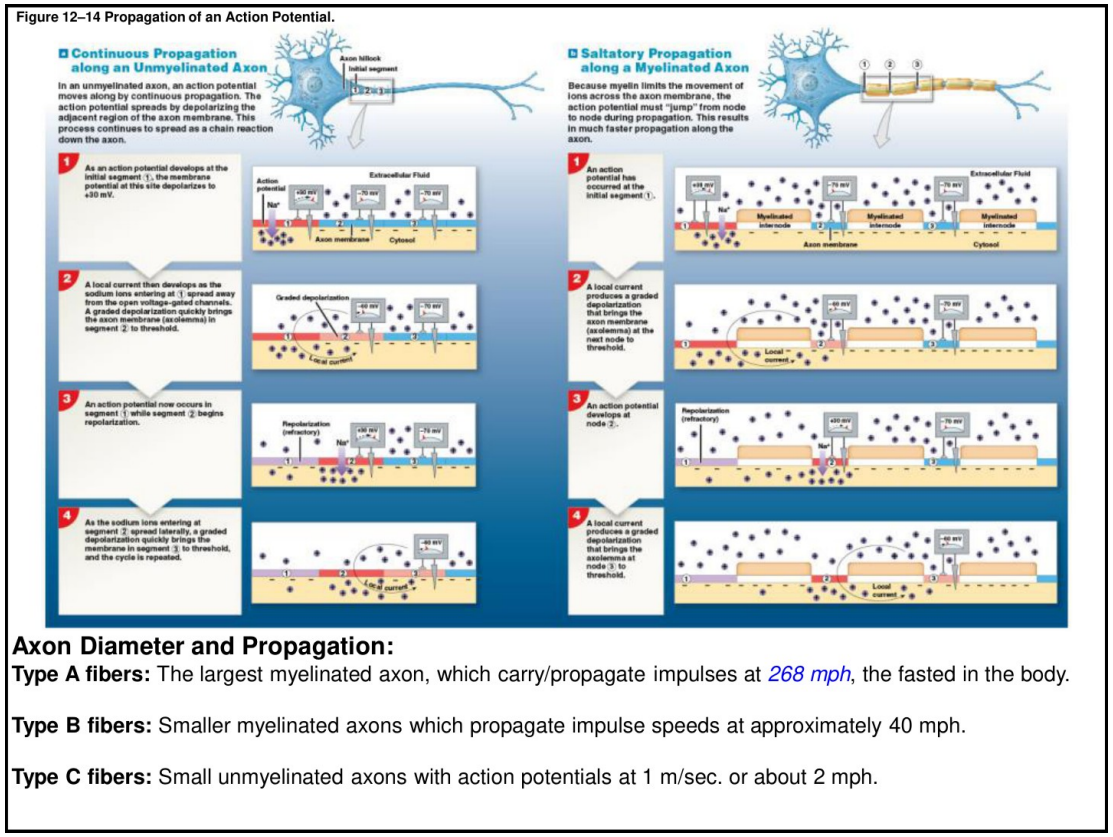
Graded versus action potential: the exam comparison

Feature	Graded potential	Action potential
Where in the neuron?	Usually dendrites / soma	Initial segment and axon
Size	Varies with stimulus	All-or-none once threshold is reached
Distance	Decreases with distance	Regenerated without decreasing amplitude
Sign	Depolarizing or hyperpolarizing	Depolarizing event followed by recovery
Summation	Can add together	Individual AP amplitudes do not sum
Threshold / refractory	No fixed AP threshold; no refractory period	Threshold and refractory periods

REMEMBER Graded signals add; action potentials repeat.

Book pp. 407–412, Table 12–3; C12 L2 slide 15; T-A/T-B.

Continuous propagation: every adjacent region



UNMYELINATED AXON

Local current from one active region depolarizes the next region of excitable membrane.

CHAIN REACTION

The next region reaches threshold, opens its voltage-gated Na+ channels, and generates another action potential.

NO FADING

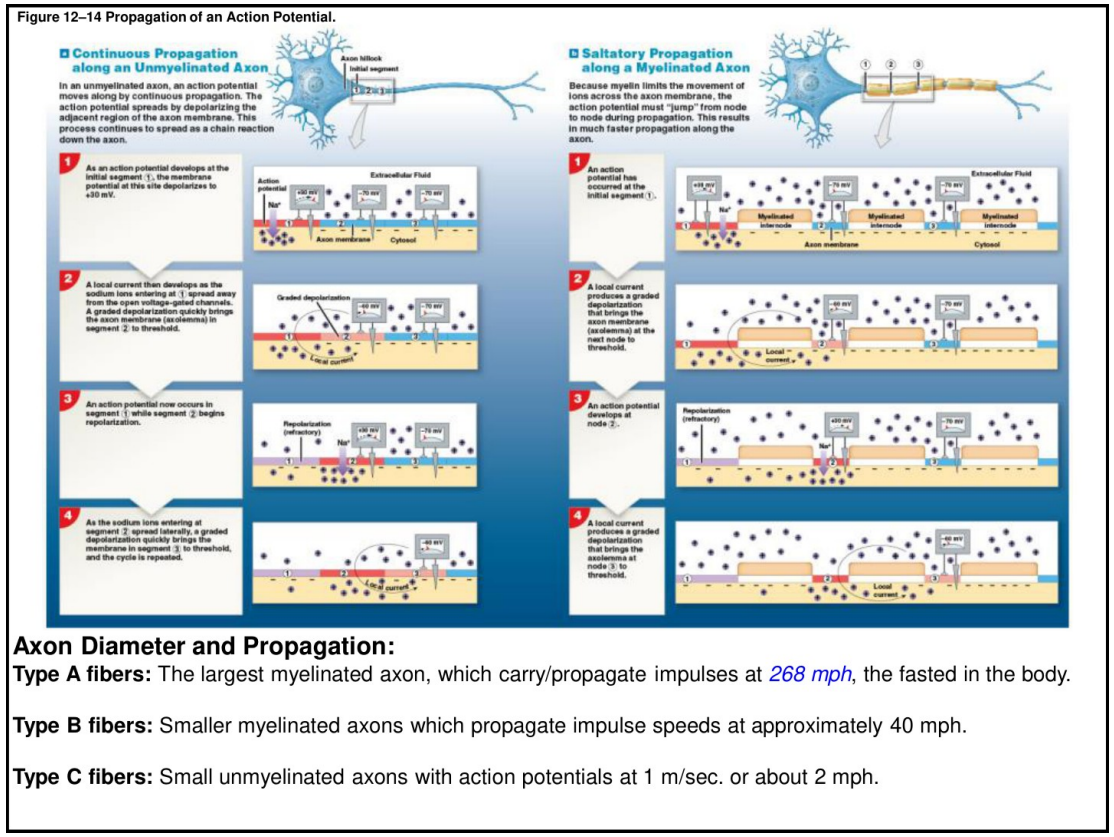
Unlike a graded potential alone, the action potential is renewed along the axon.

ANALOGY

Each person in a line repeats the complete message to the next person; the message stays intact, but each relay takes time.

REMEMBER Continuous = neighboring membrane sections take turns.

Saltatory propagation: regenerate at the nodes



Axon Diameter and Propagation:

Type A fibers: The largest myelinated axon, which carry/propagate impulses at *268 mph*, the fasted in the body.

Type B fibers: Smaller myelinated axons which propagate impulse speeds at approximately 40 mph.

Type C fibers: Small unmyelinated axons with action potentials at 1 m/sec. or about 2 mph.

MYELINATED AXON

Myelin limits ion movement across internodal membrane.

NODE TO NODE

Local current travels beneath myelin and depolarizes the next node to threshold; the action potential is regenerated there.

FASTER + EFFICIENT

Fewer membrane regions need active regeneration; proportionately less ion movement requires restoration afterward.

TWO SPEED FACTORS

Myelination and larger axon diameter increase conduction speed. The transcript emphasizes these—not simply the neuron's length.

REMEMBER Saltatory = skip the insulated stretches; fire at the gaps.

Book pp. 413–416; C12 L3 slide 1; T-A/T-C.

Fiber types: learn the class and the pattern

Fiber	Diameter / insulation	Book example speed	Typical roles in this chapter
A	Largest; myelinated; 4–20 µm	Up to 120 m/s (268 mph)	Fast position/touch information; skeletal motor commands
B	Smaller; myelinated; 2–4 µm	About 18 m/s (40 mph)	Some sensory and visceral pathways
C	Small; unmyelinated; under 2 µm	About 1 m/s (2 mph)	Slower sensory and visceral pathways

REMEMBER A = accelerated; B = between; C = crawling. Speeds are the supplied book's examples.

A stronger signal usually means more APs, not taller APs

ALL-OR-NONE

Once threshold is reached, increasing stimulus strength does not create a larger action potential in the same axon.

FREQUENCY CODE

Stronger sustained depolarization can increase the number of action potentials generated per unit time.

FACILITATION

A neuron already closer to threshold needs less additional excitatory input to fire.

CEILING

The refractory properties limit how fast action potentials can recur.

REMEMBER Amplitude stays; frequency changes.

Book pp. 409, 425, 427–428; T-B/T-C.

Checkpoint 2 | membrane signals

1. A neuron changes from −70 to −63 mV; threshold is −55 mV. What happened—and what did not?

2. Which ions move during the rising and falling phases of an action potential?

3. Why is an extra stimulus ineffective in the absolute refractory period?

4. A stronger stimulus makes an AP twice as tall. What is wrong with that statement?

REMEMBER Take turns: answer without looking, explain why, then reveal the next slide.

Checkpoint 2 | membrane signals | answers

1. Depolarization occurred, but threshold was not reached, so this change alone did not initiate an action potential.

2. Na+ moves inward during rapid depolarization. K+ moves outward during repolarization and the undershoot.

3. The relevant Na+ channels are already activated or inactivated and have not recovered their availability.

4. The same axon’s AP is all-or-none. Greater sustained input usually increases firing frequency, not AP height.

REMEMBER A correct label is only half the job. Explain the structure, mechanism, or consequence.

Book pp. 407–413, 427–428; C12 L2. Original practice questions.

3

Transfer and combine information

Follow the transmitter, identify the receptor, and predict the effect on threshold.

Use the memory cues, then explain the mechanism without reading.

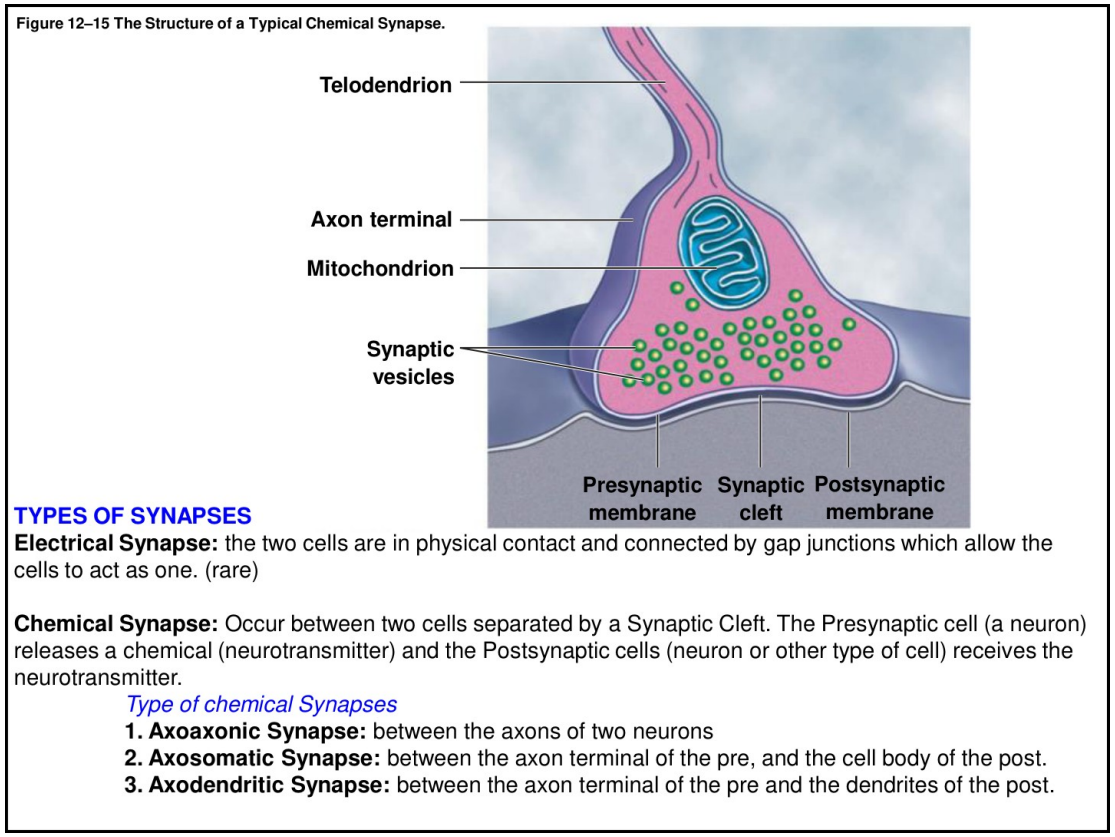
Electrical and chemical synapses

Feature	Electrical	Chemical
Connection	Gap junctions directly connect cells	Synaptic cleft separates cells
Signal transfer	Ions/current pass between cells	Neurotransmitter binds postsynaptic receptors
Speed	Rapid, with little synaptic delay	Delay for Ca2+ entry and transmitter release
Course emphasis	Described as relatively rare	Most abundant type in the supplied chapter
Direction in chapter model	Direct coupling	Usually presynaptic to postsynaptic

REMEMBER Electrical = connected. Chemical = cleft + chemical.

Book pp. 416–418; C12 L3 slide 2; T-C.

Name a synapse by where the axon ends



AXODENDRITIC

Axon terminal of one neuron contacts a dendrite of another.

AXOSOMATIC

Axon terminal contacts another neuron's cell body (soma).

AXOAXONIC

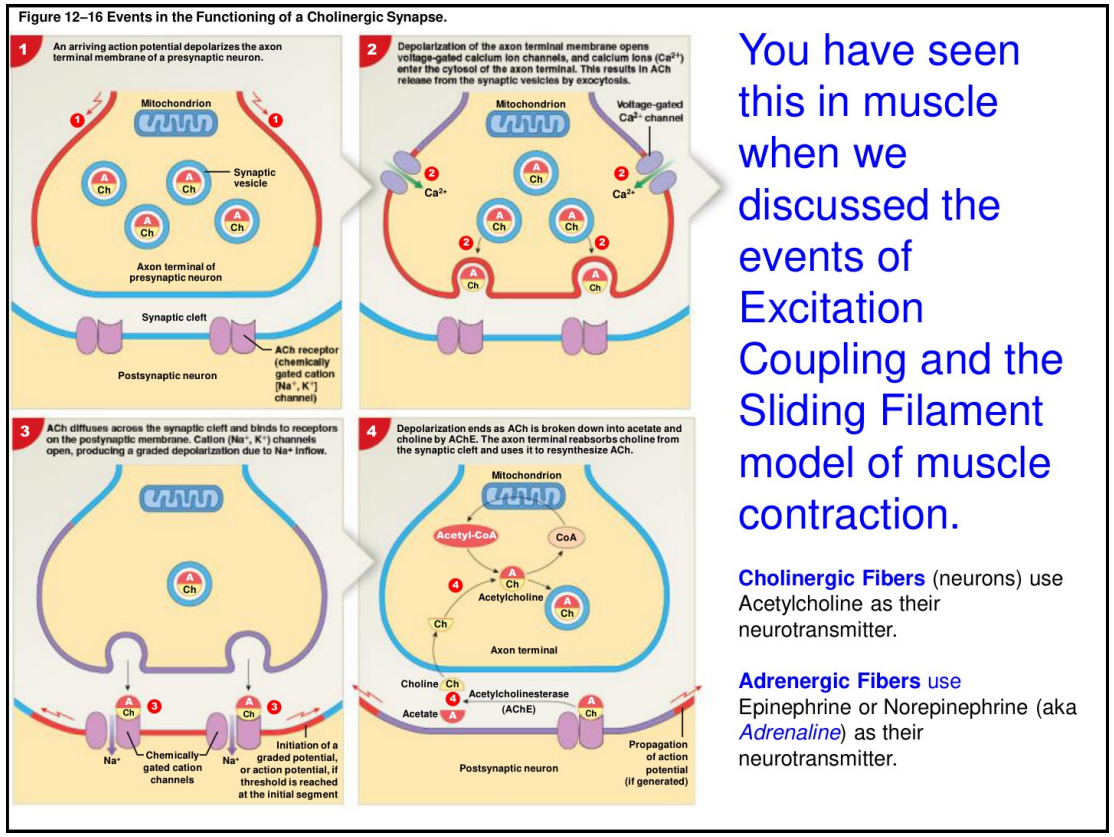
An axon contacts another axon; important in the chapter's presynaptic regulation examples.

TERMS TO LABEL

Presynaptic membrane, synaptic vesicles, synaptic cleft, postsynaptic membrane, receptor and mitochondrion.

REMEMBER Read the word in two pieces: axo–dendrite, axo–soma, axo–axon.

Cholinergic synapse: the four-event sequence



You have seen this in muscle when we discussed the events of Excitation Coupling and the Sliding Filament model of muscle contraction.

Cholinergic Fibers (neurons) use Acetylcholine as their neurotransmitter.

Adrenergic Fibers use Epinephrine or Norepinephrine (aka *Adrenaline*) as their neurotransmitter.

1 | ARRIVAL

An action potential depolarizes the presynaptic axon terminal.

2 | RELEASE

Voltage-gated Ca2+ channels open. Ca2+ entry triggers vesicle exocytosis of acetylcholine (ACh).

3 | RECEPTION

ACh binds postsynaptic chemically gated cation channels. The example produces net depolarization; an AP follows only if threshold is reached.

4 | TERMINATION

Acetylcholinesterase (AChE) breaks ACh into acetate and choline. Choline is taken back up and reused.

REMEMBER Arrival → Calcium → ACh → Receptor → AChE. Calcium triggers release; sodium helps the response.

Book pp. 418–419; C12 L3 slide 3; T-C.

Why signaling can slow down or weaken

SYNAPTIC DELAY

The book gives about 0.2–0.5 ms between presynaptic arrival and postsynaptic effect; much reflects Ca2+ entry and release steps.

MORE SYNAPSES

More successive chemical synapses can add more delay, even if the axons themselves are similar.

SYNAPTIC FATIGUE

Intense stimulation can outpace transmitter synthesis, transport and recycling; the response weakens until supplies recover.

PREDICT THE BLOCK

Blocking terminal Ca2+ entry reduces release. Blocking ACh receptors reduces the postsynaptic response. Blocking AChE delays ACh removal.

REMEMBER Fast axons can still be slowed by many synaptic handoffs.

Major neurotransmitters: first pass

Transmitter	Class / role in the supplied chapter	Remember
Acetylcholine (ACh)	Cholinergic signaling; skeletal NMJs and many other synapses	Cholinergic uses ACh
Norepinephrine (NE)	Biogenic amine; adrenergic signaling in CNS/ANS	NE = noradrenaline
Dopamine	Biogenic amine; movement control and reward-related examples	Dopamine is not an endorphin
Serotonin	Biogenic amine; mood, sleep/wake and presynaptic facilitation examples	Effect depends on receptor

REMEMBER A transmitter's name alone does not tell you every effect it can produce.

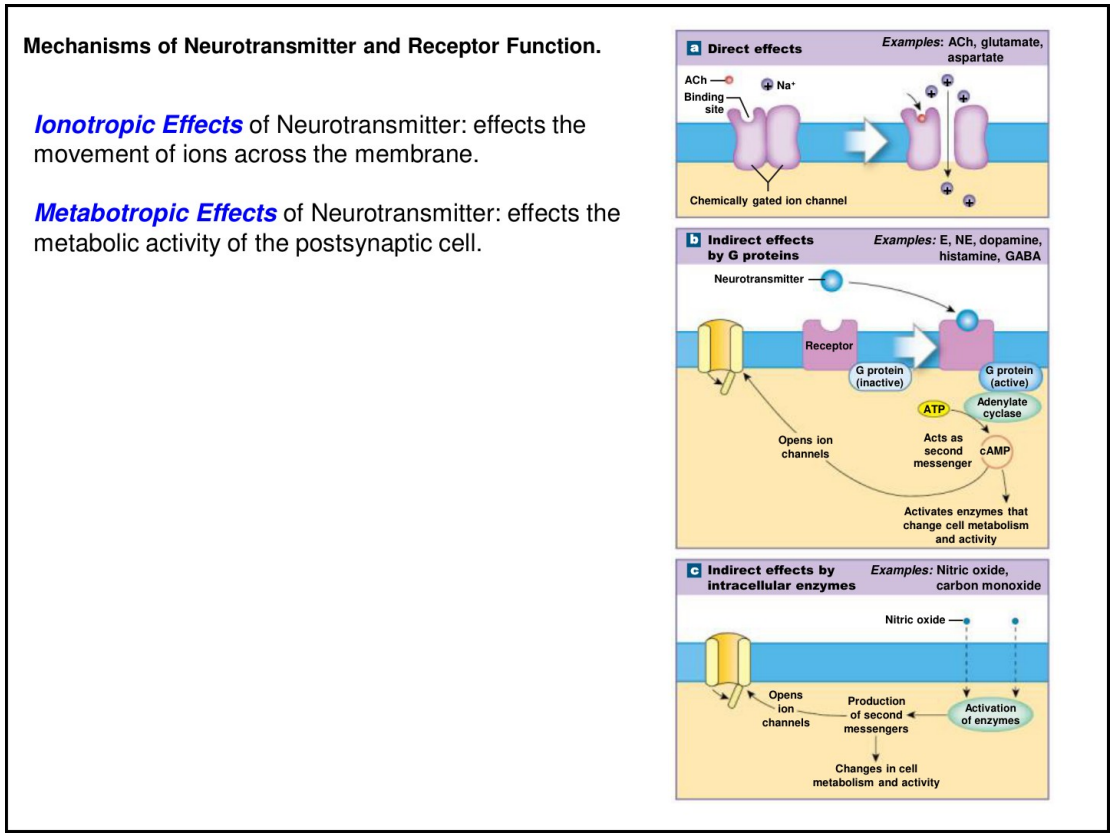
Amino acids, peptides and other messengers

Group	Examples	Key course point
Excitatory amino acids	Glutamate, aspartate	Promote excitation in the chapter's examples
Inhibitory amino acids	GABA, glycine	Can increase Cl− conductance or alter K+/Ca2+ pathways, depending on receptor
Opioid peptides	Enkephalins, endorphins, dynorphins	Pain modulation; can inhibit substance P release
Dissolved gases	Nitric oxide (NO), carbon monoxide (CO)	Can diffuse into cells and act through intracellular enzymes
Other table examples	ATP/GTP, adenosine, peptide hormones, anandamide	Recognize as additional chemical signaling categories

REMEMBER GABA/glycine brake; glutamate often excites. Receptors still determine the mechanism.

Book pp. 421–424, Table 12–4; T-C.

Receptor type determines the response



DIRECT / IONOTROPIC

Transmitter binding opens or closes an ion channel directly; membrane permeability changes.

INDIRECT / METABOTROPIC

A receptor activates a G protein and intracellular signaling steps rather than directly opening the channel.

SECOND-MESSENGER EXAMPLE

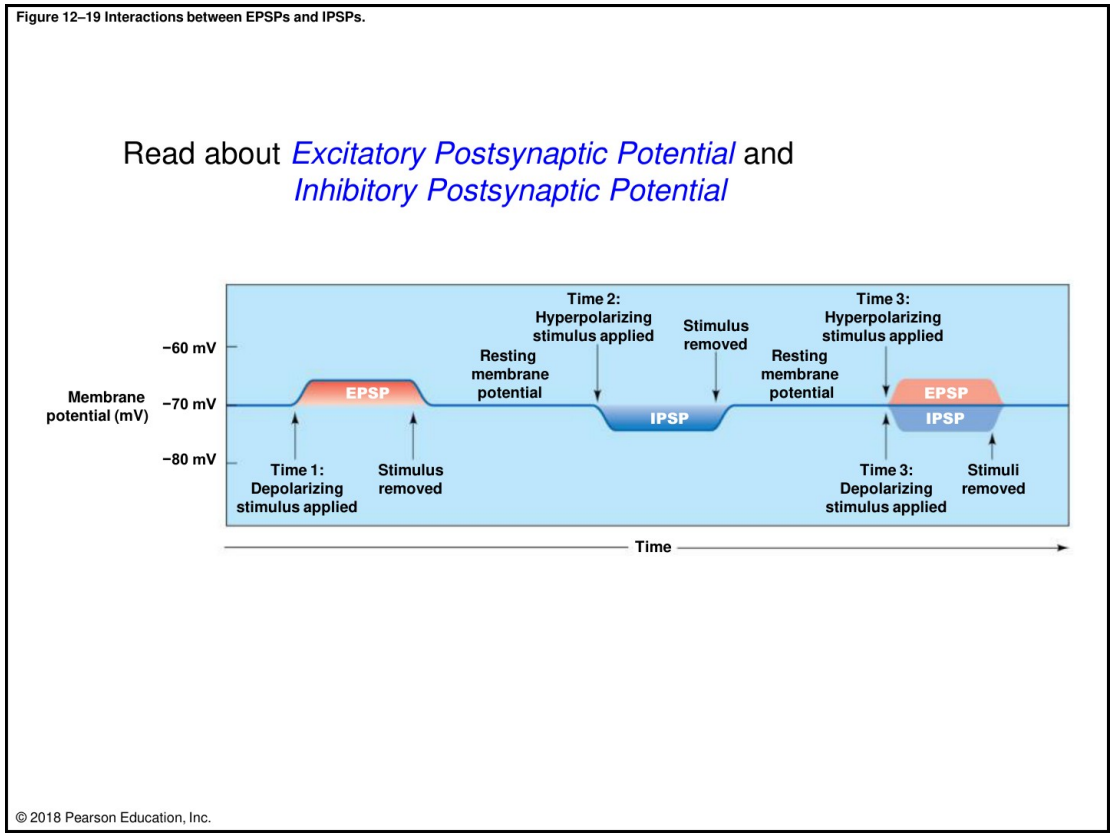
Activated G-protein signaling can activate adenylate cyclase, which converts ATP into cAMP; cAMP changes channels or enzyme activity.

FIRST VERSUS SECOND

The extracellular transmitter is the first messenger. cAMP is an intracellular second messenger.

REMEMBER Ionotropic: ion gate. Metabotropic: messenger steps.

EPSPs and IPSPs: closer to or farther from firing?



EPSP

Excitatory postsynaptic potential: a graded depolarization that makes reaching threshold more likely.

IPSP

Inhibitory postsynaptic potential: the chapter's example is graded hyperpolarization, making threshold harder to reach.

CHANNEL EXAMPLES

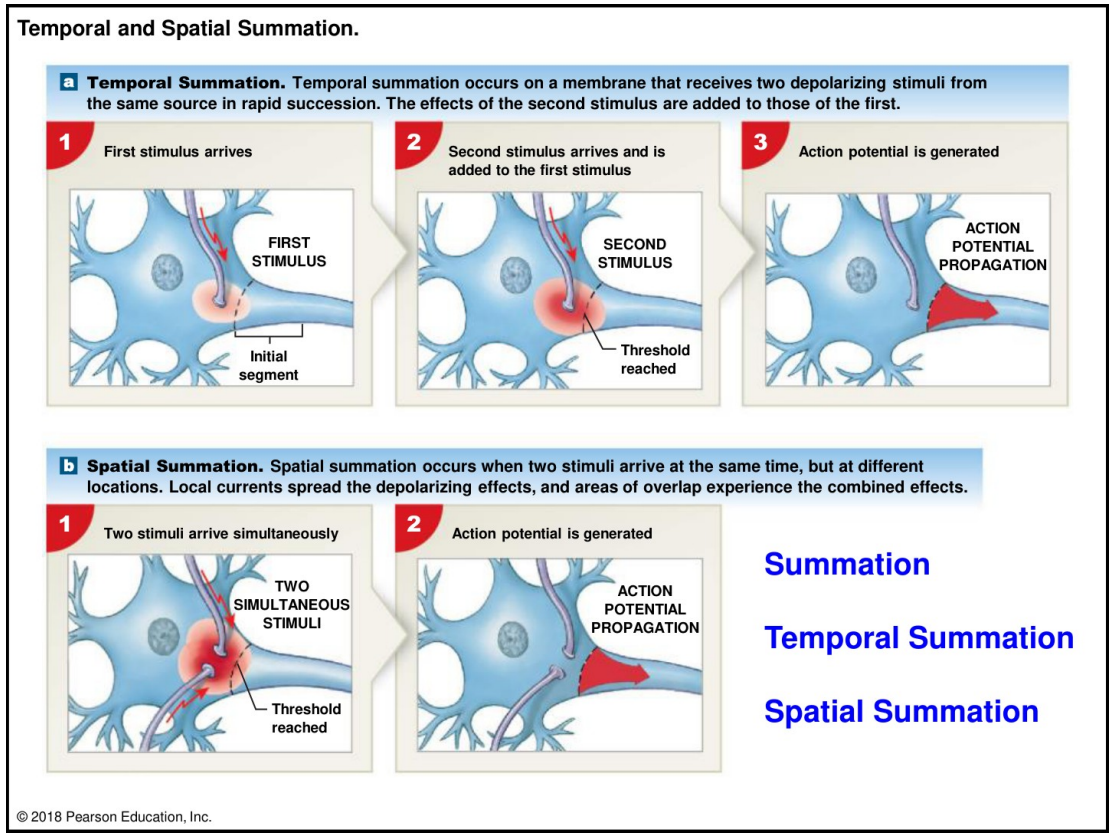
Na+ entry can depolarize. K+ exit can hyperpolarize; the book's transmitter table also includes inhibitory Cl− channels.

NET EFFECT

A neuron integrates simultaneous excitatory and inhibitory input. The balance at the trigger region matters.

REMEMBER EPSP = easier. IPSP = inhibited. Neither is itself an axonal AP.

Temporal versus spatial summation



TEMPORAL

Repeated input at one synapse arrives close enough in time that a new effect adds to the previous one.

SPATIAL

Inputs from multiple synapses arrive together and combine across locations.

WHAT SUMS?

Graded postsynaptic potentials—not the amplitude of action potentials.

THRESHOLD TEST

The combined effect must sufficiently depolarize the initial segment; inhibitory input may cancel some excitation.

REMEMBER Temporal = time at one place. Spatial = several places at once.

Do the voltage arithmetic—then interpret it

SETUP

For this simplified exercise, rest is −70 mV, threshold is −55 mV, and the stated effects reach the trigger region.

EXAMPLE A

Three simultaneous +5 mV EPSPs give −55 mV: threshold is reached.

EXAMPLE B

Three +5 mV EPSPs plus one −5 mV IPSP give −60 mV: still below threshold.

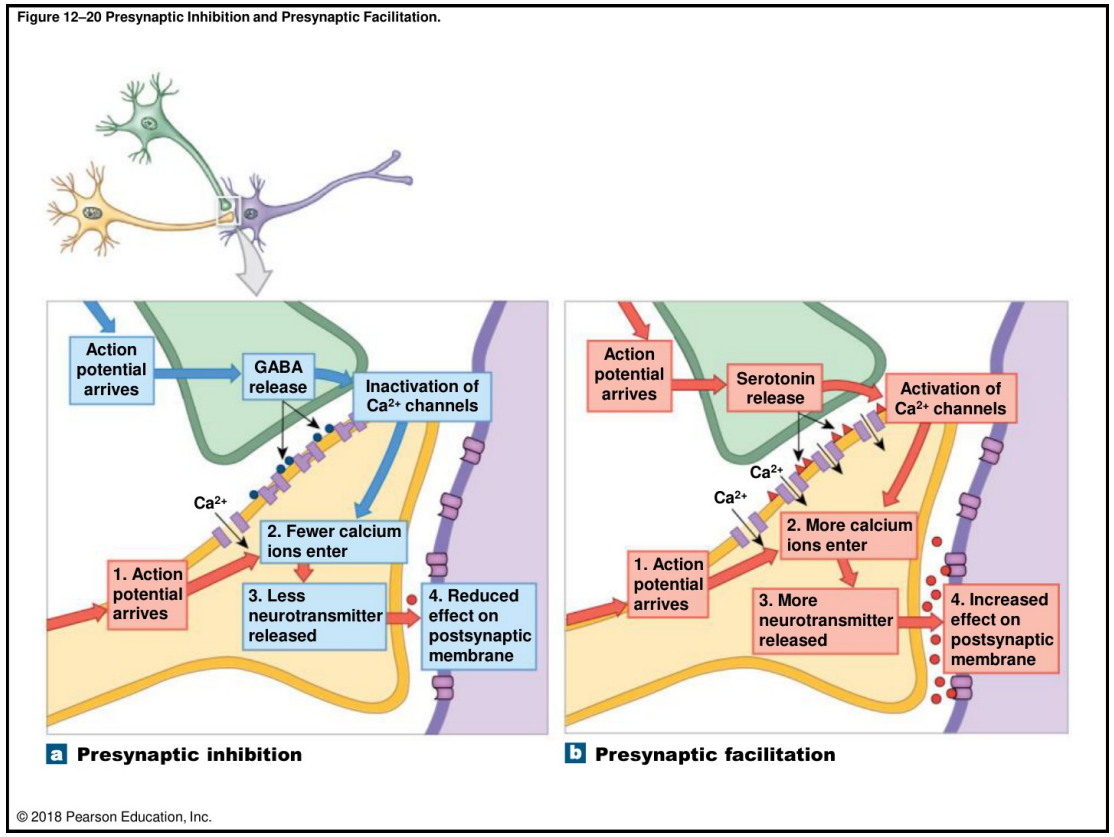
EXAMPLE C

Starting at −85 mV, a +15 mV depolarization gives −70 mV—not an action potential.

REMEMBER Calculate the final voltage, not just the size of the change.

Based on Book pp. 425–428 and T-C examples. New simplified practice calculations.

Presynaptic inhibition versus facilitation



INHIBITION EXAMPLE

An axoaxonic input releases GABA; Ca2+ entry at the target terminal is reduced, so less neurotransmitter is released.

FACILITATION EXAMPLE

An axoaxonic input involving serotonin prolongs/enhances Ca2+ entry, increasing transmitter release.

WHERE IS THE CHANGE?

At the presynaptic terminal before that terminal communicates across its main synapse.

COMPARE WITH IPSP

An IPSP describes the target cell's graded postsynaptic response. Presynaptic inhibition changes how much transmitter arrives.

REMEMBER Less Ca2+ → less release. More Ca2+ → more release.

Checkpoint 3 | synapses

1. Why does blocking terminal Ca2+ channels reduce communication across a chemical synapse?

2. Five inputs arrive at different synapses at the same time. What kind of summation is this?

3. In the cAMP example, what is the enzyme, what is its substrate, and what is the product?

4. How does presynaptic inhibition differ from a postsynaptic IPSP?

REMEMBER Take turns: answer without looking, explain why, then reveal the next slide.

Checkpoint 3 | synapses | answers

1. Less Ca2+ enters; fewer vesicles release transmitter; the postsynaptic effect decreases.

2. Spatial summation. Repeated input at one location in rapid succession is temporal summation.

3. Adenylate cyclase is the enzyme; ATP is converted into the second messenger cAMP.

4. Presynaptic inhibition reduces release from the sending terminal. An IPSP is an inhibitory graded response in the receiving membrane.

REMEMBER A correct label is only half the job. Explain the structure, mechanism, or consequence.

4

Build the spinal cord map

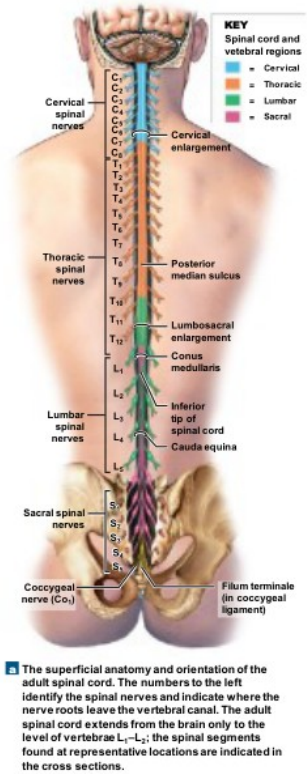
Orient the cord, protect it, locate its nuclei, and trace information through the roots.

Use the memory cues, then explain the mechanism without reading.

Spinal cord: conduit and reflex center

Figure 13–2a Gross Anatomy of the Adult Spinal Cord.

Spinal Cord:
Conduit between Brain and Spinal Nerves
Length is 18", +/- (*spinal cord proper*)
Diameter is approximately .55" Dime
Protected by *Meningeal membranes* and *bone* (vertebrae)
Four Regions of Spinal Cord
1.Cervical
2.Thoracic
3.Lumbar
4.Sacral
31 pair of spinal nerves
31 segments (correspond with nerves)



a The superficial anatomy and orientation of the adult spinal cord. The numbers to the left identify the spinal nerves and indicate where the nerve roots leave the vertebral canal. The adult spinal cord extends from the brain only to the level of vertebrae L_1–L_2; the spinal segments found at representative locations are indicated in the cross sections.

GENERAL SIZE

The supplied book gives about 45 cm / 18 inches long and a maximum width near 14 mm / 0.55 inches.

FOUR REGIONS

Cervical, thoracic, lumbar and sacral. Thirty-one segments correspond to 31 pairs of spinal nerves.

TWO ENLARGEMENTS

Cervical enlargement serves upper limbs; lumbosacral enlargement serves pelvic structures and lower limbs.

TWO BIG JOBS

Conduct sensory and motor information between brain and body; organize spinal reflex responses locally.

REMEMBER Two enlargements for two sets of limbs—not a thoracic enlargement.

Book pp. 434–437; C13 L1 slides 1–4; T-C.

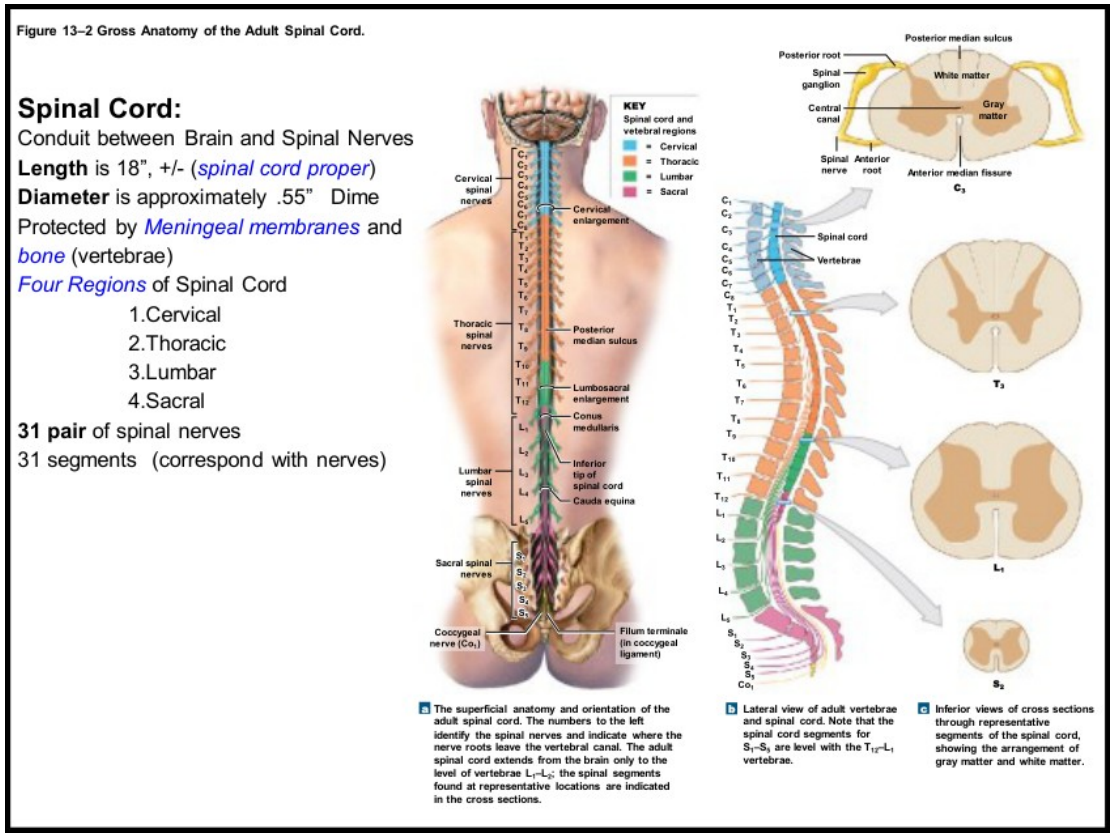
31 spinal nerve pairs: understand the naming rule

Region	Pairs	Exit / naming detail
Cervical	8 (C1–C8)	C1–C7 exit above their matching vertebrae; C8 exits between C7 and T1
Thoracic	12 (T1–T12)	Exit below the matching vertebra
Lumbar	5 (L1–L5)	Exit below the matching vertebra
Sacral	5 (S1–S5)	Sacral pairs continue the regional numbering
Coccygeal	1 (Co1)	Completes the 31-pair total

REMEMBER 8 + 12 + 5 + 5 + 1 = 31. Seven cervical vertebrae, eight cervical nerve pairs.

Book pp. 436–438; C13 L1 slides 3–8; T-C.

Conus, cauda and filum: three different things



The superficial anatomy and orientation of the adult spinal cord. The numbers to the left identify the spinal nerves and indicate where the nerve roots leave the vertebral canal. The adult spinal cord extends from the brain only to the level of vertebrae L_1–L_2; the spinal segments found at representative locations are indicated in the cross sections.

Lateral view of adult vertebrae and spinal cord. Note that the spinal cord segments for S_1–S_5 are level with the T_{12}–L_1 vertebrae.

Inferior views of cross sections through representative segments of the spinal cord, showing the arrangement of gray matter and white matter.

CONUS MEDULLARIS

Tapered inferior end of the spinal cord proper, near vertebral levels L1–L2 in the adult.

CAUDA EQUINA

Long descending nerve roots below the conus; the "horse's tail," not another solid spinal cord.

FILUM TERMINALE

Slender supporting strand extending from the conus and contributing to the coccygeal ligament.

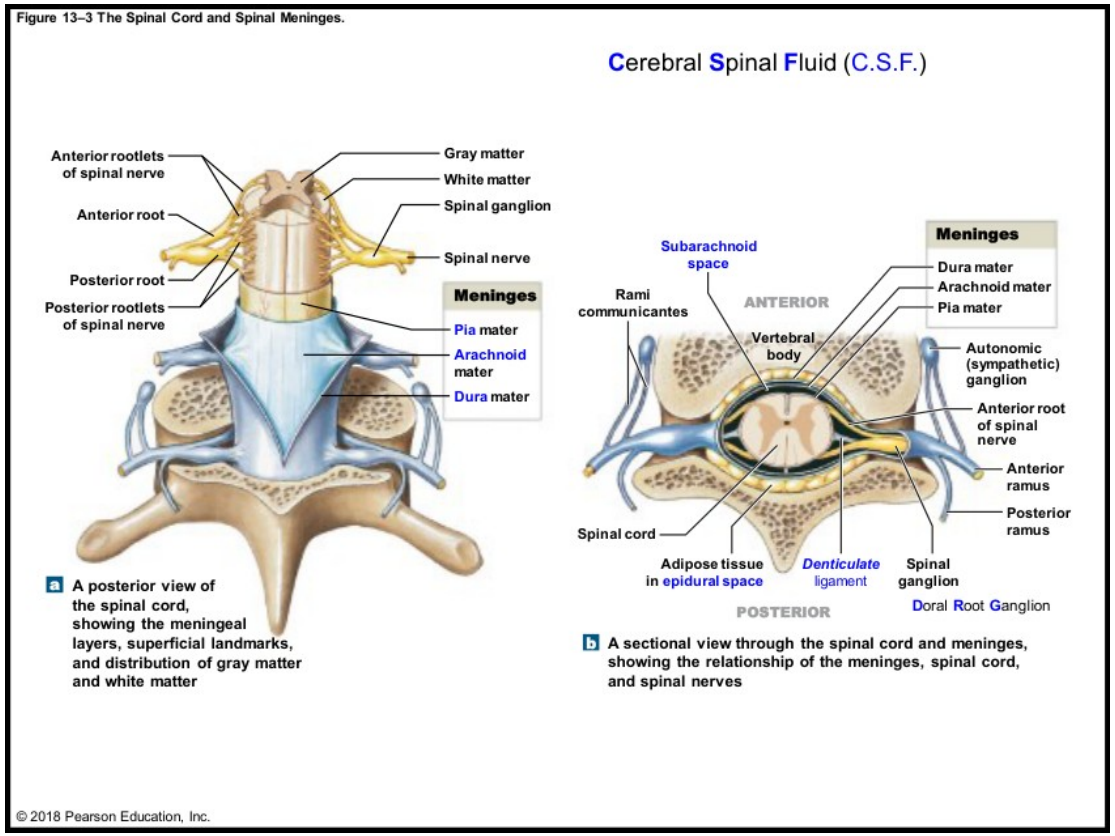
WHY THE LONG ROOTS?

The vertebral column grows longer than the cord; lower roots travel downward to their exit sites.

REMEMBER Conus = cone. Cauda = tail. Filum = thread.

Book pp. 436–438; C13 L1 slides 3–8; T-C.

Meninges: three layers from outside inward



DURA MATER

Tough outer fibrous covering. “Hard mother” is the lecture’s translation cue.

ARACHNOID MATER

Middle layer with web-like trabeculae spanning toward the pia.

PIA MATER

Delicate inner covering tightly attached to neural tissue; related to the cord’s surface blood vessels.

CONTINUITY

Spinal meninges are continuous with cranial meninges at the foramen magnum.

REMEMBER D–A–P: Dura, Arachnoid, Pia. Pia is pasted to the cord.

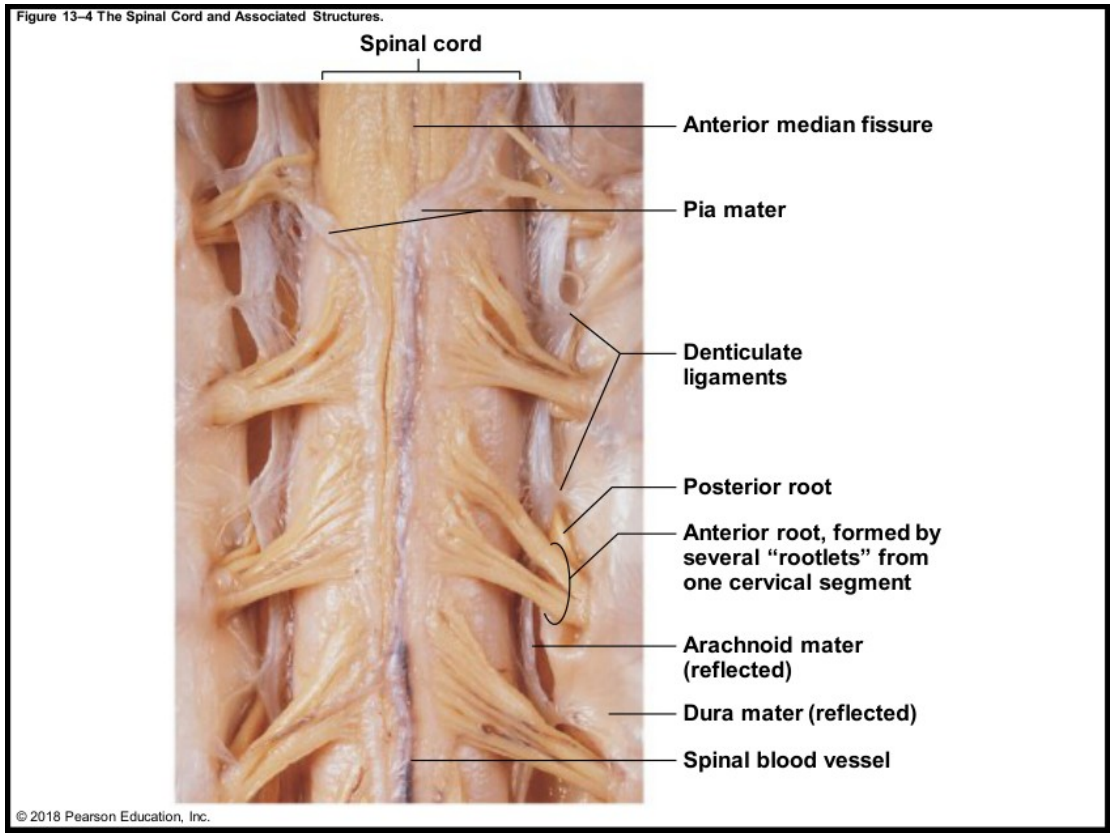
The spaces are not interchangeable

Space	Where?	What is important?
Epidural	Between vertebral canal wall and spinal dura	Fat, loose connective tissue and vessels; epidural anesthetic location
Subdural	Between dura and arachnoid	Book describes a potential/artifactual separation, not a normal open fluid space
Subarachnoid	Between arachnoid and pia	Contains CSF, trabeculae and vessels
Central canal	Inside the spinal cord	Also contains CSF; not a meningeal space

REMEMBER CSF around the cord = SUBARACHNOID, not epidural.

Book pp. 437–440; C13 L1 slide 9; T-C.

Keep the cord stable



DENTICULATE LIGAMENTS

Extensions of pia that pass through the arachnoid toward dura; resist side-to-side movement.

LONGITUDINAL SUPPORT

Dural attachments at the foramen magnum and the filum/coccygeal ligament complex resist superior–inferior shifting.

WHY IT MATTERS

Roots leave through fixed exits. Excessive motion would stretch or damage neural connections.

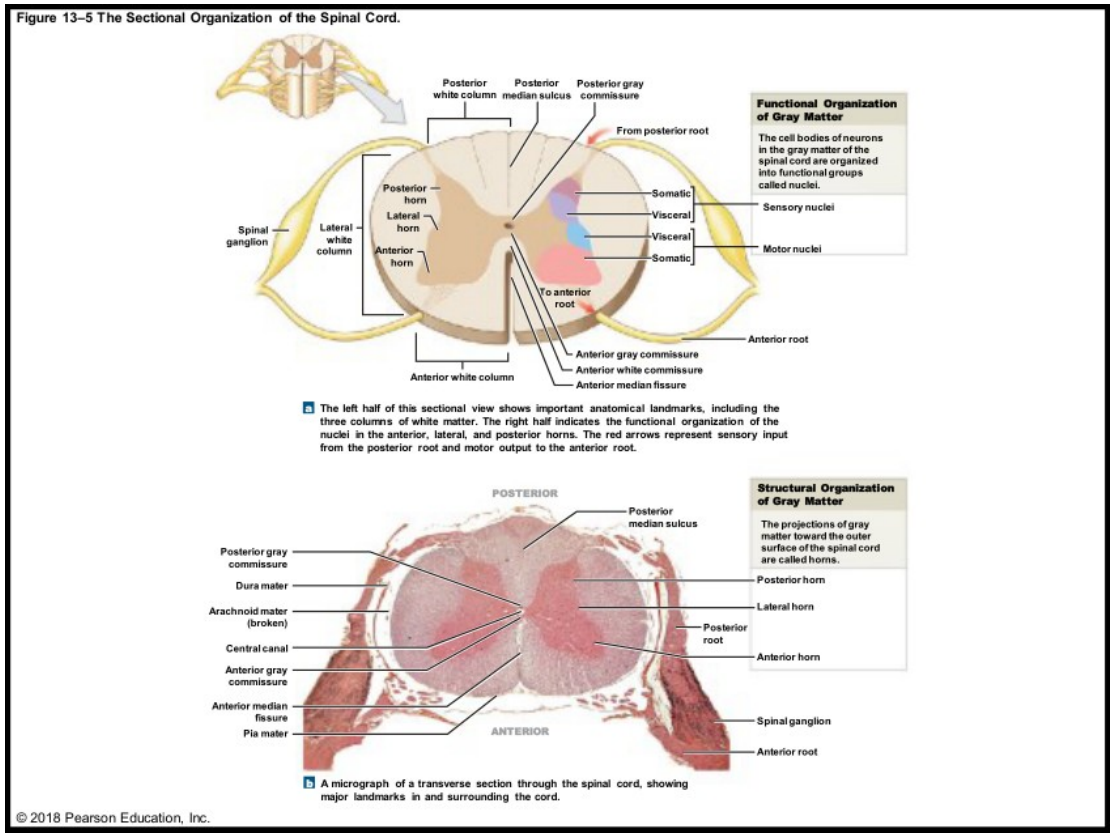
SOURCE CAUTION

The transcript/handout mention "cruciform ligaments" and vertebral attachment. The textbook describes dural attachment at the foramen magnum and pia-to-dura denticulate ligaments.

REMEMBER Denticulate = lateral support. Filum/coccygeal support = lengthwise stability.

Book pp. 438–439; C13 L1 slides 7–10; T-C.

Spinal cord cross-section: orient before labeling



POSTERIOR

Posterior median sulcus is the shallower midline groove; posterior/dorsal horns lie toward it.

ANTERIOR

Anterior median fissure is the deeper midline indentation; anterior/ventral horns are on this side.

GRAY CENTER

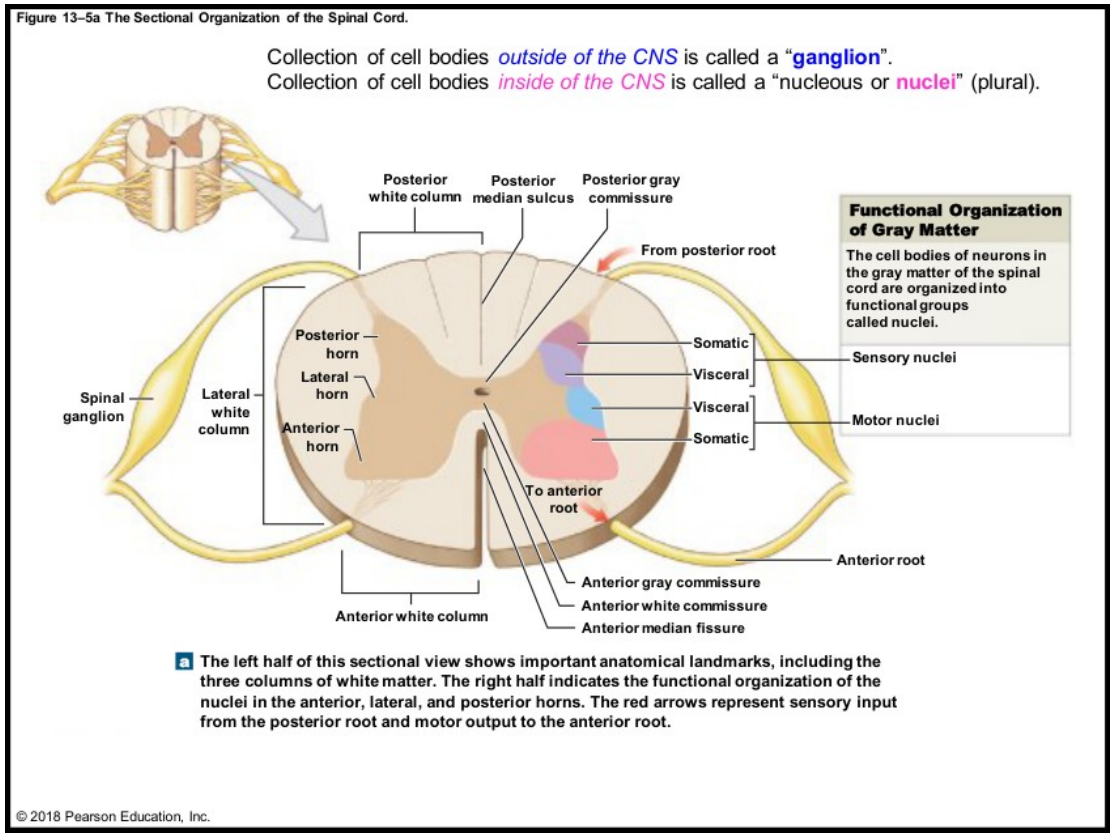
Butterfly/H-shaped gray matter surrounds the central canal. Gray commissures bridge the two halves.

WHITE OUTSIDE

Posterior, lateral and anterior white columns surround the gray matter; the anterior white commissure crosses the midline.

REMEMBER Find the deep anterior fissure first; then name every other part relative to it.

The high-priority nuclei pattern: S–V–V–S



POSTERIOR → ANTERIOR

Somatic sensory; visceral sensory; visceral motor; somatic motor.

SENSORY HALF

Posterior horn nuclei receive and integrate incoming information.

MOTOR HALF

Lateral horn nuclei contribute visceral motor output; anterior horn nuclei contribute somatic motor output.

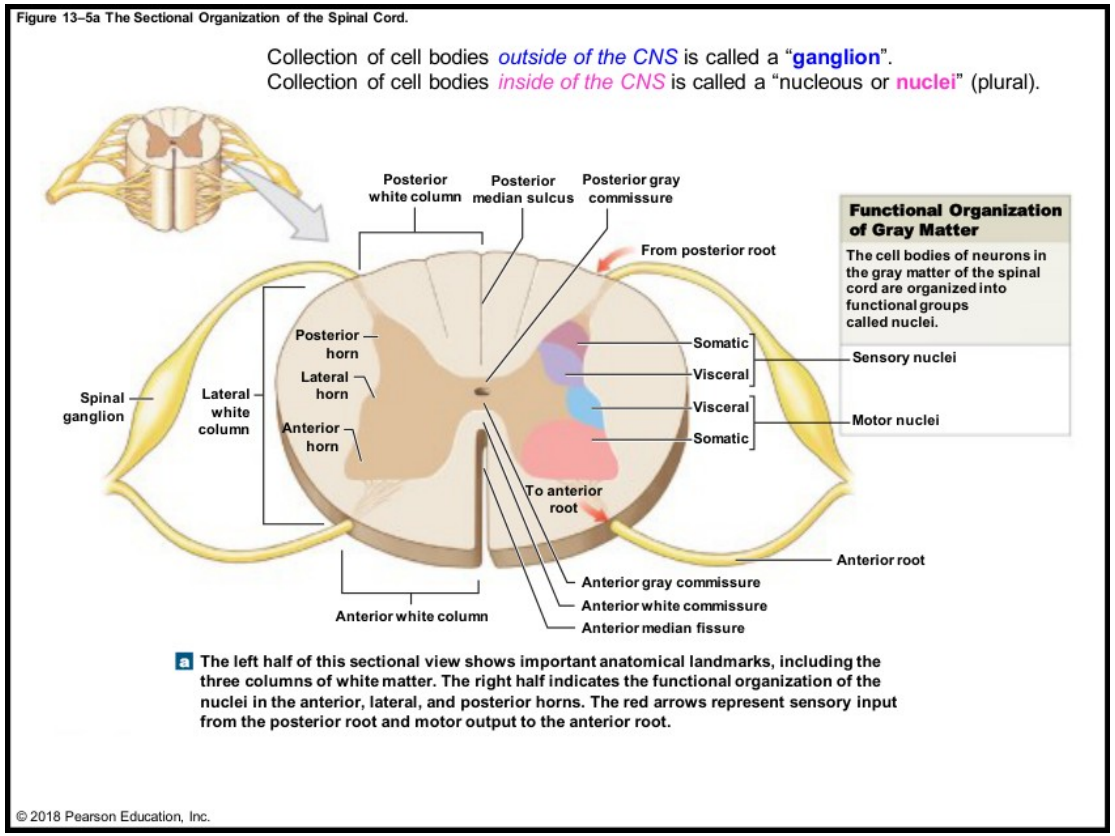
EXAM CUE

The professor explicitly flagged the organization of this diagram as something students should know.

REMEMBER S–V–V–S: the outer labels are Somatic; the middle labels are Visceral. Sensory above, motor below in this figure.

Book pp. 440–441; C13 L1 slides 12–13; T-C.

Dorsal root in; ventral root out; nerve is mixed



DORSAL / POSTERIOR ROOT

Sensory afferent fibers enter the spinal cord. The dorsal root ganglion contains their cell bodies.

VENTRAL / ANTERIOR ROOT

Motor efferent axons leave. Their cell bodies are in the CNS, so there is no equivalent ventral root ganglion.

SPINAL NERVE

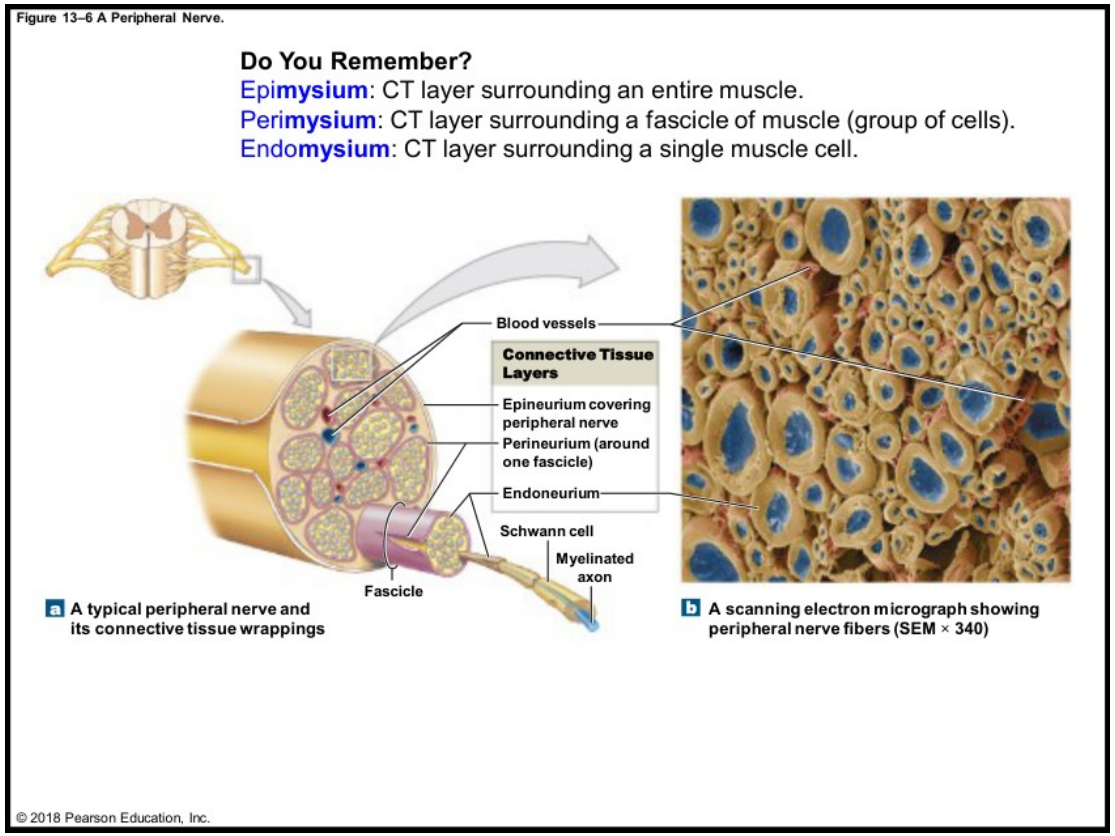
After roots join, the spinal nerve contains both sensory and motor fibers.

RAMI

Posterior and anterior rami also contain mixed fibers. A posterior ramus is not the same structure as a posterior root.

REMEMBER DAVE: Dorsal Afferent; Ventral Efferent. Root ≠ ramus.

Peripheral nerve coverings: reuse the muscle pattern



ENDONEURIUM

Delicate connective tissue around individual axons.

PERINEURIUM

Connective tissue around a fascicle—a bundle of axons.

EPINEURIUM

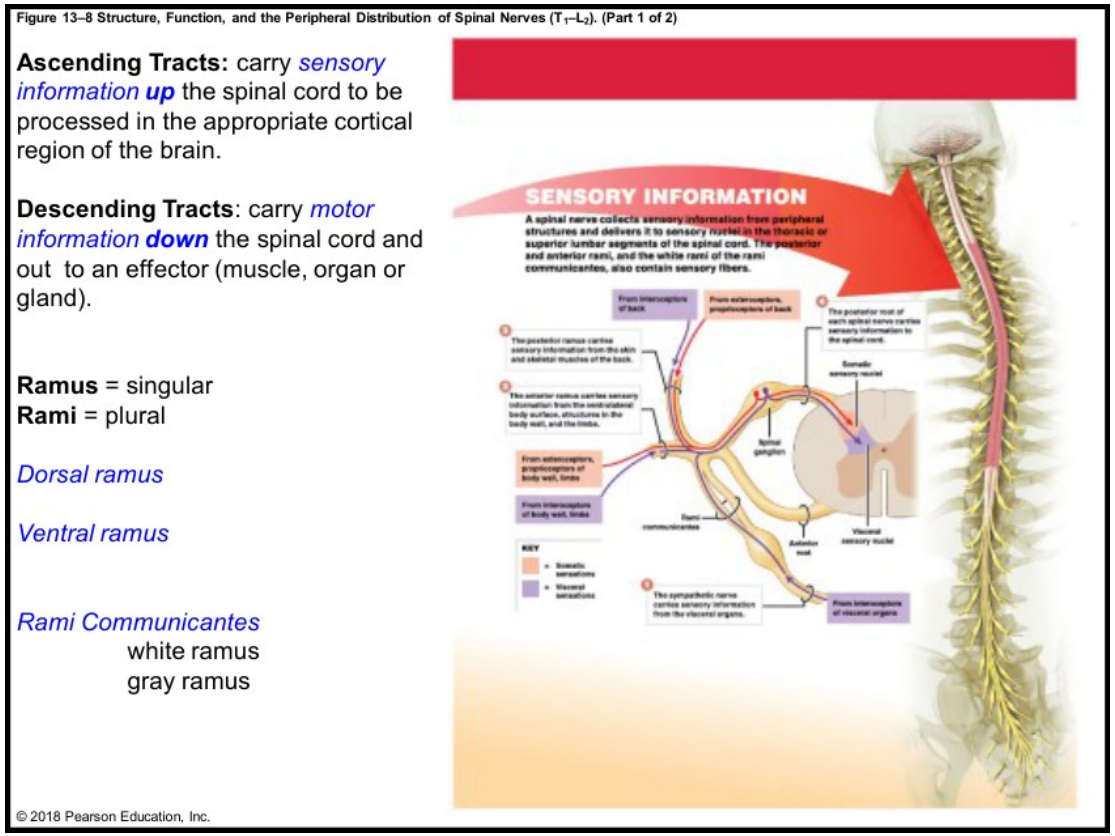
Outer connective tissue around the whole nerve.

PURPOSE

Support, protection from mechanical stress, and routes for vessels supplying the nerve.

REMEMBER Endo = each axon; peri = packet; epi = entire nerve.

White-matter tracts: direction and destination



ASCENDING

Carry sensory information toward the brain. The lecture uses spinothalamic as an example.

DESCENDING

Carry motor commands from brain toward spinal levels. The lecture uses corticospinal as an example.

NAME CLUE

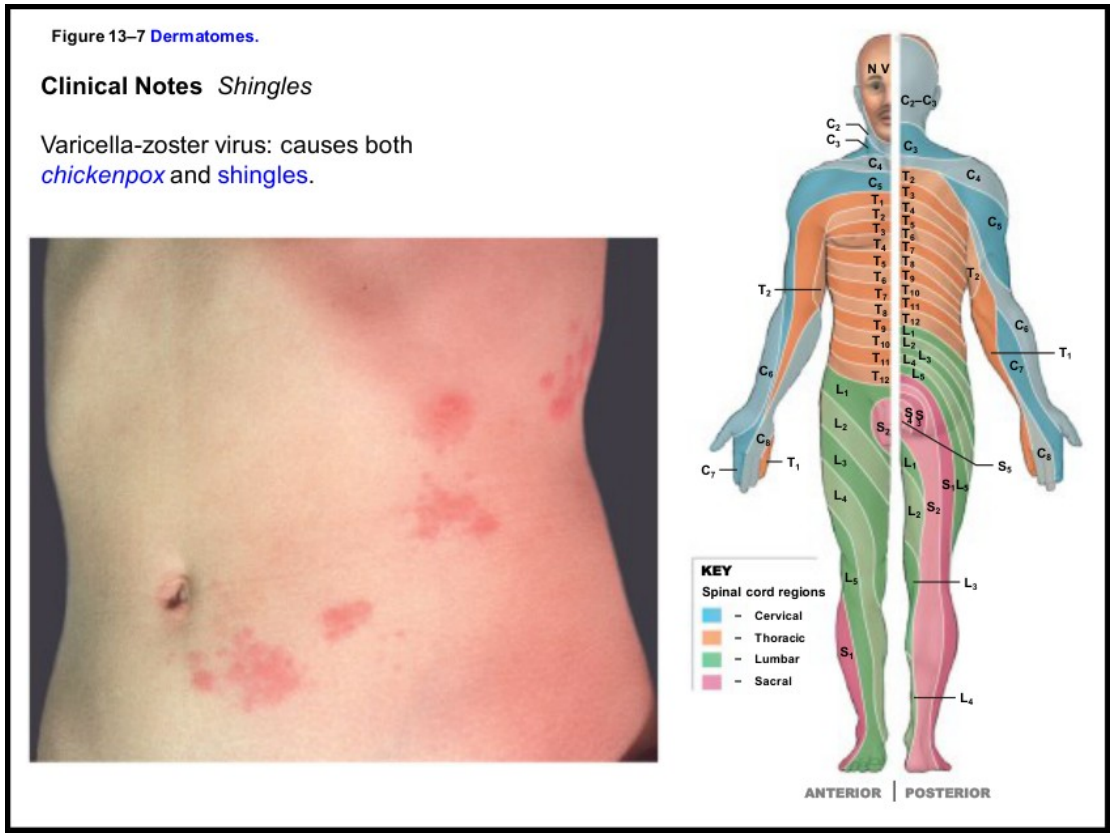
The prefix often identifies the origin and the suffix the destination: spino–thalamic; cortico–spinal.

ORGANIZATION

A tract groups axons with similar functions/direction. Shorter tracts can link spinal segments; longer ones connect cord and brain.

REMEMBER Ascending arrives at the brain; descending delivers motor commands downward.

Dermatomes: know the concept, not the whole map



DEFINITION

A skin territory supplied by a particular spinal nerve/root; the book depicts bilateral territories for a pair.

OVERLAP

Adjacent territories overlap, so the boundaries are not isolated hard lines.

APPLICATION

Sensory changes can help localize a nerve/root problem; the chapter uses shingles as a recognizable distribution example.

LECTURE LIMIT

The professor says to understand what a dermatome is, but not to memorize the complete labeled map for this exam.

REMEMBER DERM = skin; TOME = territory for this memory aid.

Checkpoint 4 | cord anatomy

1. Name the meningeal layers outside-to-inside. Which space contains CSF around the cord?

2. What changes after dorsal and ventral roots join?

3. Recite the four functional nuclei groups from posterior toward anterior.

4. Distinguish the conus medullaris, cauda equina and filum terminale.

REMEMBER Take turns: answer without looking, explain why, then reveal the next slide.

Checkpoint 4 | cord anatomy | answers

1. Dura, arachnoid, pia. CSF around the cord is in the subarachnoid space.

2. A mixed spinal nerve forms: it carries sensory and motor fibers. The separate roots have different directions/functions.

3. Somatic sensory, visceral sensory, visceral motor, somatic motor.

4. Conus: tapered end of cord proper. Cauda: descending roots. Filum: supporting fibrous strand.

REMEMBER A correct label is only half the job. Explain the structure, mechanism, or consequence.

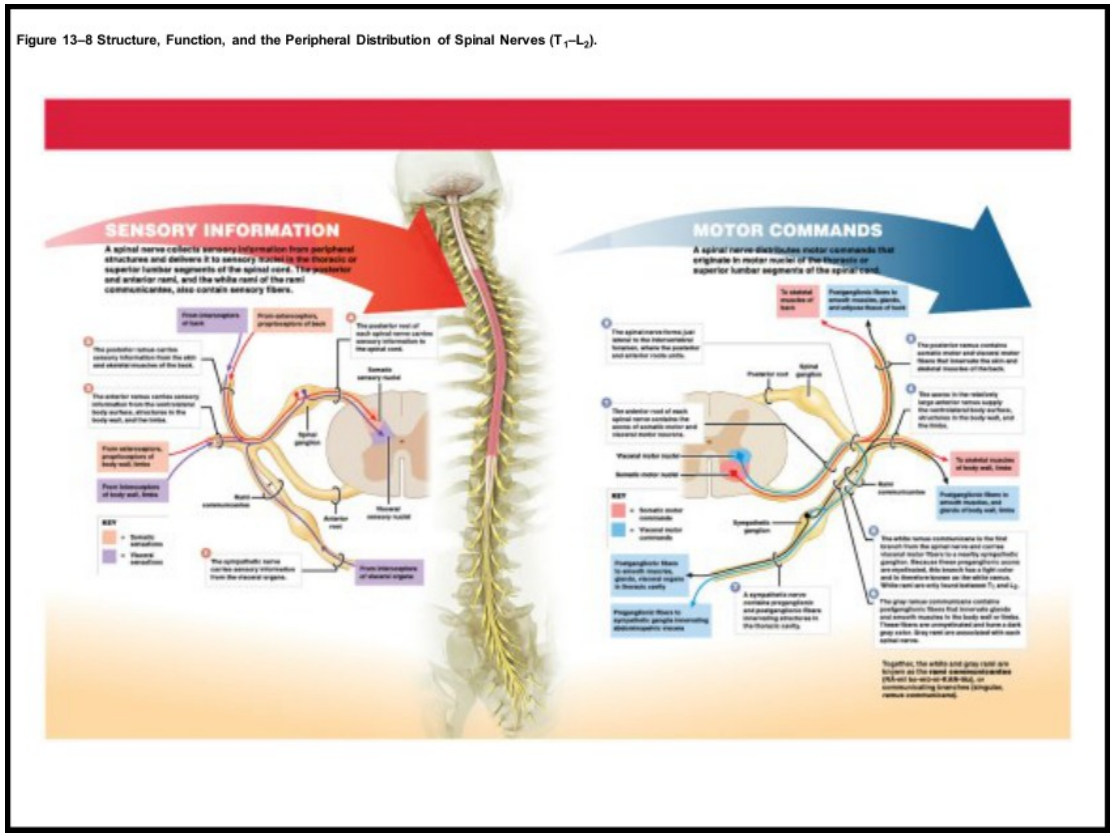
5

Trace the route and identify the target

Use the roots, rami and ganglia correctly before memorizing named nerves.

Use the memory cues, then explain the mechanism without reading.

Rami: branches after the mixed nerve forms



POSTERIOR RAMUS

Supplies sensory and motor fibers to the skin and muscles of the back.

ANTERIOR RAMUS

Supplies the ventrolateral body wall and limbs; anterior rami form the major plexuses.

WHITE RAMUS COMMUNICANS

Myelinated preganglionic sympathetic fibers travel toward a sympathetic ganglion; the illustrated white ramus also carries visceral sensory fibers.

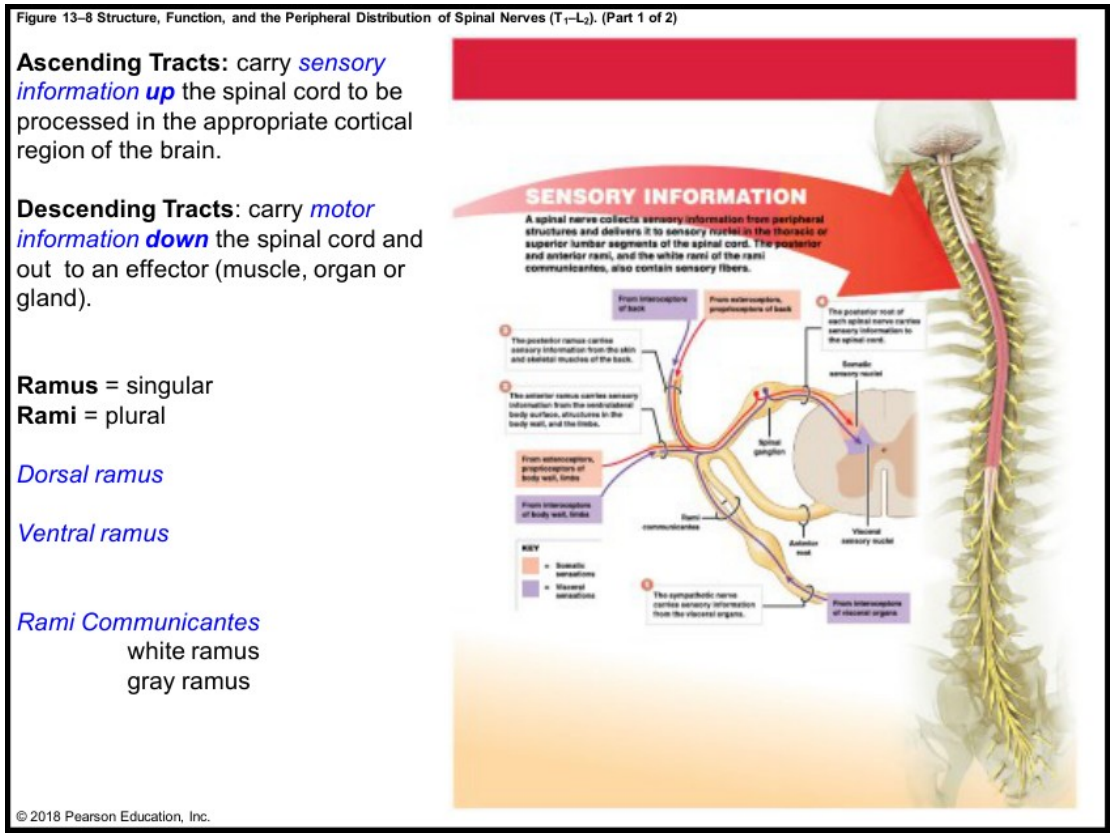
GRAY RAMUS COMMUNICANS

Unmyelinated postganglionic fibers return to spinal-nerve routes for body-wall/limb targets.

REMEMBER Root = in/out of cord. Ramus = branch to a body region. Communicans = connecting branch.

Book pp. 438, 444–447; C13 L2 slides 1–4; T-D.

Trace sensory information inward



BODY WALL / LIMB

Receptor → anterior ramus → mixed spinal nerve → dorsal root/DRG → spinal sensory nuclei.

BACK

Receptor → posterior ramus → mixed spinal nerve → dorsal root/DRG → spinal sensory nuclei.

VISCERAL EXAMPLE

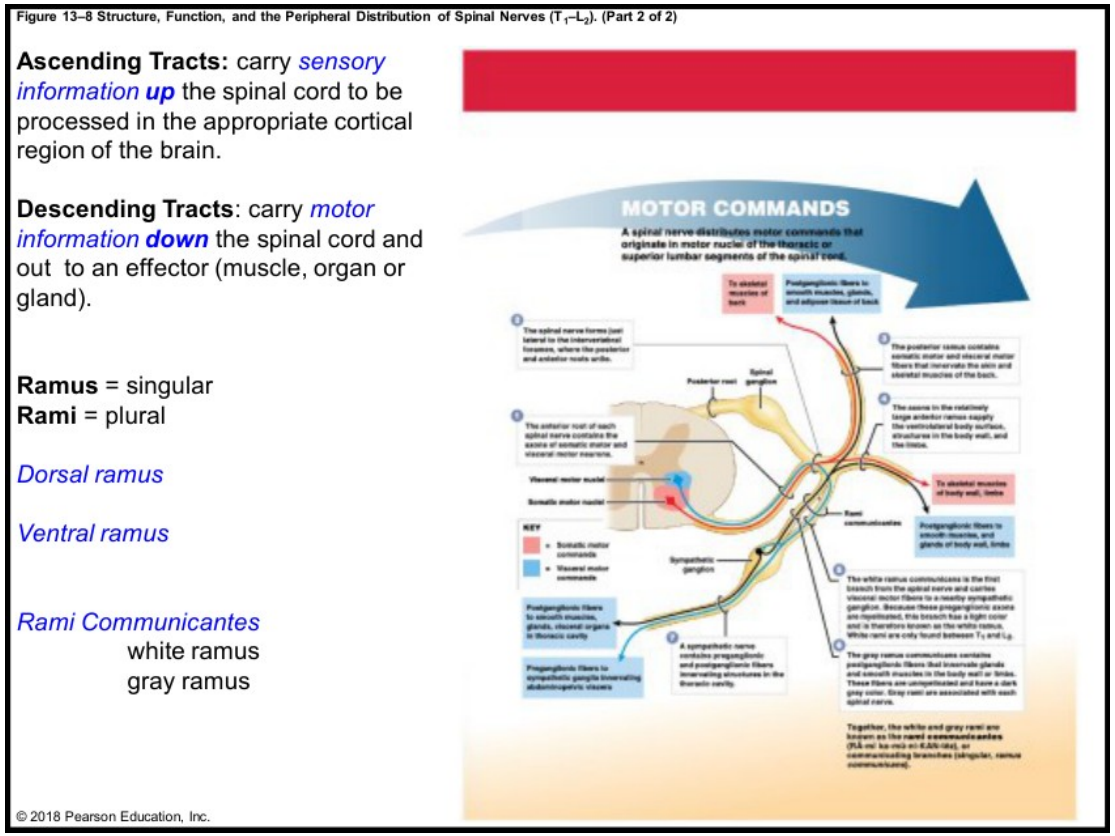
Visceral receptor → sympathetic-nerve route → white ramus → spinal nerve → dorsal root/DRG → visceral sensory nuclei.

COMMON ENDPOINT

The dorsal root carries sensory information regardless of whether its source is somatic or visceral.

REMEMBER Different origins; same sensory doorway: the dorsal root.

Trace motor commands outward



SOMATIC ROUTE

CNS somatic motor neuron → ventral root → spinal nerve → appropriate ramus → skeletal muscle.

VISCERAL ROUTE

CNS preganglionic neuron → ventral root → spinal nerve → white ramus → sympathetic ganglion pathway.

TWO-NEURON CHAIN

Visceral motor signaling in this model uses a preganglionic neuron, then a postganglionic neuron to the effector.

TARGETS

Smooth muscle, glands and adipose tissue are not voluntarily controlled just because their fibers share a mixed nerve with somatic axons.

REMEMBER Somatic: one neuron from CNS to muscle. Autonomic example: a ganglion relay.

Three sympathetic destinations in the lecture diagram

Destination	What happens at the ganglion?	How to trace it
Body wall / limbs	Preganglionic fiber synapses in chain ganglion	Postganglionic fiber returns through gray ramus, then travels in a spinal nerve/ramus
Thoracic viscera	Relay occurs in a sympathetic ganglion	Postganglionic fibers continue toward the thoracic organ
Abdominopelvic viscera	Preganglionic fiber can pass through without synapsing there	It reaches a more distant collateral ganglion; postganglionic fiber then reaches the organ

REMEMBER “Pre” and “post” refer to the autonomic ganglion synapse—not distance from the spinal cord.

Book p. 445, Fig. 13–8; C13 L2 slide 4; T-D.

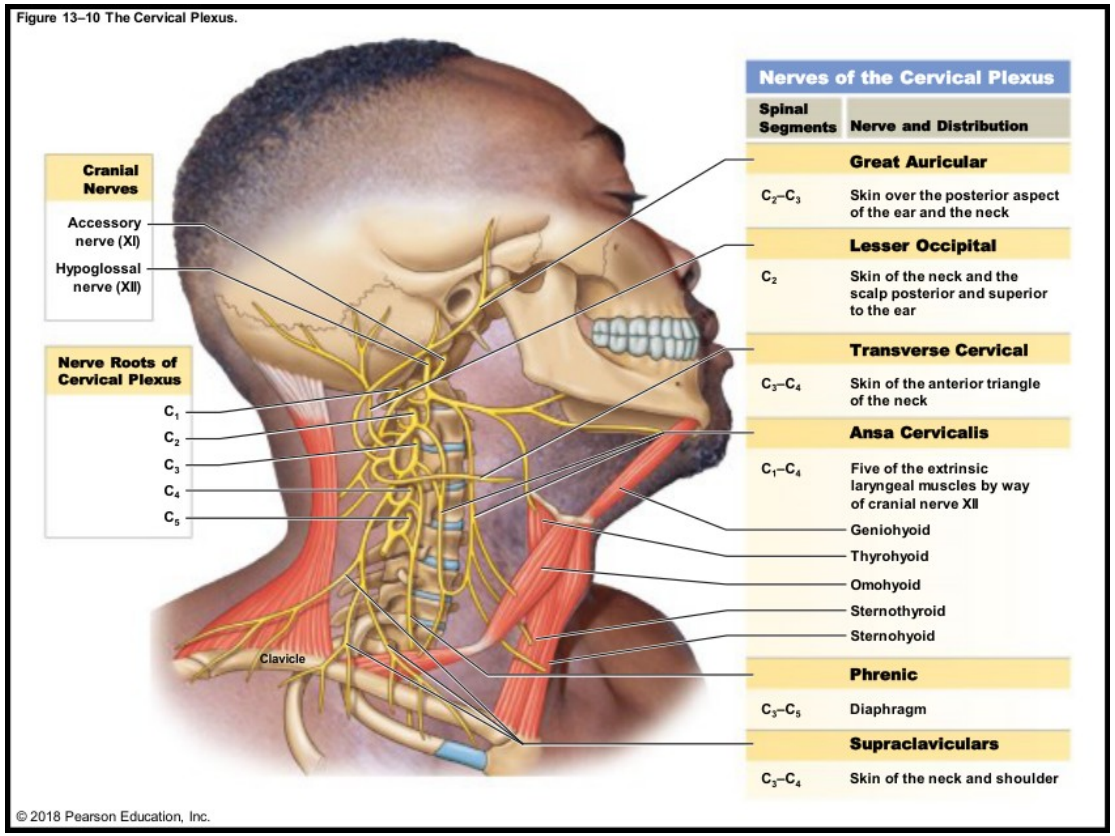
Four plexuses: memorize your course's ranges

Plexus	Range used in your sources	Broad region
Cervical	C1–C5	Neck and diaphragm
Brachial	C5–T1	Pectoral girdle and upper limb
Lumbar	T12–L4	Pelvic/lower-limb contributions
Sacral	L4–S4	Pelvis and lower limb

REMEMBER C5 overlaps cervical/brachial; L4 overlaps lumbar/sacral. No separate thoracic plexus in this list.

Book pp. 446–451; C13 L2 slides 5, 9–12; T-D.

Cervical plexus: the phrenic nerve is the priority



BOUNDARIES

Know C1–C5 for the plexus in these course sources.

KEY NERVE

Phrenic nerve receives contributions from C3, C4 and C5.

KEY TARGET

The diaphragm: a critical breathing muscle.

TEST CUE

The professor specifically says to know this nerve's contributing roots, unlike most detailed individual root lists in the other plexuses.

REMEMBER "C3, C4, C5 keep the diaphragm alive."

Book pp. 447–448; C13 L2 slides 5–6; T-D.

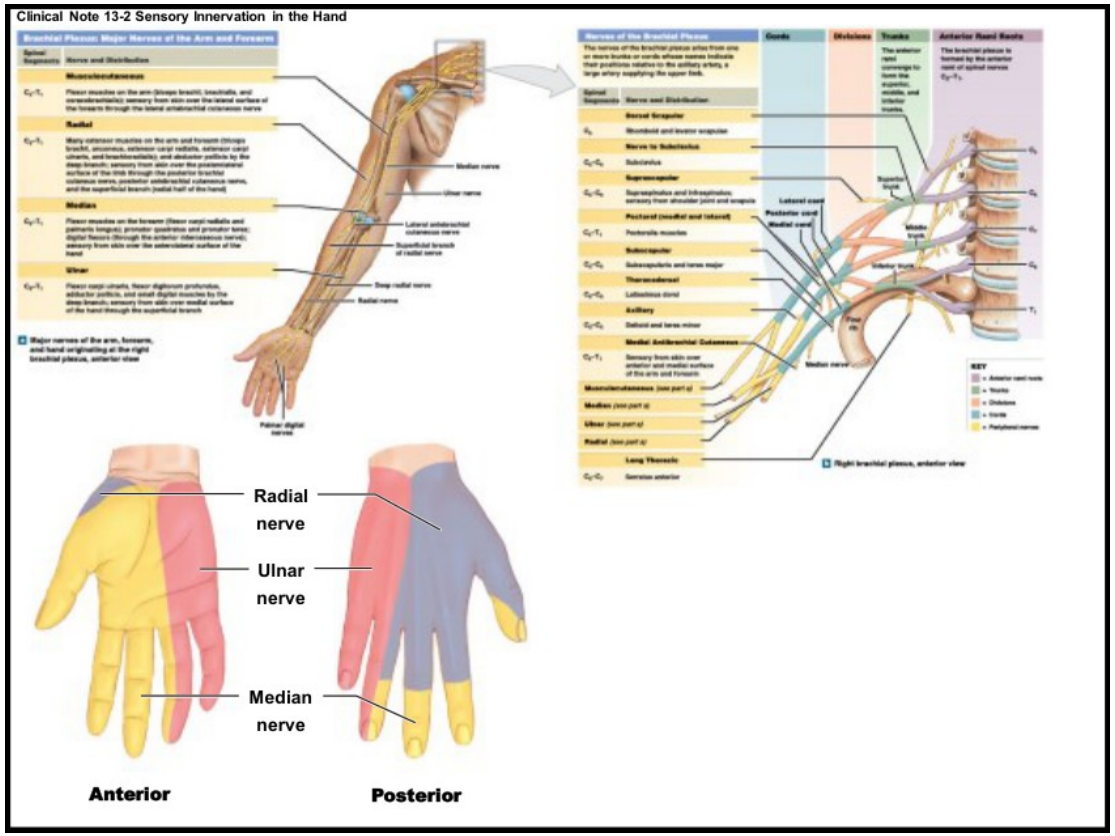
Brachial plexus: four major test targets

Nerve	Lecture-level association	Representative muscles / target
Musculocutaneous	Flexors of the arm	Biceps brachii, brachialis, coracobrachialis
Median	Forearm/digital flexion; pronation	Flexor carpi radialis, palmaris longus, pronators; digital flexors
Ulnar	Additional forearm/hand flexors and intrinsic hand actions	Flexor carpi ulnaris; parts of deep digital flexors; small hand muscles
Radial	Extensors of arm and forearm	Triceps brachii and the wrist/finger extensor group

REMEMBER Musculocutaneous bends the arm; median/ulnar help flex the forearm/hand; radial extends.

Book p. 448; C13 L2 slides 7–8; T-D.

Brachial wiring: understand without overmemorizing



SEQUENCE

Anterior rami (“roots” of the plexus) → trunks → divisions → cords → named peripheral nerves.

TRUNKS + CORDS

The source names superior/middle/inferior trunks and lateral/posterior/medial cords.

MULTIPLE CONTRIBUTIONS

A peripheral nerve may contain fibers originating at several spinal levels. A single muscle can therefore receive overlapping contributions.

SCOPE

Know what a plexus does. The professor does not ask this class to memorize every root-to-cord-to-branch detail.

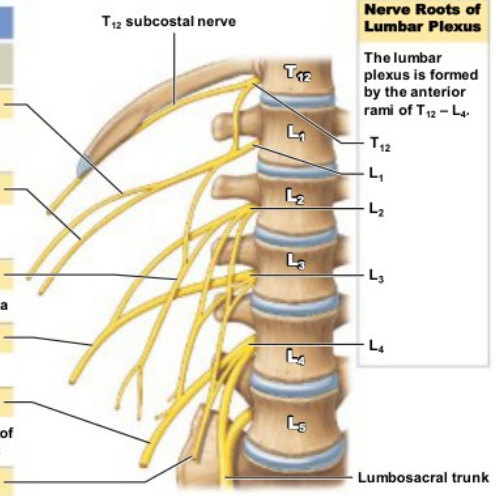
REMEMBER Roots, Trunks, Divisions, Cords, Branches: “Read The Diagram Carefully, Buddy.”

Book pp. 447–449; C13 L2 slides 7–8; T-D.

Lumbar plexus: femoral and obturator

Figure 13–12a The Lumbar and Sacral Plexuses.

Lumbar Plexus	
Spinal Segments	**Nerve and Distribution**
	Iliohypogastric
T_{12}–L_1	External and internal oblique and transverse abdominis; skin over the inferior abdomen and buttocks
	Ilio-inguinal
L_1	Abdominal muscles (with iliohypogastric nerve); skin over superior, medial thigh and portions of external genitalia
	Genitofemoral
L_1–L_2	Skin over anteromedial thigh and portions of external genitalia
	Lateral Femoral Cutaneous
L_2–L_3	Skin over anterior, lateral, and posterior thigh
	Femoral
L_2–L_4	Quadriceps femoris, sartorius, pectineus, and iliopsoas; skin of the anteromedial thigh, and medial surface of the leg and foot
	Obturator
L_2–L_4	Gracilis, and adductor magnus, brevis and longus; skin from the medial surface of the thigh



a Lumbar plexus, anterior view

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PLEXUS RANGE

T12–L4 in the handout and lecture.

FEMORAL

The supplied table lists quadriceps, sartorius, pectineus and iliopsoas. The professor emphasizes the anterior-thigh/hip-flexion group.

OBTURATOR

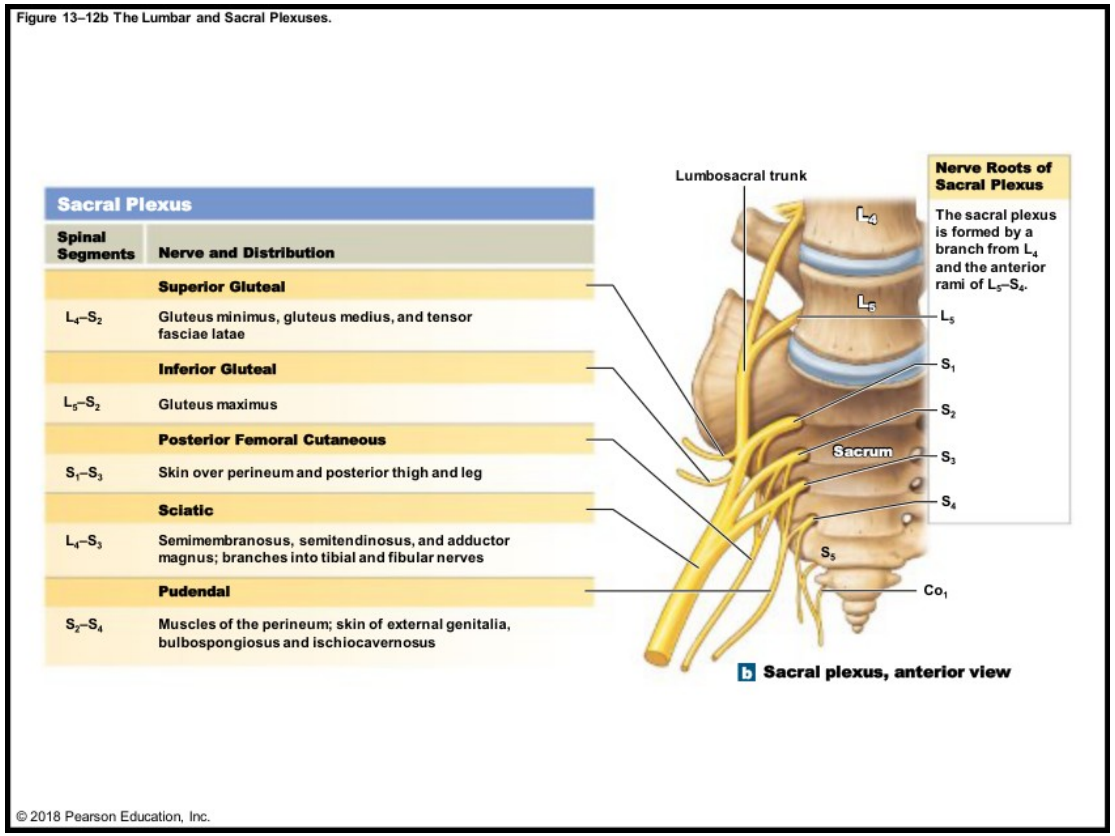
Gracilis and adductor magnus, brevis and longus: remember the thigh-adductor association.

DON'T MIX THE ACTION

Adduction moves toward the midline. A speech-recognition "abduction" near the adductor list should not replace the slide's muscle names.

REMEMBER Femoral = front-thigh group. Obturator = adductors bring the thighs together.

Sacral plexus: gluteal nerves and sciatic nerve


Figure 13–12b The Lumbar and Sacral Plexuses.

Sacral Plexus

Spinal Segments	Nerve and Distribution
	Superior Gluteal
L_4–S_2	Gluteus minimus, gluteus medius, and tensor fasciae latae
	Inferior Gluteal
L_5–S_2	Gluteus maximus
	Posterior Femoral Cutaneous
S_1–S_3	Skin over perineum and posterior thigh and leg
	Sciatic
L_4–S_3	Semimembranosus, semitendinosus, and adductor magnus; branches into tibial and fibular nerves
	Pudendal
S_2–S_4	Muscles of the perineum; skin of external genitalia, bulbospongiosus and ischiocavernosus

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PLEXUS RANGE

L4–S4 in these sources.

SUPERIOR GLUTEAL

Gluteus medius, gluteus minimus and tensor fasciae latae in the table.

INFERIOR GLUTEAL

Gluteus maximus. The lecturer especially wants the gluteal-muscle association.

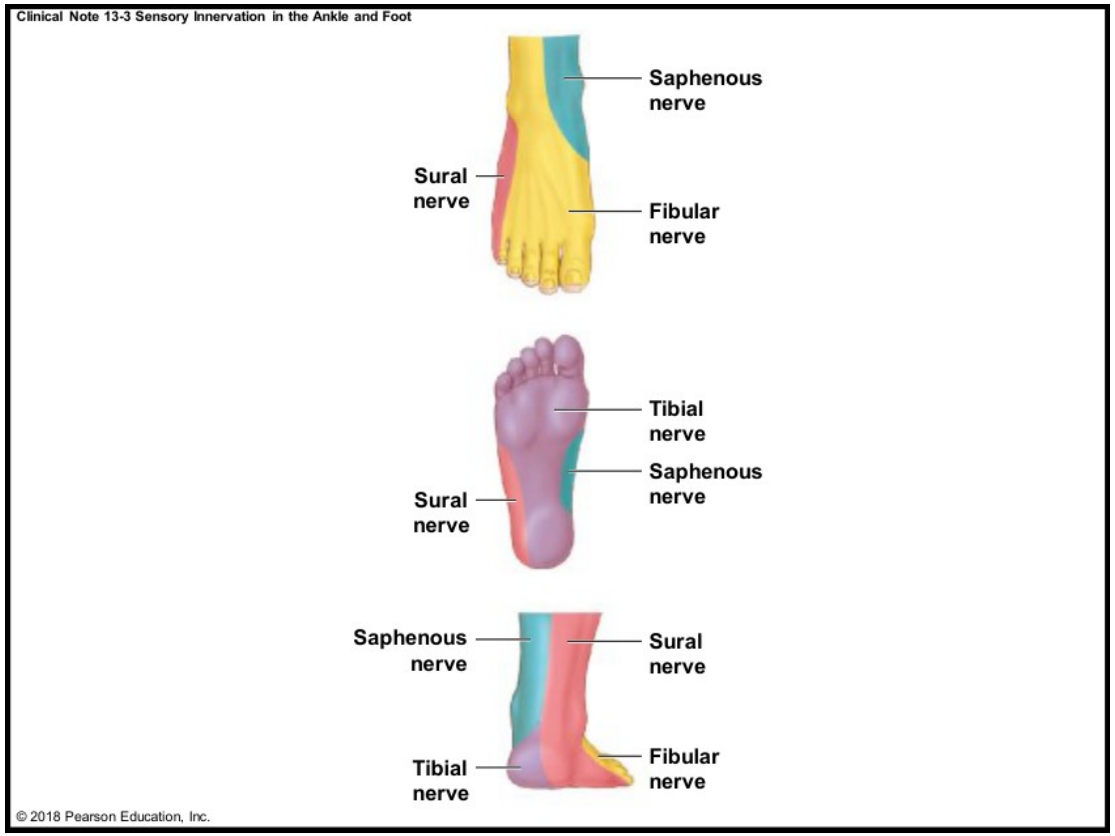
SCIATIC

The highlighted table includes semimembranosus, semitendinosus and adductor magnus; it divides into tibial and fibular branches.

REMEMBER Inferior gluteal → maximus. Sciatic → the highlighted hamstring group and major lower-limb branches.

Book pp. 450–451; C13 L2 slides 10, 12; T-D.

Hand and foot sensory maps: recognize the application



HAND

The source image maps median, ulnar and radial territories.

FOOT

The image distinguishes tibial, fibular, sural and saphenous territories.

PURPOSE

A pattern of sensory change, combined with muscle findings, can help identify a nerve pathway involved.

LECTURE LIMIT

The professor says not to memorize the detailed foot map. Treat these as localization examples and reference visuals.

REMEMBER Map the symptom to a pathway—but do not mistake a peripheral-nerve map for a dermatome map.

Checkpoint 5 | pathways and plexuses

1. A sensory impulse from the back enters through which ramus and which root?

2. Does a preganglionic fiber become postganglionic merely by passing through a ganglion?

3. Name the phrenic roots and its target. Which plexus contains it in this course?

4. Match the nerves: arm flexion, arm/forearm extension, thigh adduction, gluteus maximus.

REMEMBER Take turns: answer without looking, explain why, then reveal the next slide.

Checkpoint 5 | pathways and plexuses | answers

1. Posterior ramus, then mixed spinal nerve and dorsal/posterior root.

2. No. The terms refer to the neuron before versus after the ganglionic synapse.

3. C3–C5; diaphragm; cervical plexus (course range C1–C5).

4. Musculocutaneous; radial; obturator; inferior gluteal, respectively.

REMEMBER A correct label is only half the job. Explain the structure, mechanism, or consequence.

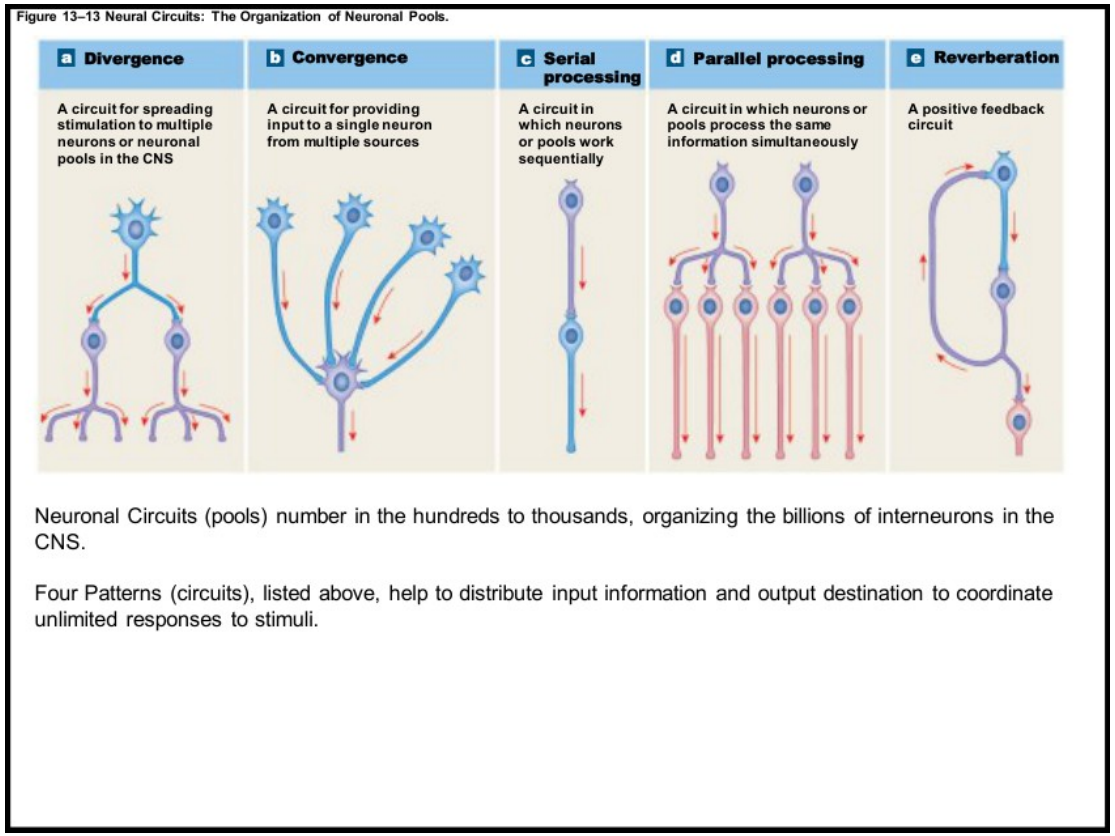
6

From neuronal pools to automatic responses

Explain how many neurons cooperate—and why a spinal reflex can happen before conscious pain.

Use the memory cues, then explain the mechanism without reading.

Five patterns of neural circuitry



DIVERGENCE / CONVERGENCE

Divergence distributes one input to many outputs. Convergence combines many inputs at one target.

SERIAL / PARALLEL

Serial processing is a sequence of relays. Parallel processing sends information through multiple simultaneous processing pathways.

REVERBERATION

Feedback connections reactivate earlier neurons; activity can continue until inhibition or fatigue ends the loop.

NEURONAL POOL

A functional group of interconnected neurons with particular input sources and output destinations.

REMEMBER Diverge = distribute. Converge = collect. Serial = sequence. Parallel = simultaneous. Reverberate = repeat.

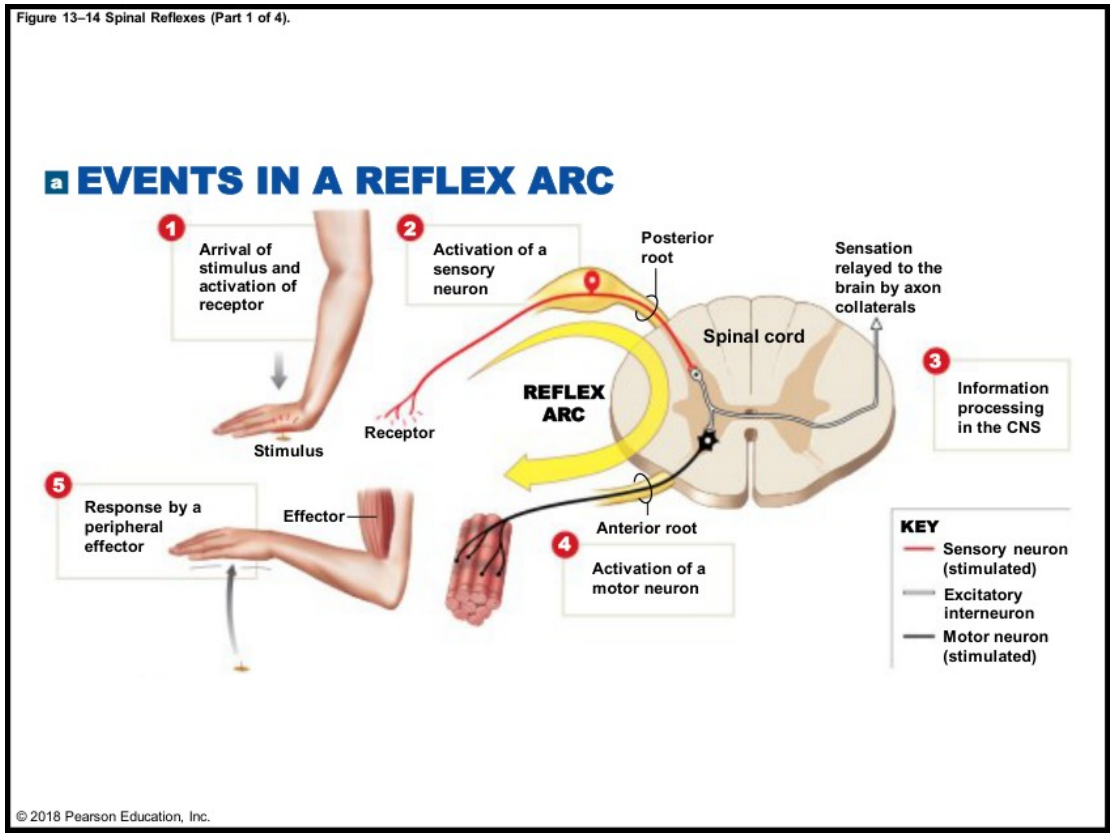
Book pp. 452–453; C13 L3 slide 1; T-D.

Recognize the circuit from a scenario

Scenario	Circuit pattern	Why?
One sensory input informs several brain regions	Divergence	One source spreads to multiple targets
Several sources influence a single motor neuron	Convergence	Many inputs collect at one target
Information moves through relay 1, then 2, then 3	Serial	Sequential processing
Withdrawal, weight shift and awareness occur together	Parallel	Several paths process information at once
A response persists briefly after its initiating input	Reverberation	Feedback can maintain activity

REMEMBER Ask “how many inputs, how many paths, and does the circuit loop?”

The five steps of a reflex arc



1 + 2 | INPUT

A stimulus activates a receptor; a sensory neuron carries the signal toward the CNS.

3 | INTEGRATION

Information is processed in the CNS. A polysynaptic arc uses interneurons; a monosynaptic arc can connect sensory directly to motor.

4 + 5 | OUTPUT

A motor neuron carries the command to a peripheral effector, which produces the response.

FAST PROTECTION

A spinal withdrawal response can start before conscious awareness of pain; collateral pathways also relay information to the brain.

REMEMBER Receptor → Sensory → Integration → Motor → Effector. “Receive, Send, Integrate, Move, Execute.”

Book pp. 453–456; C13 L3 slides 2–4; T-D.

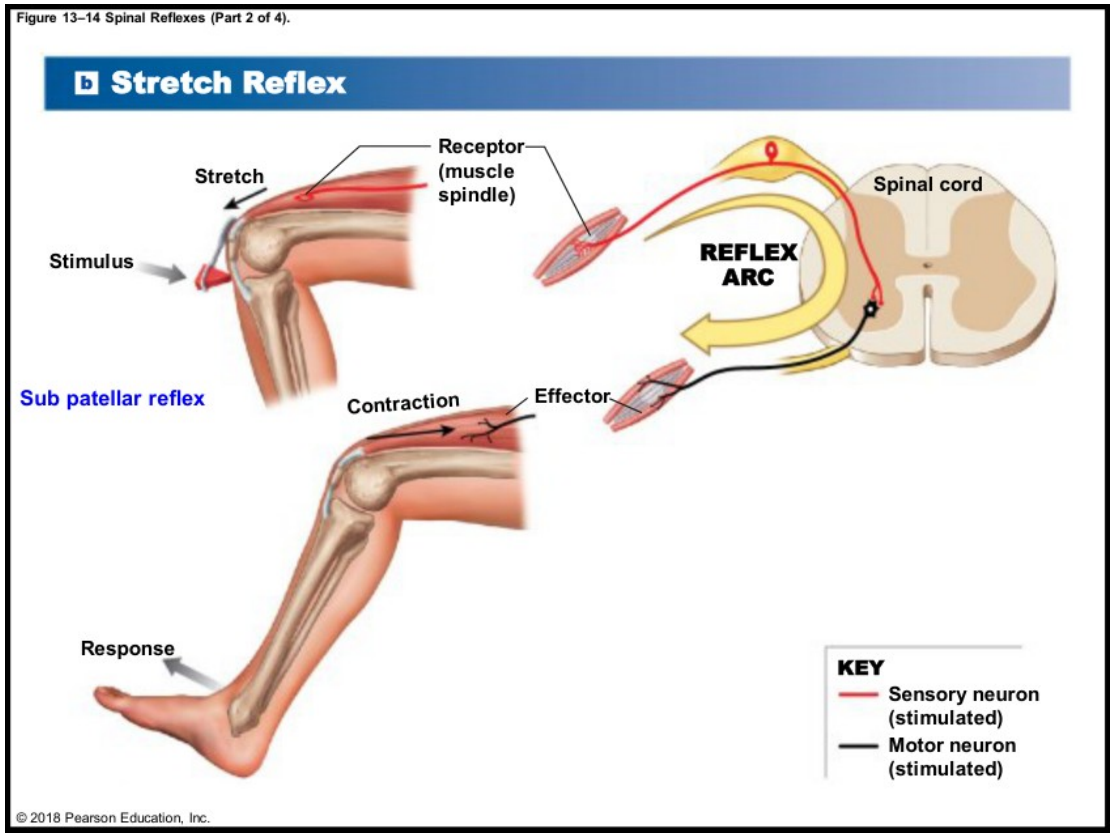
Classify one reflex four different ways

Question	Two categories	What separates them?
How did it develop?	Innate / acquired	Inborn circuitry versus learned motor pattern
What is the effector?	Somatic / visceral	Skeletal muscle versus visceral effectors
How many synapses?	Monosynaptic / polysynaptic	One versus more than one synapse
Where is it processed?	Spinal / cranial	Spinal cord versus brain

REMEMBER Development, response, complexity, site: one reflex can have a label in all four categories.

Book pp. 456–457; C13 L3 slide 7; T-D.

Monosynaptic versus polysynaptic



MONOSYNAPTIC

Sensory neuron synapses directly on a motor neuron. Minimum two neurons, one CNS synapse.

POLYSYNAPTIC

One or more interneurons participate; multiple synapses enable excitation/inhibition of multiple muscle groups.

SPEED TRADEOFF

Fewer synapses minimize delay. More synapses allow more complex and adjustable responses.

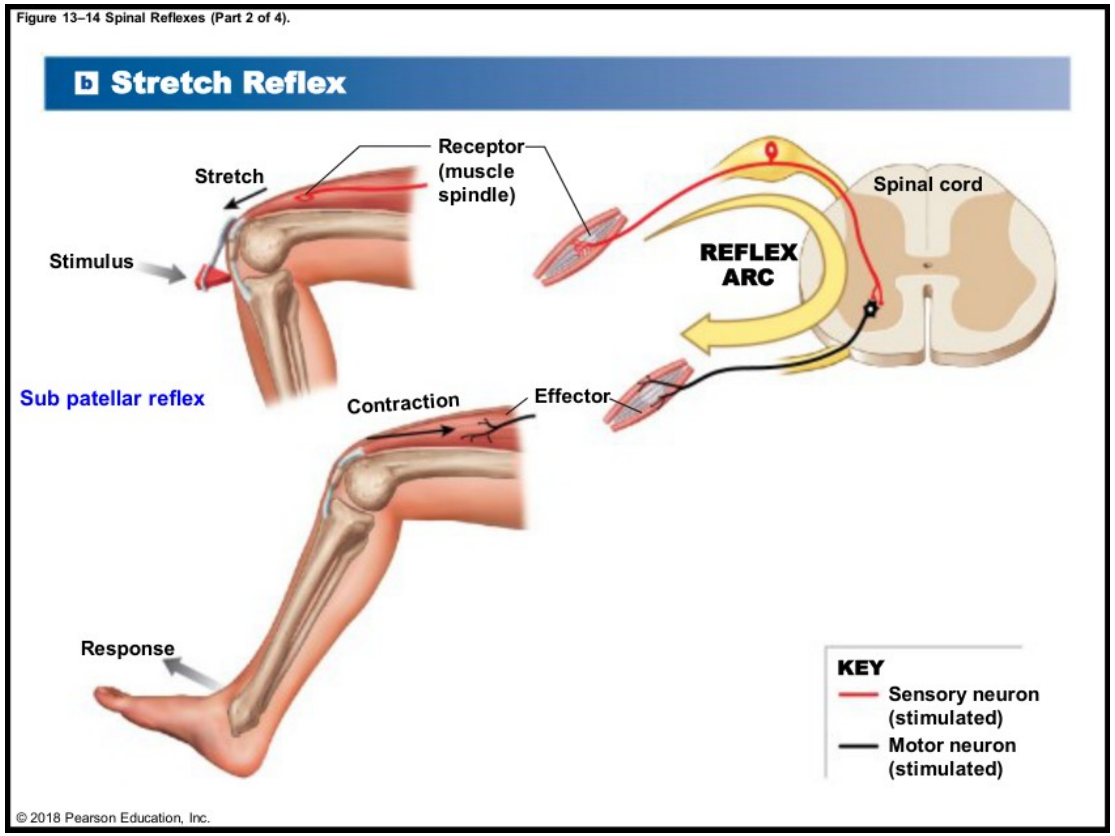
COMMON TRAP

Five steps in a reflex arc do not mean five neurons or five synapses.

REMEMBER Mono counts synapses—not neurons, muscles or steps.

CORE

Stretch reflex: the patellar example



STIMULUS

A patellar-tendon tap briefly stretches quadriceps and its muscle spindles.

SENSORY ARM

The spindle's sensory neuron enters through the dorsal root and excites a motor neuron.

MOTOR ARM

Motor output returns to the stretched muscle; quadriceps contraction extends the knee.

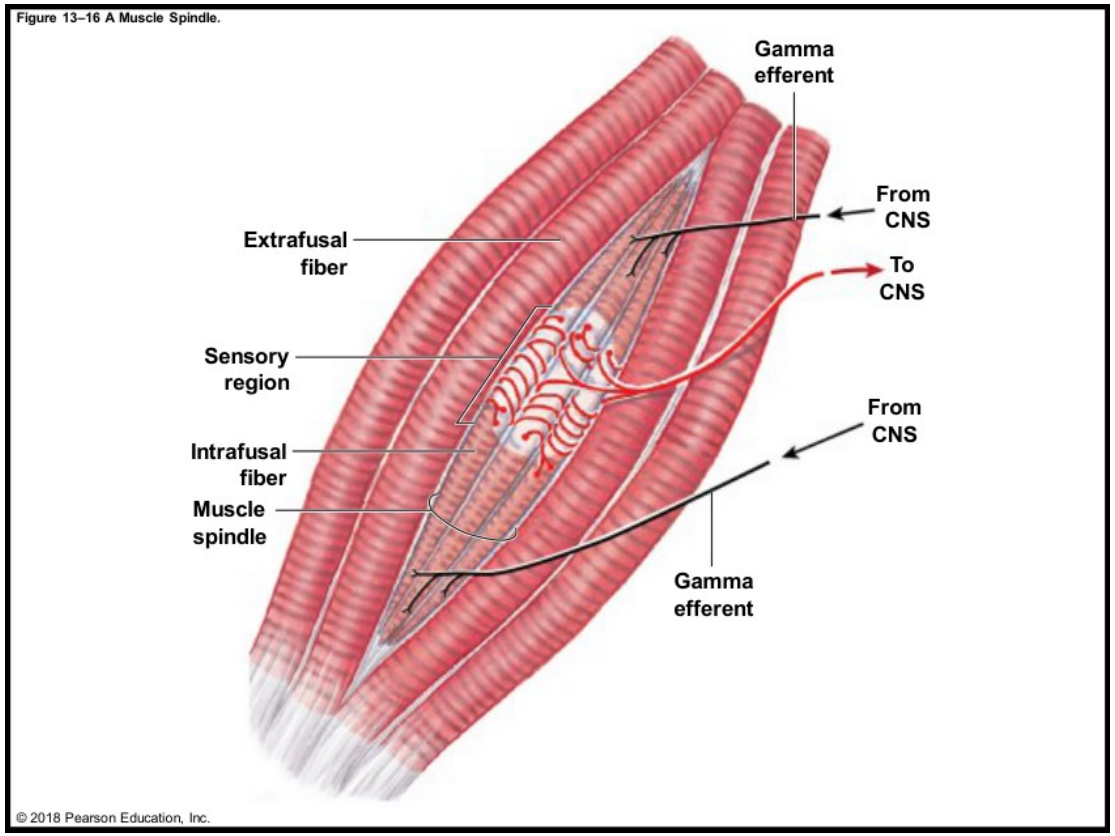
FUNCTION

Automatic opposition to stretch helps regulate muscle length and posture. The core excitatory arc is monosynaptic.

REMEMBER Stretch a muscle → that muscle contracts. The receptor is the muscle spindle.

Book pp. 454, 457–458; C13 L3 slide 5.

Muscle spindle: intrafusal, extrafusal and gamma



INTRAFUSAL

Specialized small fibers inside the spindle; a central sensory region is monitored by sensory endings.

EXTRAFUSAL

The surrounding ordinary skeletal muscle fibers that generate most muscle force.

GAMMA EFFERENTS

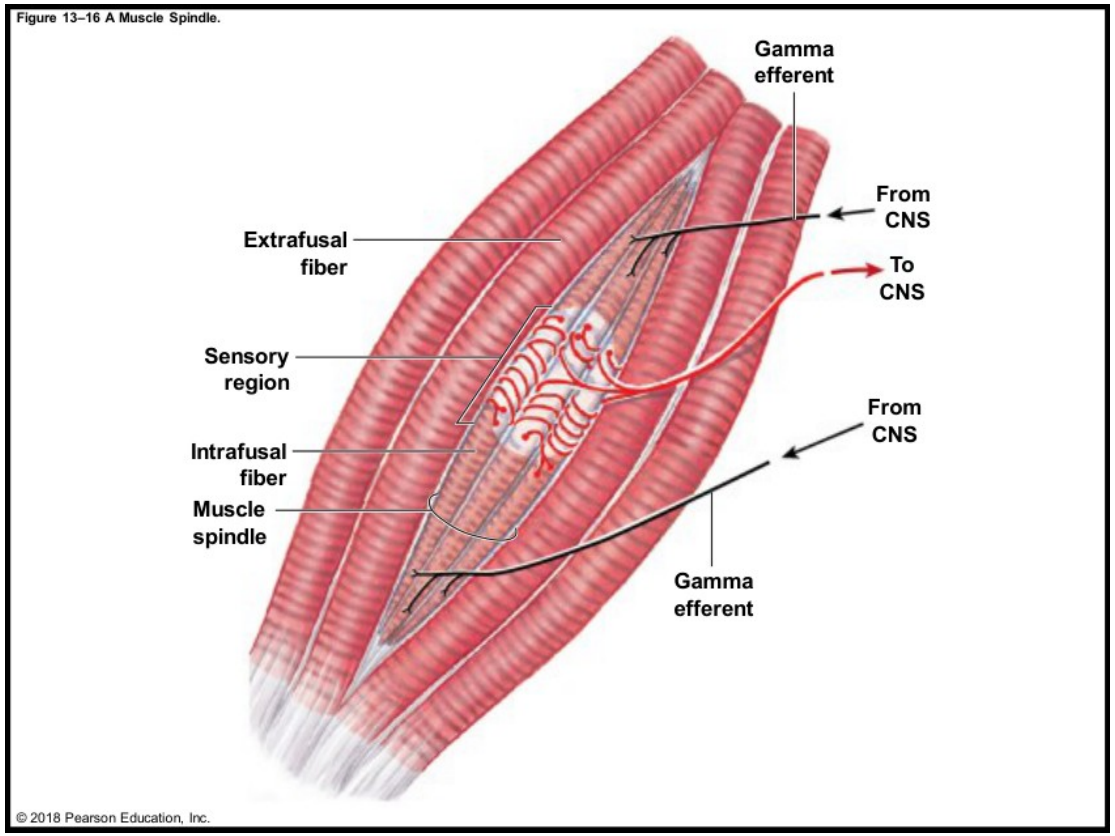
Motor axons to the contractile ends of intrafusal fibers; adjust spindle sensitivity.

WHY ADJUST?

As a muscle shortens, gamma activity helps keep the sensory region responsive to a newly imposed stretch or load.

REMEMBER INTRA is IN the spindle; EXTRA is the external working muscle.

Postural reflexes: small corrections all the time



UPRIGHT STANCE

Multiple muscle groups continually make small adjustments to keep the body balanced.

FORWARD LEAN

Stretching postural muscles can increase their contraction and counter the change in position.

COORDINATION

If correction overshoots, different muscle groups adjust; this is not one isolated, permanent contraction.

CONNECTION

The lecture connects proprioceptive input, neural pools and subconscious balance control to these ongoing responses.

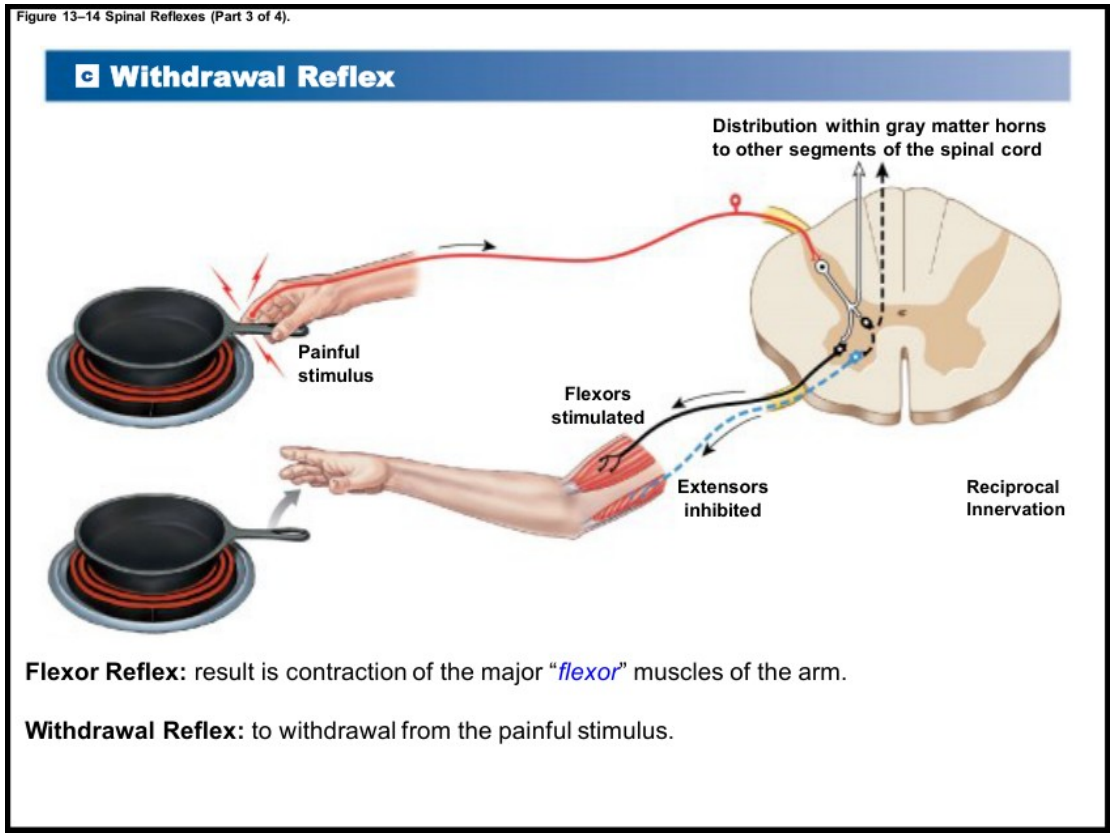
REMEMBER Standing still is active control—not the absence of nervous-system activity.

Stretch reflex versus Golgi tendon reflex

Feature	Stretch reflex	Tendon reflex
Monitored feature	Muscle length/stretch	Tension in the tendon
Receptor	Muscle spindle	Golgi tendon organ
Typical source response	Contract the stretched muscle	Inhibit the muscle generating excessive tension
Core circuit	Monosynaptic excitation	Polysynaptic; inhibitory interneuron
Purpose emphasized	Control length, tone and posture	Limit excessive tension / tendon damage

REMEMBER Spindle senses stretch; tendon organ senses tension.

Withdrawal reflex: flexors on, extensors off



TRIGGER

A painful or otherwise threatening stimulus activates sensory input.

FLEXION

Spinal interneurons stimulate motor neurons to the appropriate flexor muscles, pulling the affected part away.

RECIPROCAL INHIBITION

Interneurons inhibit antagonistic extensors so they do not fight the withdrawal.

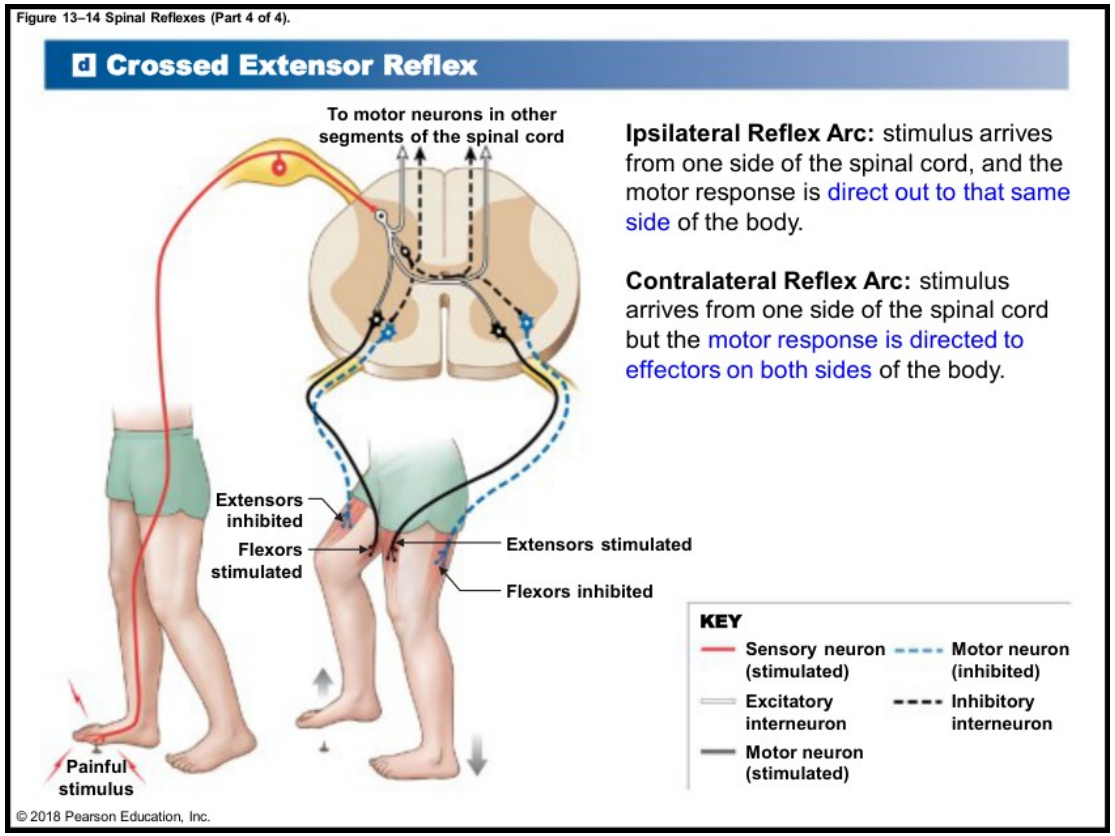
COORDINATION

Stronger stimuli can recruit additional segments and muscles. Signals also ascend toward conscious perception.

REMEMBER Pull away with flexors; let go with extensors.

Book pp. 455, 459–460; C13 L3 slide 4; T-D.

Crossed extensor reflex: withdraw and stay standing



STIMULATED SIDE

Flexors contract and extensors are inhibited, withdrawing the foot from the painful stimulus.

OPPOSITE SIDE

Extensors contract and flexors are inhibited, supporting body weight on the other leg.

CROSSING

Interneurons cross the spinal midline to coordinate the contralateral response.

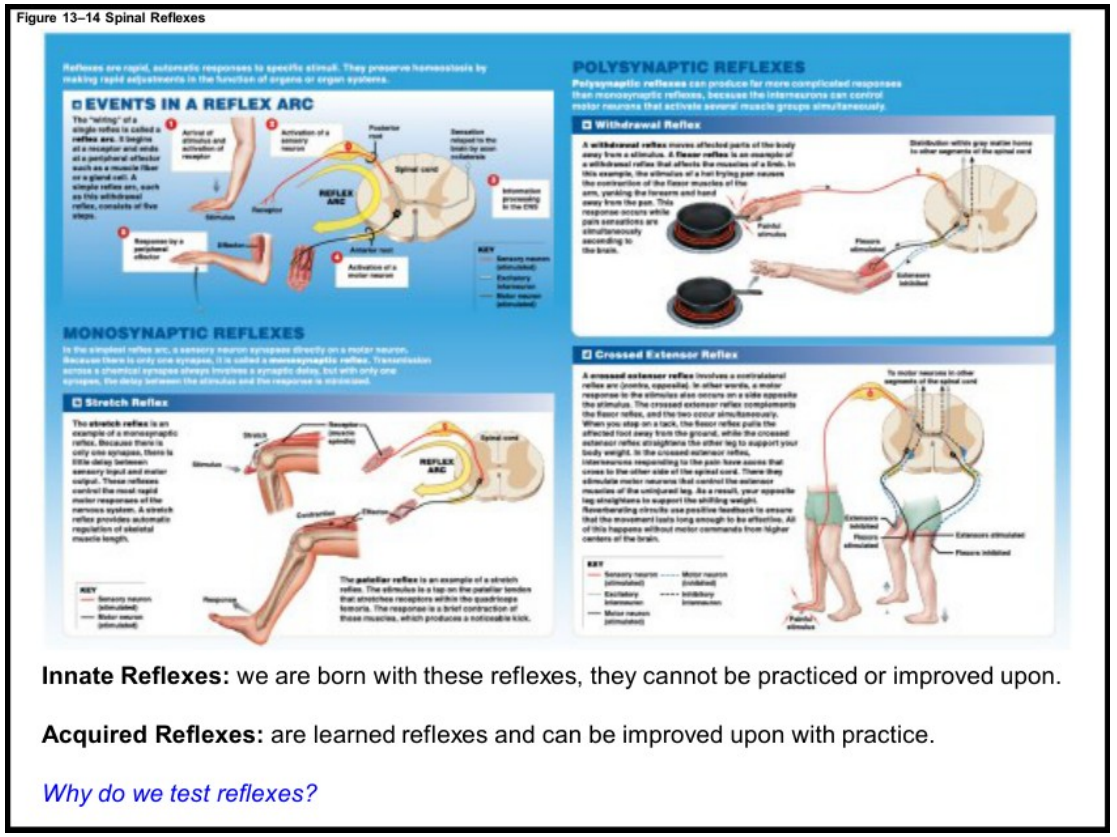
TERMS

Ipsilateral = same side as the stimulus. Contralateral = opposite side. The combined withdrawal/crossed-extensor behavior involves both sides.

REMEMBER One leg lifts; the other leg holds.

Book pp. 455, 459; C13 L3 slide 6.

Polysynaptic responses: the textbook’s five features



INTERNEURON POOLS

Processing can excite one set of motor neurons and inhibit another.

MULTIPLE SEGMENTS

Intersegmental connections distribute responses to additional muscle groups.

RECIPROCAL INHIBITION

Opposing muscles are coordinated rather than simply contracting together.

REVERBERATION + COOPERATION

Feedback can prolong a response; several reflexes cooperate to produce coordinated movement.

REMEMBER Pools, segments, opposites, persistence, cooperation.

Book pp. 459–460, 462.

The brain can tune a spinal reflex

LOCAL INITIATION

A spinal reflex does not need a new conscious command before it begins.

DESCENDING INFLUENCE

Brain pathways can facilitate or inhibit spinal interneurons and motor neurons through EPSPs and IPSPs.

REINFORCEMENT

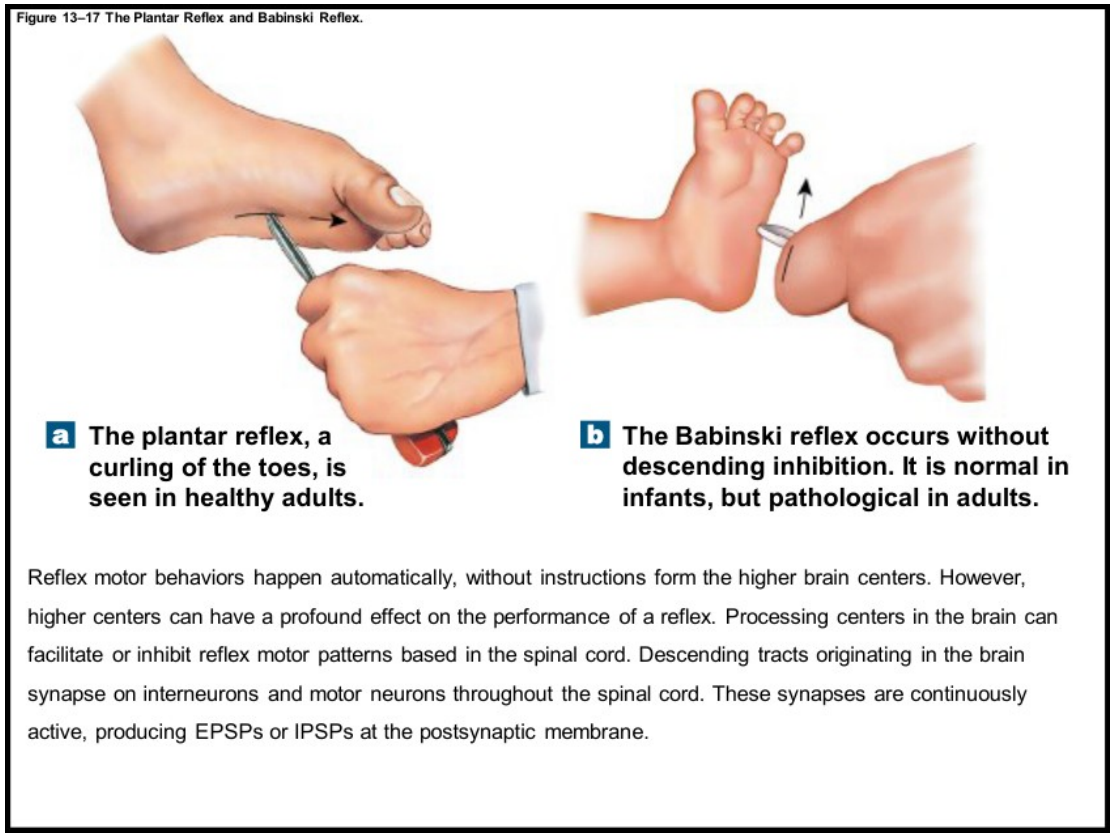
Generalized facilitation enhances a reflex; the book uses the Jendrassik maneuver as an example.

PATTERN REUSE

Higher centers can recruit and adjust existing spinal motor patterns instead of specifying every individual muscle contraction.

REMEMBER A spinal reflex is locally organized—not disconnected from the brain.

Plantar versus Babinski response



ADULT PLANTAR RESPONSE

In the chapter's description, stroking the lateral sole normally produces toe curling.

INFANT BABINSKI RESPONSE

Toe fanning/extension is normal early in development while descending pathways are immature.

ADULT BABINSKI SIGN

Its presence can indicate impaired descending inhibition from CNS disease or injury.

WHY THIS IS HERE

It demonstrates how higher centers change the expression of a spinal reflex.

REMEMBER Babinski belongs in the baby example; its reappearance in an adult is a CNS warning sign in this chapter.

Checkpoint 6 | circuits and reflexes

1. List the five reflex-arc steps. Must every reflex have an interneuron?

2. A muscle suddenly stretches. Which receptor starts the stretch reflex, and what is the response?

3. What happens to flexors and extensors on each side during a crossed-extensor response?

4. How can a spinal reflex occur without a conscious command yet still be affected by the brain?

REMEMBER Take turns: answer without looking, explain why, then reveal the next slide.

Book pp. 453–461; C13 L3. Original practice questions.

Checkpoint 6 | circuits and reflexes | answers

1. Receptor, sensory neuron, CNS integration, motor neuron, effector. No: a monosynaptic arc connects sensory directly to motor.

2. Muscle spindle; the stretched muscle contracts. Contrast a Golgi tendon organ's tension-monitoring role.

3. Stimulated side: flexors on, extensors off. Opposite side: extensors on, flexors off.

4. Spinal circuitry can initiate the response locally while descending pathways facilitate or inhibit that circuitry.

REMEMBER A correct label is only half the job. Explain the structure, mechanism, or consequence.

7

Turn understanding into recall

Finish with mixed cases, drawing rehearsal, a precise glossary, and a one-sided exam sheet.

Use the memory cues, then explain the mechanism without reading.

Mixed cases | connect the chapters

1. Terminal Ca2+ entry is reduced, but the presynaptic AP still arrives. Predict transmitter release and the postsynaptic effect.

2. A ventral root is selectively damaged. Would you first expect a sensory-input or motor-output deficit?

3. An injured neuron cannot conduct through a badly damaged myelinated segment. Why is “all-or-none means nothing can stop it” misleading?

4. An image shows a gray butterfly with a central canal. Name the anterior motor region and the adjacent outward route.

REMEMBER Take turns: answer without looking, explain why, then reveal the next slide.

Book pp. 399–401, 409–419, 437–441. Original practice questions.

Mixed cases | connect the chapters | answers

1. Less vesicle release and a smaller postsynaptic effect are expected in the chapter's synapse model.

2. Motor-output deficit: ventral roots carry efferent axons. Dorsal roots carry sensory input.

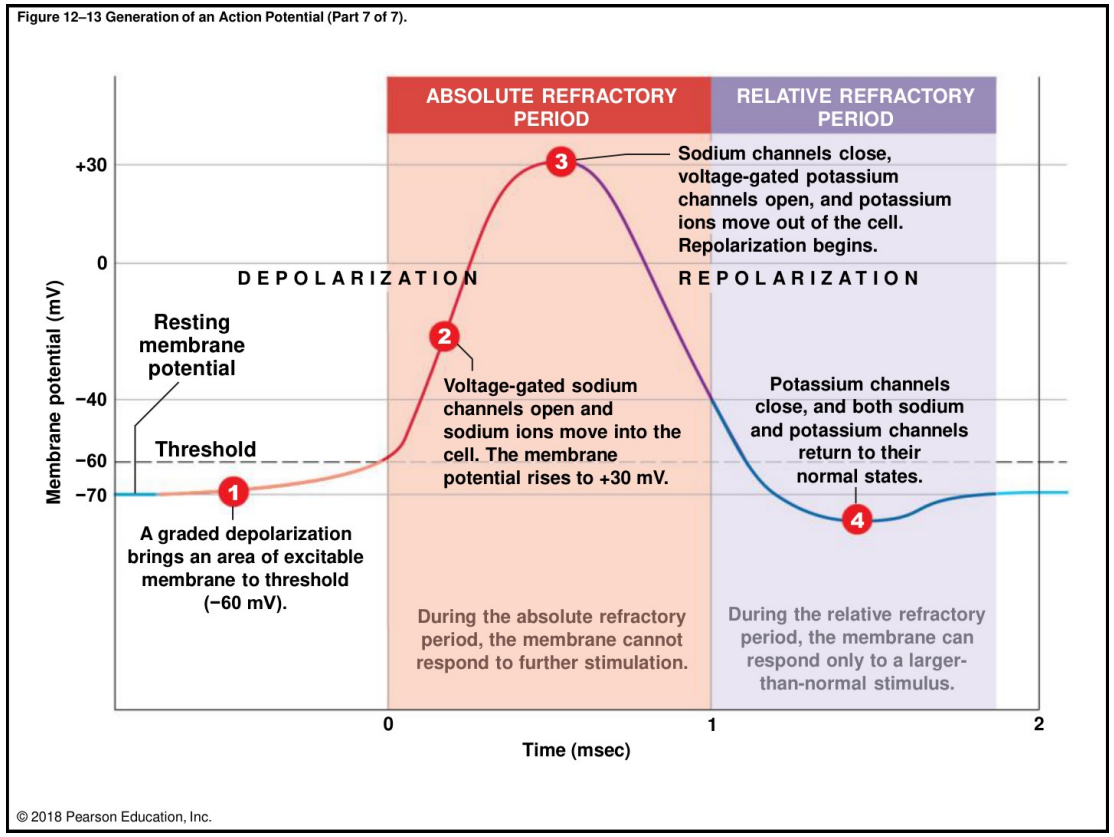
3. All-or-none describes firing once threshold is reached in excitable membrane. Successful propagation still requires a functioning pathway.

4. Anterior/ventral horn contains somatic motor nuclei; their axons leave through the ventral/anterior root.

REMEMBER A correct label is only half the job. Explain the structure, mechanism, or consequence.

Book pp. 399–401, 409–419, 437–441. Original practice questions.

Hand-draw practice 1: neuron and action potential



NEURON

Sketch soma, dendrites, axon hillock, initial segment, myelinated axon, nodes and terminals. Add the direction of signaling.

LOCAL VS LONG DISTANCE

Label graded potentials near the receiving region and AP propagation along the axon.

VOLTAGE CURVE

Draw −70 mV rest, −55 mV lecture threshold, +30 mV peak, falling phase and undershoot.

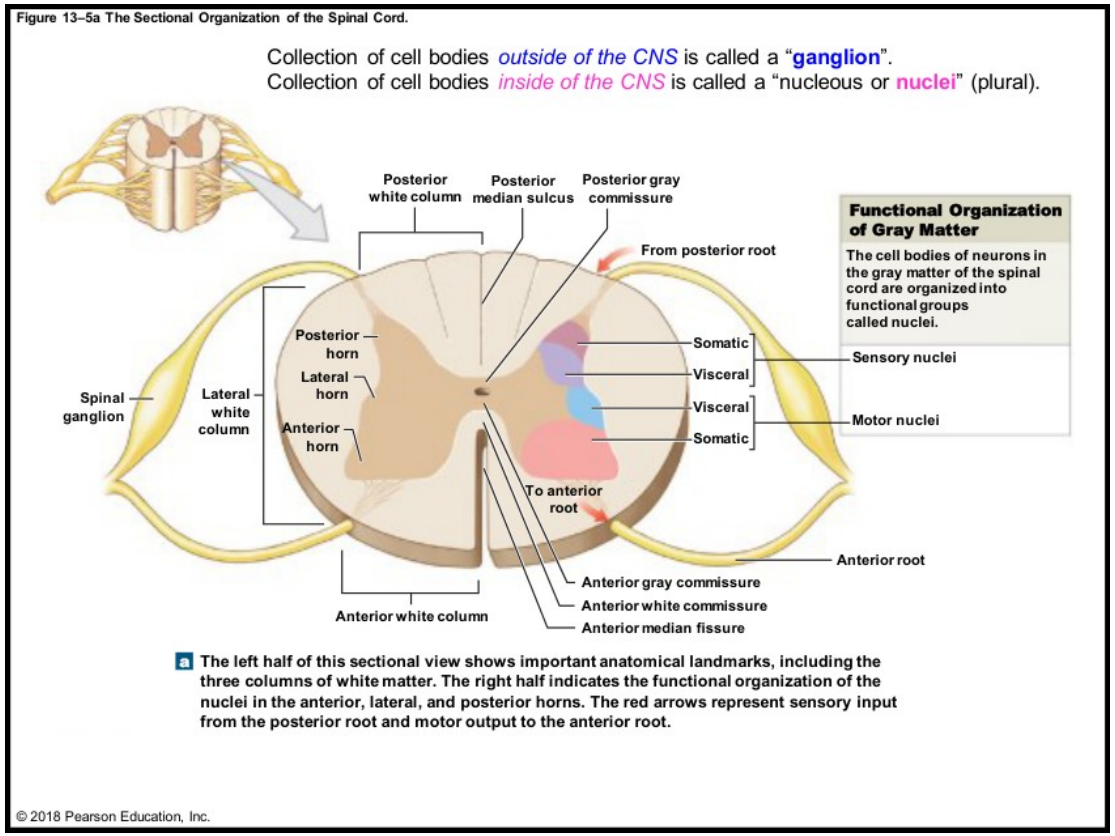
GATES

Write Na+ in on the rise and K+ out on the fall. Mark absolute versus relative refractory intervals.

REMEMBER Draw it from memory first; compare afterward. This slide is for practice—not an image to print on your exam sheet.

Book Figs. 12–2, 12–13; C12 L1/L2.

Hand-draw practice 2: cord and reflex arc



CORD

Draw a circle with a gray butterfly and central canal. Mark posterior sulcus and anterior fissure.

PATHWAYS

Add dorsal root + DRG, ventral root, mixed spinal nerve and two mixed rami.

FUNCTIONS

Place the S–V–V–S nuclei order and label sensory in / motor out.

REFLEX

Draw receptor, sensory neuron, integration, motor neuron and effector. Add antagonist inhibition or a crossing branch for a more complex reflex.

REMEMBER A simple, correct sketch beats a tiny copied picture—and printed pictures are not allowed.

Book Figs. 13–5, 13–14; C13 L1/L3; T-C exam rule.

Study Assistance Sheet: the recorded rule

PAPER

One side of a standard sheet of paper; do not use the reverse side.

TEXT

Typed or handwritten text is permitted in the recording.

VISUALS

Any image must be hand-drawn. Do not print, photocopy or shrink a diagram onto the sheet.

USE IT WELL

The professor recommends a few hard-to-retain items—not an unreadable ocean of text. The supplied PDF is text-only and leaves a blank area for your own sketches.

REMEMBER Print one side only, at actual size. Add any desired diagrams by hand.

T-C: Pasted text (3).txt, closing exam instructions.

What the professor explicitly prioritizes

Memorize / explain	De-emphasized details	Keep in the review
Graded vs AP; ion gates; pump; glia; synapses	Not a license to skip spoken explanations	Statements made in class can be testable
S–V–V–S, roots, mixed nerves, pathway routes	Complete dermatome map memorization	Definition and localization purpose
Four plexus ranges; phrenic C3–C5; named nerve targets	Every individual nerve's root range; full PT-level wiring	Phrenic roots are the explicit exception
Reflex steps, circuitry and classification	Detailed foot sensory-map memorization	Later handout/book reflexes still included

REMEMBER A priority marker is evidence from the recording—not a promise that other assigned material is excluded.

T-B, T-C, T-D; all six handouts.

Source cautions 1 | do not memorize transcription errors

Issue	Safer source-grounded distinction
Multipolar = always motor	Motor examples are multipolar, but interneurons can be multipolar too (Book pp. 394–395).
Perikaryon includes nucleus	Book p. 392: perikaryon is cytoplasm around the nucleus; soma is the cell body.
Pump off = always −90 mV	The book's −90 mV example requires K+-selective permeability assumptions (pp. 402–405).
Adenylate cyclase turns into cAMP	Adenylate cyclase catalyzes ATP conversion to cAMP (p. 424).
Dopamine is an endorphin	Book separates dopamine (amine) from opioid peptides such as endorphins (pp. 420–423).

REMEMBER Where sources differ, learn the mechanism and retain the source distinction.

Book pp. 392–395, 402–405, 420–424; T-A–T-C.

Source cautions 2 | preserve the limits of the evidence

Issue	How this deck handles it
Innate reflexes can never change	Innate means inborn. Book p. 457 and section 13-8 allow modulation/suppression.
Contralateral means both sides	Contralateral means opposite side. The combined reflex behavior involves both sides (pp. 455, 459).
Four neural circuit patterns	The diagram and textbook actually list five (pp. 452–453).
Expanded antidepressant list / “laughing gas”	Those transcript identifications are not established by the supplied book; do not use them as pharmacology facts.
Cruciform / vertebral attachment wording	Book pp. 438–439 describes pia-to-dura denticulate ligaments and dural/coccygeal longitudinal support.

REMEMBER The transcript is evidence of class emphasis, not an error-free scientific reference.

Book pp. 438–439, 452–461; C13 L3; T-C/T-D.

Key terms | cells and signaling structures

Term	Definition to be able to say aloud
Soma / perikaryon	Cell body / cytoplasm of the cell body around its nucleus
Nissl bodies	Clusters of rough ER and free ribosomes involved in protein synthesis
Axolemma / axoplasm	Axon plasma membrane / axon cytoplasm
Collateral / telodendria	Side branch of an axon / fine terminal branches
Neurolemma / internode / node	Outer Schwann-cell covering / myelinated segment / gap between segments
Synapse / effector	Communication junction / target that produces a response

REMEMBER Pair the word with a location on a drawing—not just another word.

Key terms | electrical and chemical mechanisms

Term	Meaning
Electrochemical gradient	Combined electrical and concentration forces acting on an ion
Threshold	Membrane voltage at which an AP is initiated in the excitable region
Refractory period	Interval of absent or reduced responsiveness after AP initiation
Summation	Combination of graded potentials across time and/or space
Cholinergic / adrenergic	ACh signaling / NE or related adrenergic signaling in the lecture
Neuromodulator / second messenger	Chemical altering synaptic activity / intracellular signal generated after receptor activation

REMEMBER For every mechanism, identify the ion, the channel, and the membrane location.

Key terms | spinal structures and pathways

Term	Meaning
Nucleus / ganglion	CNS / PNS collection of neuronal cell bodies
Horn / column / tract	Gray projection / white-matter region / functionally related axon bundle
Commissure	Connection crossing the spinal midline
Rootlet / root / ramus	Small root contribution / cord connection / nerve branch
Plexus / dermatome	Interwoven nerve network / spinal-nerve skin territory
Preganglionic / postganglionic	Before / after the autonomic ganglion relay

REMEMBER Structure names tell you location, scale, or the path a signal takes.

Key terms | reflex vocabulary

Term	Meaning
Neuronal pool	Functional group of interconnected neurons
Reflex arc	Neural pathway producing a reflex response
Reciprocal inhibition	Excitation of one muscle group paired with inhibition of its antagonist
Ipsilateral / contralateral	Same side / opposite side relative to the stimulus
Intrafusal / extrafusal	Within the muscle spindle / surrounding force-producing muscle fibers
Reinforcement	Facilitation that enhances a reflex response

REMEMBER Never answer “mono” or “poly” without saying what you are counting.

Additional branch names | cervical and brachial

Branch(es)	Associated region or muscle in the textbook
Great auricular, lesser occipital	Skin around ear, neck and posterior/superior scalp
Transverse cervical, supraclavicular	Anterior neck; neck/shoulder skin
Ansa cervicalis	Listed neck/hyoid-associated muscles
Axillary; suprascapular	Deltoid/teres minor; supraspinatus/infraspinatus
Dorsal scapular; long thoracic	Rhomboids/levator scapulae; serratus anterior
Pectoral; subscapular; thoracodorsal	Pectoral muscles; subscapularis/teres major; latissimus dorsi

REMEMBER Reference only: the lecture does not require every branch or its individual root range.

Book pp. 447–449; C13 L2.

Additional branch names | lumbar and sacral

Branch(es)	Associated region or role in the textbook
Iliohypogastric / ilio-inguinal	Abdominal-wall muscles and selected lower-abdominal/groin skin
Genitofemoral	Anteromedial thigh and parts of external genital skin
Lateral femoral cutaneous	Cutaneous territory of the thigh
Posterior femoral cutaneous	Posterior thigh/perineal cutaneous territory
Pudendal	Perineal muscles and external genital skin
Tibial / fibular / sural / saphenous	Named lower-limb branches shown in the sensory distribution figures

REMEMBER Recognize these names; prioritize femoral, obturator, gluteal and sciatic targets first.

Book pp. 450–451; C13 L2 slides 10–13.

Clinical vocabulary | Chapter 12

Term	Source-based meaning
Anesthetic / neurotoxin	Agent reducing sensation / compound interfering with nervous function
Anticholinesterase	Blocks ACh breakdown by AChE
Atropine / d-tubocurarine	The text's examples of interfering with ACh receptor actions in different target tissues
Excitotoxicity	Excessive neurotransmitter stimulation that can damage neurons
Neuropathy / neuroblastoma	Peripheral sensory symptoms / malignant tumor of neuroblasts
Tay–Sachs / dysthymia	The chapter's lipid-metabolism/neuron-deterioration example / persistent depressive-condition term

REMEMBER These are assigned-text vocabulary, not treatment instructions or confirmed exam emphasis.

Clinical vocabulary | Chapter 13

Term	Source-based meaning
Areflexia / spinal shock	Absent reflexes / transient depressed reflex activity below a cord injury
Paraplegia / quadriplegia	Lower-limb paralysis / upper- and lower-limb involvement
Hemiparesis	Weakness affecting one side
Brown-Séquard syndrome	Unilateral cord lesion pattern: source describes same-side weakness/proprioceptive loss and opposite-side pain/temperature loss
Erb palsy / Kernig sign	Brachial-plexus stretch injury example / meningeal irritation sign described in the text
Myelography / nerve conduction study	Contrast imaging of spinal structures / testing conduction through a nerve pathway

REMEMBER Match the term to the structure or function affected; do not turn a single sign into a diagnosis.

Final teach-back: can you explain the whole route?

FROM STIMULUS

Describe a receptor, a graded potential, threshold and the resulting sensory AP.

THROUGH THE CORD

Trace the mixed nerve, dorsal root and spinal integration; distinguish a local reflex from an ascending pathway.

TO THE RESPONSE

Trace motor output through a ventral root, identify an appropriate nerve/effector, and describe the chemical synapse.

IF SOMETHING CHANGES

Predict the effect of blocked Na+ channels, reduced Ca2+ entry, myelin damage, a root lesion or loss of descending inhibition.

REMEMBER The goal is a connected explanation—not a stack of disconnected definitions.

Coverage map | every learning-outcome section

CHAPTER 12

12-1 Divisions	**5–7**
12-2 Neurons	**8–12**
12-3 Neuroglia	**13–17**
12-4 Membrane potential	**21–27**
12-5 Action potentials	**28–34**
12-6 Synapses	**38–41**
12-7 Transmitters / receptors	**42–44**
12-8 Integration	**45–48**

CHAPTER 13

13-1 Organization	**52**
13-2 Cord + meninges	**52–57**
13-3 Gray/white matter	**58–62**
13-4 Nerves + plexuses	**60–77**
13-5 Neuronal pools	**81–82**
13-6 Reflex arc / categories	**83–85**
13-7 Reflex mechanisms	**86–92**
13-8 Brain influence	**93–94**

REMEMBER Use chapter sections to find any concept you need to revisit.

Book learning outcomes on pp. 389 and 433.

Source index | the exact files used

TEXTBOOK

chapters12-13.pdf: Chapter 12, pp. 389–432 (PDF pages 1–44); Chapter 13, pp. 433–464 (PDF pages 45–76).

CHAPTER 12 HANDOUTS

233 Ch12_Lecture_01_Student HO.pdf (11 slides); _02_ (15 slides); _03_ (7 slides).

CHAPTER 13 HANDOUTS

233 Ch13_Lecture 01 Handout.pdf (15 slides); Lecture 02 (13 slides); Lecture 03 (10 slides).

FIGURES

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REMEMBER Source footers use printed textbook page numbers and original handout slide numbers.

The seven uploaded PDFs.

Transcript index | distinguish the similarly named files

T-A | ORGANIZED SUMMARY

Pasted text (4).txt — Chapter 12 foundations, glia, membrane potentials, propagation and synaptic overview. Not a verbatim transcript.

T-B | RAW TRANSCRIPT

Pasted text (2).txt — Chapter 12 recap, ion gradients, gated channels and graded/action potentials.

T-C | RAW TRANSCRIPT

Pasted text (3).txt — Synapses and integration, early spinal-cord anatomy, plus the one-sided note-sheet rule.

T-D | RAW TRANSCRIPT

Pasted text(2).txt — Spinal pathways, plexuses, neural circuits and the beginning of reflexes. Note the missing space before “(2)”.

REMEMBER No claim is made that these files contain every class session or the complete actual exam scope.

The four uploaded text files; exact filenames preserved.